CountBCF: Bayesian Causal Forests for Count and Zero-Inflated Count Outcomes

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Abstract

Count outcomes, such as hospital admissions, infection incidences, insurance claims, and single-cell read counts, are pervasive in causal research. A significant portion of these outcomes are also zero-inflated, featuring structural zeros from units that never enter the count-generating process. Estimating average and heterogeneous treatment effects on these responses is statistically demanding: traditional parametric count models impose rigid, easily violated covariate-outcome assumptions, while flexible continuous-outcome learners ignore discreteness, mean-variance coupling, and the structural-zero channel. This paper introduces **CountBCF**, a flexible Bayesian model that makes no parametric assumptions on how covariates relate to the outcome. From a single posterior fit, CountBCF delivers both the average treatment effect (ATE) and the per-unit conditional average treatment effect (CATE). In zero-inflated settings, the CATE decomposes in closed form into a count-rate channel and a structural-zero channel, exposing sign-changing heterogeneity that single-channel methods cannot detect. We release the open-source textttcountbcf R package and show in simulations that CountBCF provides better-calibrated CATE inference than continuous-outcome and high-flexibility baselines while remaining competitive for ATE estimation, with correctly specified parametric models retaining their efficiency in simple settings. In an applied analysis of solid-cancer incidence in the Radiation Effects Research Foundation Life Span Study, it reproduces the established average radiation-risk pattern and recovers its principal age-at-exposure and attained-age gradients without pre-specifying their interaction form. The same analysis illustrates how the forest can identify a departure from a canonical model and how epidemiological sensitivity analyses can distinguish a reference-group artifact from a new radiation effect.

Keywords: Bayesian causal forests; Conditional average treatment effects; Count regression; Heterogeneous treatment effects; Log-linear BART; Poisson and negative binomial regression; Zero-inflated regression; Structural zeros.

1 Introduction

Many of the outcomes that motivate modern causal inference are counts [Choi and Ni, 2023a]. Hospital admissions and emergency-department visits [DEB and TRIVEDI, 1997], infection incidences [Chen et al., 2019], traffic crashes [Shankar et al., 1997], insurance claims [Shi and Valdez, 2012], citation totals [Yan et al., 2011], recidivism events [LAND et al., 1996], and many more are all

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integer-valued and frequently bunched at zero. The applied literatures that produced these outcomes – epidemiology [Lee and Neocleous, 2010], economics [Winkelmann and Zimmermann, 1994], ecology [Richards, 2007], social sciences [Hilbe, 2025], etc – share an interest not just in whether an intervention works on average: the average treatment effect (ATE), but in for whom it works, and by how much: the conditional average treatment effect (CATE). The past decade of heterogeneous treatment effect (HTE) methodology has answered that demand with flexible nonparametric tools – Bayesian Additive Regression Trees [Chipman et al., 2010, Hill, 2011], Bayesian Causal Forests [BCF; Hahn et al., 2020], generalised random forests [Athey et al., 2019], meta-learners [Künzel et al., 2019], and double machine learning [Chernozhukov et al., 2018] – but the bulk of that methodology assumes a continuous, approximately Gaussian response.

Treating counts as if they were Gaussian has three well-documented failure modes [Cameron and Trivedi, 2013, Murray, 2021]. Count distributions place positive mass at zero and at small integers, often on a large fraction of the support; the conditional variance grows with the conditional mean, often super-linearly under overdispersion; and the substantive scientific quantity is almost always a rate or a relative risk, both of which live on the log scale. These pathologies bias point estimates, distort uncertainty quantification, and degrade the heterogeneity signal that HTE methods are designed to recover. A second, sharper failure occurs when the count outcome is also zero-inflated (ZI): a non-trivial fraction of units never enter the count-generating process at all (a structural zero), while the rest accumulate strictly positive counts at heterogeneous intensities. A canonical example is annual healthcare utilisation [Chen et al., 2019]: in many provider panels and insurance cohorts, a sizeable fraction of enrollees record zero visits in a given year because they are effectively out of the care-seeking system, while the remaining enrollees record strictly positive visit counts at heterogeneous intensities. The same pattern arises in manufacturing defect counts, automobile insurance claims, ecological species-abundance surveys, sentencing and recidivism counts, and single-cell RNA sequencing, where a large excess-zero mass reflects technical dropout on top of a count process.

What makes zero-inflation especially awkward for causal inference is that an intervention can move both mechanisms, and the two movements need not point in the same direction [Choi and Ni, 2023b]. Continuing the healthcare example: a workplace wellness programme aimed at preventive care could plausibly pull some enrollees into the system – shrinking the structural-zero fraction – while simultaneously lowering the visit intensity of those already engaged because preventive care substitutes for acute episodes. The response-scale CATE is then a mixture of the two channel effects; whether it ends up positive, negative, or near zero depends on which channel dominates for which unit. A method that ignores the structural-zero mechanism necessarily mis-attributes its movement: if the treatment shifts the structural-zero probability but leaves the count rate untouched, a Poisson-only working model has to absorb the entire change into the rate estimate, biasing it; the same mis-attribution flows in the opposite direction when the treatment moves the count rate but not the structural-zero probability [Choi and Ni, 2023b].

Faced with a count outcome and a CATE question, an applied analyst today reaches for one of three default tools, each of which gives up something the analysis arguably needs. The first is a Gaussian HTE method, most commonly BCF or causal BART, applied to the raw counts as if they were continuous; this preserves the regularisation and uncertainty quantification practitioners trust, but commits all three Gaussian pathologies above. The second is a parametric Poisson, negative binomial, ZIP, or ZINB generalised linear model with treatment-by-covariate interactions [Cameron and Trivedi, 2013, Lambert, 1992, Greene, 1994], possibly stabilised by propensity weighting [Rosenbaum and Rubin, 1983] or a doubly robust correction [Chernozhukov et al., 2018]; this respects the count likelihood but recovers heterogeneity only through pre-specified interaction terms, defeating the nonparametric premise that motivated the move to HTE methods in the first

place. The third is a generalised random forest with a Poisson moment condition [Athey et al., 2019], or a related relative-risk-targeted estimator [Kennedy, 2024]; this is honest about the count structure but discards the two structural priors that distinguish BCF from other forest-based HTE methods, leaving heterogeneity estimates exposed to the high-variance behaviour that motivated those priors to begin with. Two-part and hurdle learners [Mullahy, 1986] separate the channels of a zero-inflated outcome but do not share covariate information across them and do not implement BCF differential shrinkage.

The structural priors in question are the distinguishing features of Hahn et al. [2020]. Their construction decomposes the conditional mean of the response into a prognostic function – what the outcome would be in the absence of treatment, as a smooth function of the covariates – and a moderating function – by how much treatment shifts that baseline at each point in covariate space. The two functions are fit by separate BART ensembles with deliberately different priors: the prognostic forest is large and flexible; the moderating forest is small and tightly shrunk toward zero, reflecting the prior belief that treatment effect heterogeneity is typically smoother and weaker than the prognostic surface itself. This differential shrinkage stabilises CATE estimates in finite samples without forcing them to be constant. The second structural ingredient is the inclusion of an estimated propensity score among the inputs to the prognostic forest only. Hahn et al. [2020] show that omitting this control creates regularisation-induced confounding (RIC): the moderating forest absorbs prognostic signal that the regularised prognostic forest underfits, biasing CATE estimates in the direction of the propensity score. Both pathologies are at least as severe under a log-link count likelihood as under a Gaussian one: the multiplicative geometry of the log scale amplifies the consequences of prognostic underfitting, and the count likelihood's tight mean-variance coupling leaves the moderating forest doubly exposed to RIC bias. The same arguments apply per channel in the zero-inflated case.

Our contribution. This paper develops **CountBCF**, a unified Bayesian causal forest framework for count and zero-inflated count outcomes that retains both of BCF's structural ingredients while replacing its Gaussian backbone with a log-linear count likelihood. The framework covers two regimes within a single model and a single sampler:

1. **A count-faithful BCF.** For non-zero-inflated count outcomes, we extend the BCF of Hahn et al. [2020] from a Gaussian to a Poisson and a negative binomial likelihood by routing both the prognostic and the moderating forest through the log-linear BART backbone of Murray [2021]. The result is a BCF that respects discreteness, mass at zero, and mean-variance coupling without abandoning the prognostic/moderating decomposition that defines the BCF family.

2. **A six-forest BCF decomposition for zero-inflated outcomes.** Murray's log-linear BART parameterises a zero-inflated count likelihood through three function groups: the count log-rate f, the structural-zero log-odds f_0, and the not-structural-zero log-odds f_1, with the structural-zero probability identified as $\sigma(f_0 - f_1)$. We replace each group's single forest with the BCF propensity-adjusted prognostic / moderator pair, yielding six forests in total: (μ_f, τ_f) for the count channel and $(\mu_{f_0}, \tau_{f_0}), (\mu_{f_1}, \tau_{f_1})$ for the ZI channel. The non-zero-inflated case is recovered as the three-forest sub-model.

3. **RIC protection on the log scale, per channel.** We retain the propensity-as-covariate fix and the differential-shrinkage prior structure of BCF, with the prognostic forest of each function group receiving more trees and a weaker zero-bias prior than the corresponding moderating forest. The leaf prior is the GIG mixture of Murray [2021], conjugate under a latent-variable augmentation; concentration hyperparameters are calibrated separately for

the count and ZI groups to reflect their different natural scales, and within each group the moderating forest, with fewer trees and a stub-biased tree-shape prior, is shrunk strictly more aggressively than its prognostic counterpart.

4. **Closed-form response-scale heterogeneity and channel decomposition.** The log-linear parameterisation means that, given posterior draws of the forests at any covariate value, the response-scale conditional treatment effect is available without any additional Monte Carlo step: it is a deterministic function of the stored forest values. We report per-unit posterior means, 50% and 95% credible intervals, and population-averaged ATE summaries, all on the response scale. In the zero-inflated case the same posterior draws produce a per-unit decomposition of the CATE into a count-rate channel $\Delta\lambda(x)$ and a structural-zero channel $\Delta p_{\text{zi}}(x)$.

5. **Open software and a pre-registered simulation study.** We release `countbcf`, an open-source R package built on the `countbart` MCMC backend of Wikle and Zigler [2023], with a single user-facing entry point that consumes a treated/control split and returns the posterior draw matrices needed for the closed-form CATE. The package is accompanied by a pre-registered simulation study (Section 6) that crosses linear and nonlinear data-generating processes with Poisson, NB, ZIP, and ZINB likelihoods and sweeps one-at-a-time over sample size, covariate dimension, and target average treatment effect, scoring all methods against a rate-scale truth.

The remainder of the paper proceeds as follows. Section 2 places CountBCF in the literature on count and zero-inflated causal inference and identifies the specific gap it fills. Section 3 fixes notation, including the latent structural-zero indicator and the response-scale potential-outcome means that are the paper's target estimand. Section 4 reviews the log-linear BART machinery, the GIG-mixture leaf prior, and the redundant (f_0, f_1) parameterisation that makes the ZI leaf updates conditionally conjugate. Section 5 introduces CountBCF proper: the likelihood and parameterisation in both the count and the zero-inflated regimes, the differential-shrinkage prior structure, the augmented Gibbs sampler, and the closed-form response-scale CATE and channel-decomposition summaries. Section 6 registers the simulation study used to evaluate CountBCF against a Gaussian-BCF baseline. Sections 7 and 8 are reserved for the simulation studies results and the application of CountBCF in a real-world setting.

2 Relationship to previous literature

This section reviews the literatures that bear on CATE estimation for count and zero-inflated count outcomes. The cell of the methodological grid in which CountBCF sits – causal × count-aware (with or without zero-inflation) × nonparametric Bayesian – is sparsely populated; the subsections that follow walk the adjacent cells in turn before Section 2.4 states the gap that CountBCF fills.

2.1 Causal inference for count outcomes: parametric and semiparametric eras

Causal analysis of count outcomes has its longest tradition in the parametric era of generalised linear models. The Poisson and negative binomial GLMs with a log link [Cameron and Trivedi, 2013] deliver interpretable rate-ratio summaries of the average causal effect under an unconfoundedness assumption and a correctly specified mean function. These models extend cleanly to the negative binomial when equi-dispersion fails, and when unconfoundedness is plausible only conditionally on covariates, the prescription is essentially the same as for continuous outcomes: condition on

the covariates or on the propensity score [Rosenbaum and Rubin, 1983]. The two-stage residual inclusion of Terza et al. [2008] and the marginal structural models of Robins et al. [2000] relax the unconfoundedness assumption in different ways while keeping the parametric log-linear-mean structure. Their common feature is that they target the average effect under a fixed functional form: heterogeneity, if permitted at all, enters through pre-specified treatment-by-covariate interactions, which is workable for a small number of binary moderators but not for a moderate-dimensional covariate vector with unknown interaction structure.

The semiparametric and doubly robust literature relaxes the parametric programme's commitment to a single correct outcome model. Funk et al. [2011] survey doubly robust estimators across continuous and discrete outcomes, including Poisson outcome models paired with logistic propensity models. van der Laan and Gruber [2010] and van der Laan and Rose [2011] develop targeted maximum likelihood estimation (TMLE) as a general semiparametric framework whose construction extends naturally to count outcomes via a Poisson or negative binomial submodel for the targeted update step. Chernozhukov et al. [2018] introduce the double / debiased machine learning (DML) framework: the outcome regression and propensity score are fit by arbitrary high-quality machine learners, cross-fitted across folds to avoid overfitting bias, and combined in a Neyman-orthogonal score whose root yields an asymptotically normal ATE estimate. DML is outcome-model-agnostic: a Poisson-loss gradient boosting machine, a neural network, or BART can sit in the outcome slot without changing the orthogonality property. By design these methods target the ATE rather than the per-unit CATE; heterogeneity, when reported at all, is typically delivered by stratifying across pre-specified subgroups or by a separate post-hoc fit of the outcome regression – neither of which preserves the orthogonality property that motivated the doubly robust construction. We adopt the spirit of the doubly robust programme – propensity adjustment of the outcome regression – via the route of Hansen [2008] and Hahn et al. [2020], but reject the restriction to the average effect.

2.2 Classical and causal zero-inflated count models

The starting point for any zero-inflated analysis is the ZIP regression of Lambert [1992], which writes the response as a two-component mixture: with probability $\pi(x)$ the unit is a structural zero, and otherwise it is drawn from a Poisson with rate $\lambda(x)$. The hurdle alternative [Mullahy, 1986] factors the joint density into a binary "any vs. none" model and a zero-truncated count model; hurdle treats all zeros as structural while ZIP allows sampling zeros from the count component as well. Greene [1994] extends both constructions to the negative binomial case. Hall [2000] develops a random-effects ZIP for repeated-measures data, and Li [2012] settles the identifiability question: in the absence of constraints, the (π, λ) pair is not jointly identified from the marginal distribution of Y, but a logistic link on π together with a log link on λ and at least one continuous covariate restore identifiability. The `pscl` package of Zeileis et al. [2008] consolidates the count GLM toolkit – Poisson, NB, ZIP / ZINB, hurdle – in a single `R` treatment.

When interest shifts from describing Y to estimating a treatment effect, the two-component structure of ZI data complicates matters in two ways. First, the response-scale estimand is itself a sum of two interpretable contrasts: a shift in the structural-zero probability and a shift in the count rate. Second, identifying each contrast separately requires that the ignorability assumption be combined with the link-function restrictions of Li [2012] or with an instrument acting on only one channel. The naive plug-in strategy – estimate a ZIP / ZINB regression with treatment-covariate interactions and report the implied mean contrast – inherits all of the distributional rigidity of the underlying GLM. The two-part learner – fit a binary CATE for $\mathbf{1}\{Y_i > 0\}$ and a count CATE for $Y_i \mid Y_i > 0$, then combine – correctly separates the channels but loses joint efficiency across them and does not share covariate information. van der Laan and Rose [2011]

discuss how doubly robust / TMLE constructions adapt to these working models; Kennedy [2024] formalises the closely related mismatch between absolute-risk partitioning criteria and relative-risk targets. Guo et al. [2017] develop an IV mediation analysis for ZI counts that defines indirect effect ratios suitable for count outcomes. Yu et al. [2022] propose a Multiplicative Structural Nested Mean Model for ZI nonnegative longitudinal outcomes. Oganisian et al. [2021] build a Bayesian nonparametric model using Dirichlet process mixtures that jointly predict structural zeros, individualised treatment effects, and latent subgroups. Choi and Ni [2023b] approach ZI outcomes from the causal-discovery direction. Lukusa et al. [2017] review the closely related literature on zero-inflated models with missing data. Each of these methods commits either to a particular longitudinal structural form, to a single response-scale estimand, or to a likelihood that does not isolate the structural-zero and count components for inference.

2.3 Heterogeneous-effects methods and the BART line

Methodological work on HTE has been dominated by Gaussian or binary outcomes. Hill [2011] introduced BART [Chipman et al., 2010, 1998] into the causal inference literature as a flexible learner for the outcome regression under unconfoundedness; her construction estimates a single BART model on (X, Z) and reads the CATE off as the difference of the model's predictions at $Z = 1$ and $Z = 0$. This estimator inherits BART's strong empirical performance and produces a fully Bayesian posterior over per-unit treatment effects, but it ties the prognostic and moderating components of the conditional mean to a single tree-structure prior, making no provision for the asymmetric regularisation that distinguishes heterogeneity estimation from outcome estimation, and – applied to a count outcome – it inherits the three Gaussian failure modes catalogued above.

Athey et al. [2019] introduce generalized random forests (GRF), a non-Bayesian alternative built on random forests with locally weighted moment conditions. GRF accommodates a Poisson moment for count outcomes; the asymptotic theory of Wager and Athey [2018] establishes pointwise consistency and asymptotic normality. GRF with a Poisson moment shares a feature CountBCF aims to escape: the splitting criterion targets local likelihood improvement, not relative-risk heterogeneity. Kennedy [2024] formalises this observation. Künzel et al. [2019] catalogue the standard meta-learners (S, T, X, DR), which are estimator-agnostic recipes that delegate the regression and propensity work to a user-chosen base. In principle a meta-learner can be deployed with a count-aware base, but the published meta-learner studies use Gaussian or binary base learners; meta-learners do not by themselves address RIC, and they offer no off-the-shelf solution for ZI counts.

The fourth strand – and the one from which CountBCF draws its computational backbone – is the line of work on BART and its count-aware extensions. Murray [2021] extends BART to count, binomial, and multinomial outcomes by replacing the Gaussian likelihood with a log-linear count likelihood (Poisson or negative binomial) and replacing the Gaussian leaf prior with a generalised inverse Gaussian (GIG) mixture leaf prior whose hyperparameters are calibrated so that, after exponentiation, the implied prior on the forest sum has zero mean and a user-controlled marginal variance. The mixture structure is engineered so that, after a Gamma latent-variable augmentation, the GIG mixture is conjugate. Section 4.2.2 of Murray [2021] extends the construction to zero-inflation through the redundant parameterisation $p_{\mathrm{zi},z}(x) = \sigma\big(f_0(x) - f_1(x)\big)$, in which each of f_0 and f_1 carries its own log-linear BART prior, and the data drive symmetry-breaking through latent indicators. The redundancy is precisely what makes the leaf updates conditionally conjugate under a symmetric GIG-mixture prior. Wikle and Zigler [2023] package both constructions into the `countbart` R/C++ implementation that the present paper extends.

Hahn et al. [2020] introduce BCF as the BART-based answer to the CATE-estimation problem

under unconfoundedness, with the differential-shrinkage and propensity-as-covariate ingredients catalogued in Section 1. The construction has antecedents in the prognostic score literature of Hansen [2008] and the RIC argument of Hahn et al. [2018]. Wikle and Zigler [2023] restrict their count BART implementation to the predictive setting; they do not pursue the BCF decomposition. The combination of the BCF decomposition with the log-linear count likelihood (with or without zero-inflation), which is the content of the present paper, has not appeared in the literature.

2.4 Identified gap

To summarise: the predictive count and ZI-count cells are well populated, anchored by Murray [2021] and the `countbart` implementation of Wikle and Zigler [2023]. The causal Gaussian cell is well populated, anchored by Hahn et al. [2020] on the Bayesian side and by generalized random forests [Athey et al., 2019, Wager and Athey, 2018] on the frequentist side. The causal count and causal ZI-count cells are sparsely populated, and within them no existing method combines (i) the propensity-adjusted (μ_f, τ_f) decomposition that has made BCF the de-facto Bayesian benchmark for heterogeneous causal effects, with (ii) a count-aware (and where applicable, zero-inflation-aware) likelihood that respects the discreteness, overdispersion, and structural-zero process of integer outcomes. The closest prior work either retains the Gaussian likelihood (BCF; Hahn et al., 2020), drops the BCF decomposition (log-linear BART, with or without ZI; Murray, 2021), separates the two ZI channels and loses joint efficiency (two-part / hurdle learners; Mullahy, 1986; Guo et al., 2017; Yu et al., 2022; Oganisian et al., 2021), or replaces the Bayesian machinery with frequentist random forests targeting an absolute-risk or relative-risk criterion [Athey et al., 2019, Kennedy, 2024]. **CountBCF fills this gap.** It routes the prognostic and moderating forest of each log-linear BART function group through the backbone of Murray [2021] while preserving Hahn et al.'s propensity-adjusted differential-shrinkage prior structure, yielding a count-aware Bayesian causal forest with closed-form per-unit response-scale CATEs, a per-unit channel decomposition in the zero-inflated case, and full posterior uncertainty.

3 Problem statement and notation

Let $Y_i \in \mathbb{Z}_{\geq 0}$ denote a count-valued response, $Z_i \in \{0, 1\}$ a binary treatment indicator, and $X_i \in \mathbb{R}^p$ a length-p vector of pre-treatment covariates. We observe an iid sample $\{(Y_i, Z_i, X_i)\}_{i=1}^n$ from some joint distribution. Capital Roman letters denote random variables, with realisations in lower case. When Y or Z appear without a subscript they denote length-n column vectors; $\mathbf{X}$ is the $n \times p$ design matrix of covariates.

In the potential outcomes framework [Imbens and Rubin, 2015], write $Y_i(0), Y_i(1) \in \mathbb{Z}_{\geq 0}$ for the counterfactual responses under control and treatment, with observed outcome $Y_i = Z_i Y_i(1) + (1 - Z_i) Y_i(0)$ (SUTVA). Throughout we assume strong ignorability:

$$Y_i(0), Y_i(1) \perp\!\!\!\perp Z_i \,\big|\, X_i, \tag{1}$$

$$0 \;<\; \pi(X_i) \;\equiv\; \Pr(Z_i = 1 \mid X_i) \;<\; 1, \tag{2}$$

the second of which (overlap) is required to identify treatment effects everywhere in covariate space. The estimated propensity score is denoted $\widehat{\pi}(X_i)$; for the simulation study of Section 6 we set $\widehat{\pi} = \pi$.

Latent structural-zero indicator. In the zero-inflated regime, the response splits into a structural-zero component and a count component. Let $S_i \in \{0, 1\}$ be the latent indicator that unit i is a structural zero:

$$S_i = 1 \implies Y_i = 0, \qquad S_i = 0 \implies Y_i \mid X_i, Z_i \sim \text{Poisson}(\lambda(X_i, Z_i)) \quad (\text{or NB}(\lambda, \kappa)), \tag{3}$$

and write the conditional structural-zero probability as

$$p_{\mathrm{zi},\,z}(X_i) \;\equiv\; \Pr(S_i = 1 \mid X_i, Z_i = z), \qquad z \in \{0,1\}. \tag{4}$$

The marginal count-component mean is $\lambda(x,z) = \mathbb{E}[Y_i \mid X_i = x, Z_i = z, S_i = 0]$; we parameterise $\log\lambda$ via a sum-of-trees prior in Section 5. The non-zero-inflated regime is recovered by setting $p_{\mathrm{zi},\,z} \equiv 0$ (no structural-zero process); equivalently, $S_i \equiv 0$ for all i.

Response-scale potential-outcome means. Combining the two channels yields the response-scale potential-outcome means

$$\mu_z(x) \;=\; \mathbb{E}[Y_i(z) \mid X_i = x] \;=\; \big(1 - p_{\mathrm{zi},\,z}(x)\big)\,\exp\big(\log\lambda_z(x)\big), \qquad z \in \{0,1\}. \tag{5}$$

Our target estimands are the response-scale CATE and ATE:

$$\tau(x) \;=\; \mu_1(x) - \mu_0(x), \qquad \mathrm{ATE} \;=\; \mathbb{E}_X\big[\tau(X)\big]. \tag{6}$$

When $p_{\mathrm{zi},\,z} \equiv 0$, (5) reduces to $\mu_z(x) = \exp(\log\lambda_z(x))$ and (6) becomes the plain count CATE.

Two-channel decomposition. In the zero-inflated regime, the response-scale CATE in (6) mixes a change in the count intensity and a change in the structural-zero probability. Methods that fit only one channel necessarily miss part of the effect. In Section 5.4 we therefore report – alongside τ – the two per-unit channel decompositions

$$\Delta\lambda(x) \;=\; \exp(\log\lambda_1(x)) - \exp(\log\lambda_0(x)), \qquad \Delta p_{\mathrm{zi}}(x) \;=\; p_{\mathrm{zi},\,1}(x) - p_{\mathrm{zi},\,0}(x). \tag{7}$$

Scales. Treatment effects naturally live on up to three scales:

- the count log-rate scale, on which the moderating forest $\tau_f(x)$ is parameterised; this is a relative-risk-like quantity, $\tau_f(x) \approx \log\{\lambda_1(x)/\lambda_0(x)\}$ in the non-zero-inflated case;
- the ZI log-odds scale, on which the treatment shift in η_{zi} is parameterised (zero-inflated regime only);
- the response (rate) scale, $\tau(x)$ in (6), against which all reported metrics are evaluated.

All relevant quantities are reported. The simulation study (Section 6) is scored against the rate-scale truth.

4 Background: log-linear BART

This section gives the condensed restatement of the log-linear BART construction of Murray [2021] – including the zero-inflation extension of §4.2.2 – that we need to define CountBCF in Section 5. The prognostic and moderating forests of each function group are individually log-linear BART regressions, so the leaf prior, the conditional sufficient statistics, and the latent-variable augmentation recapped here reappear unchanged inside the CountBCF Gibbs sampler. A reader already fluent in Murray [2021] §§3–4.2.2 may skip to Section 5.

4.1 The BART prior

A single BART regression [Chipman et al., 2010] expresses an unknown function $f : \mathbb{R}^p \to \mathbb{R}$ as a sum of m shallow regression trees,

$$f(\mathrm{x}) \;=\; \sum_{h=1}^{m} g(\mathrm{x};\; T_h, M_h), \tag{8}$$

where T_h is a binary tree partitioning $\mathbb{R}^p$ and $M_h = (\mu_{h,1}, \ldots, \mu_{h,b_h})'$ collects the leaf parameters at the b_h terminal nodes of T_h; the operator $g(\mathrm{x};\; T_h, M_h)$ returns $\mu_{h,t}$ when x falls in terminal node t of T_h. The tree prior of Chipman et al. [1998] assigns each internal node at depth d a split probability $\alpha\,(1+d)^{-\beta}$, with $(\alpha, \beta) = (0.95, 2)$ biasing trees toward stubs. With a Gaussian response, a Gaussian leaf prior is conjugate, and the posterior is sampled by a backfitting Gibbs scheme that updates one tree at a time using grow / prune / change / swap MH moves on T_h and a conjugate draw of $M_h \mid T_h$.

4.2 The log-linear extension

For non-negative responses one writes the same sum-of-trees representation on the log scale,

$$\log f(\mathrm{x}) \;=\; \sum_{h=1}^{m} g(\mathrm{x};\; T_h, M_h), \tag{9}$$

and reparameterises each leaf parameter as $\lambda_{h,t} = \exp(\mu_{h,t}) > 0$, so that $f(\mathrm{x}) = \prod_h \lambda_{h,t(\mathrm{x};T_h)}$. The Poisson and negative binomial regression models take the form

$$Y_i \mid \mathbf{X}_i \;\sim\; \mathrm{Poisson}\big(\mu_{0i}\, f(\mathbf{X}_i)\big), \qquad Y_i \mid \mathbf{X}_i \;\sim\; \mathrm{NB}\big(\mathrm{size} = \kappa,\; \mathrm{mean} = \mu_{0i}\, f(\mathbf{X}_i)\big), \tag{10}$$

with μ_{0i} a unit-specific offset and κ the NB dispersion, satisfying $\mathrm{Var}(Y_i) = \mu_{0i} f(\mathbf{X}_i) + \mu_{0i}^2 f(\mathbf{X}_i)^2/\kappa$. The Poisson model is the $\kappa \to \infty$ limit. Both likelihoods reappear inside the zero-inflated construction of Section 4.5: the not-structural-zero component of the ZI mixture is exactly the model in (10), while the structural-zero process is governed by an independent pair of log-linear BART components introduced below.

After collecting all but the hth tree's fit into $f^{-h}(\mathbf{X}_i) = \prod_{l \neq h} g(\mathbf{X}_i; T_l, \Lambda_l)$, the contribution of a single observation to the joint posterior over (T_h, Λ_h) has the conditionally Gamma-shaped form

$$L\big((T_h, \Lambda_h);\; T_{-h}, \Lambda_{-h}, y\big) \;\propto\; \prod_{t=1}^{b_h} \lambda_{h,t}^{r_{h,t}}\, \exp\!\big[-s_{h,t}\, \lambda_{h,t}\big], \tag{11}$$

with conditional sufficient statistics

$$r_{h,t} \;=\; \sum_{i:\, \mathbf{X}_i \in \mathcal{A}_{h,t}} u_i, \qquad s_{h,t} \;=\; \sum_{i:\, \mathbf{X}_i \in \mathcal{A}_{h,t}} v_i\, f^{-h}(\mathbf{X}_i), \tag{12}$$

$\mathcal{A}_{h,t}$ the leaf-t partition cell of T_h, and (u_i, v_i) likelihood-specific summaries that depend on the function group (see Sections 4.4 and 4.5).

4.3 The GIG-mixture leaf prior

Murray [2021] constructs a leaf prior conjugate to (11) from two pieces. The generalised inverse Gaussian (GIG) family has density

$$p_{\mathrm{GIG}}(\lambda \mid \eta, \chi, \psi) \;\propto\; \lambda^{\eta-1} \exp\Big[-\tfrac{1}{2}\big(\chi/\lambda + \psi\lambda\big)\Big], \qquad \lambda > 0, \tag{13}$$

covering the Gamma ($\chi = 0$) and inverse-Gamma ($\psi = 0$) families as boundary cases. The CountBCF leaf prior is the equal mixture

$$p\big(\lambda_{h,t} \mid c, d\big) \;=\; \tfrac{1}{2}\, p_{\mathrm{GIG}}\big(\lambda_{h,t} \mid -c,\, 2d,\, 0\big) \;+\; \tfrac{1}{2}\, p_{\mathrm{GIG}}\big(\lambda_{h,t} \mid +c,\, 0,\, 2d\big), \tag{14}$$

with stochastic representation $\lambda_{h,t} = W$ with probability $1/2$ and $\lambda_{h,t} = 1/W$ otherwise, $W \sim \mathrm{Gamma}(c, d)$. This construction makes $\log \lambda_{h,t}$ exactly symmetric about zero and keeps (14) conjugate to (11): the leaf full conditional is again a GIG mixture, and the integrated likelihood for the tree MH step is a ratio of GIG normalising constants [Murray, 2021, §3.1].

The pair (c, d) is reparameterised by a user-facing concentration – a_0 for the count component, z_{conc} for the ZI components (Section 4.5) – and the number of trees m, by setting $\psi_1(c) = a_0^2/m$ and $d = \exp(\psi(c))$, where ψ and ψ_1 are the digamma and trigamma functions. This calibration gives $\mathbb{E}[\log \lambda_{h,t}] = 0$ and $\mathrm{Var}(\log \lambda_{h,t}) = a_0^2/m$, so the implied prior on $\log f(\mathbf{X}) = \sum_h \log \lambda_{h,t(\mathbf{X};T_h)}$ is approximately $\mathrm{N}(0, a_0^2)$ at any $\mathbf{X}$. The concentration therefore plays the same role for $\log f$ that the prior leaf scale plays for f in Gaussian BART. For large m or small a_0 the calibration admits the closed-form approximation $c \approx m/a_0^2 + 1/2$, $d \approx m/a_0^2$ [Murray, 2021, Proposition 3].

4.4 Latent-variable augmentation for the count component

To put the count likelihood into the form (11), Murray [2021] introduces one Gamma latent variable ξ_i per observation, with full conditional $\xi_i \mid - \sim \mathrm{Gamma}(\kappa + y_i,\ \kappa + \mu_{0i} f(\mathbf{X}_i))$ under the NB likelihood, and $\xi_i \equiv 1$ under the Poisson likelihood. Conditioning on ξ_i, the per-observation contribution of f collapses to a Poisson-type kernel $f(\mathbf{X}_i)^{y_i} \exp[-\xi_i \mu_{0i} f(\mathbf{X}_i)]$, which integrated against the GIG-mixture prior (14) gives (11) with $u_i = y_i$ and $v_i = \xi_i \mu_{0i}$. Under the NB likelihood the dispersion κ is updated by a random-walk MH step on $\log \kappa$ with a Beta-prime prior implied by $\kappa/(1+\kappa) \sim \mathrm{Beta}(\kappa_a, \kappa_b)$. We defer the specific update to Section 5.3.

4.5 Zero-inflation and the redundant (f_0, f_1) parameterisation

For a zero-inflated count outcome the likelihood is a two-component mixture: with conditional probability $\omega(\mathbf{X}_i)$ the response is a structural zero, and otherwise it is drawn from the Poisson or NB law (10). Murray [2021] §4.2.2 writes the mixing probability through a pair of log-linear BART forests f_0, f_1,

$$\omega(\mathbf{X}_i) \;=\; \frac{\exp(f_0(\mathbf{X}_i))}{\exp(f_0(\mathbf{X}_i)) + \exp(f_1(\mathbf{X}_i))} \;=\; \sigma\big(f_0(\mathbf{X}_i) - f_1(\mathbf{X}_i)\big), \tag{15}$$

where each of f_0, f_1 has its own log-linear BART prior (9) with an independent GIG-mixture leaf prior (14) (with concentration z_{conc}). The parameterisation is redundant: adding a common constant to both f_0 and f_1 leaves ω unchanged. The redundancy is identified through the prior – both forests are centred at zero with symmetric GIG-mixture leaf priors – and confers two concrete benefits. Conditionally Gamma-shaped leaf updates with closed-form GIG-mixture posteriors are obtained via a pair of latent variables. Let $z_i \in \{0, 1\}$ be the augmented mixture indicator that resolves whether an observed $Y_i = 0$ came from the structural process ($z_i = 0$) or from the count

process with a zero draw ($z_i = 1$). When $Y_i > 0$, $z_i \equiv 1$; when $Y_i = 0$, z_i has the Bernoulli full conditional

$$\Pr(z_i = 1 \mid -) \;=\; \frac{(1-\omega(\mathbf{X}_i))\, p_{\text{count}}(0 \mid \mathbf{X}_i)}{\omega(\mathbf{X}_i) + (1-\omega(\mathbf{X}_i))\, p_{\text{count}}(0 \mid \mathbf{X}_i)}, \tag{16}$$

where $p_{\text{count}}(0 \mid \mathbf{X}_i)$ is the count-component zero probability ($\exp(-\lambda)$ for Poisson / ZIP, $(\kappa/(\kappa+\lambda))^\kappa$ for NB / ZINB). Let further $\phi_i > 0$ be an Exponential latent decoupling f_0 and f_1, with full conditional

$$\phi_i \;\mid\; - \;\sim\; \text{Exponential}\big(\exp(f_0(\mathbf{X}_i)) + \exp(f_1(\mathbf{X}_i))\big). \tag{17}$$

Conditioning on (z_i, ϕ_i), the joint conditional likelihood of (f_0, f_1) factorises into independent terms of the form (11): the f_0 leaves have $(u_i, v_i) = (1 - z_i, \phi_i)$ and the f_1 leaves have $(u_i, v_i) = (z_i, \phi_i)$ [Murray, 2021, §A.5, Proposition 1]. The redundant sampler is also numerically better behaved than a strictly identified parameterisation: Murray [2021, Supplement §A] report substantially better effective-sample-size performance across the full probability range. In the body of this paper we write the structural-zero log-odds as $\eta_{\text{zi}} = f_0 - f_1$ and report inference for the identified functionals only.

4.6 Implementation backbone

We inherit the C++ implementation of the log-linear BART sampler – the GIG-mixture leaf draw, the grow / prune / change / swap tree MH moves, the latent-variable updates (z_i, ξ_i, ϕ_i), the soft-mixture-weight recomputation, and the dispersion MH step – from the `countbart` R package of Wikle and Zigler [2023]. CountBCF adds the BCF (μ_f, τ_f) decomposition on top of this backend by replacing the single forest of log-linear BART (per function group) with a forest pair and introducing a per-observation weight that routes treated and control units to the correct forest; we describe that change next.

5 CountBCF: BCF decomposition for count and ZI count outcomes

This section defines CountBCF. Section 5.1 states the likelihood and parameterisation in both the count and the zero-inflated regimes; Section 5.2 collects the priors and the differential shrinkage applied symmetrically across function groups; Section 5.3 writes out the MCMC sweep in Algorithm 1 and the associated Gibbs and MH updates; Section 5.4 explains how posterior draws are post-processed into the estimands of Section 3, including the per-unit channel decomposition in the ZI case.

5.1 Likelihood and parameterisation

CountBCF models the conditional mean of a count response (and, in the ZI regime, the structural-zero probability) on the log scale using a treatment-aware BCF decomposition. We treat the count and zero-inflated cases as two sub-models of a common framework, selected by the user via a `count_model` switch that takes values in {`poisson`, `nb`, `zipoisson`, `zinb`}.

Count regime: three forests. For unit i with covariates X_i and treatment indicator $Z_i \in \{0, 1\}$, the Poisson / NB sub-model writes

$$\boxed{\log \lambda(X_i, Z_i) \;=\; \mu_f\big(X_i,\, \widehat{\pi}(X_i)\big) \;+\; Z_i \cdot \tau_f(X_i),} \tag{18}$$

Table 1: The six forests of CountBCF in the zero-inflated regime. The function group g labels the log-linear BART target inherited from Murray [2021]; the BCF role s labels the prognostic (μ) vs. moderating (τ) piece introduced by the (μ, τ) decomposition. The non-ZI sub-model retains only the $g = 0$ row.

g	Function group	Forest pair $(\mu^{(g)}, \tau^{(g)})$	Trees (m_μ, m_τ)	Weight ω_i on $\tau^{(g)}$
0	Count log-rate $\log \lambda$	$(\mu_f,\ \tau_f)$	(250, 50)	Z_i
1	ZI log-odds f_0	$(\mu_{f_0},\ \tau_{f_0})$	(100, 50)	Z_i
2	not-ZI log-odds f_1	$(\mu_{f_1},\ \tau_{f_1})$	(100, 50)	Z_i

with response

$$Y_i \mid X_i, Z_i \ \sim\ \text{Poisson}\big(\lambda(X_i, Z_i)\big) \ \text{ or } \ \text{NB}\big(\text{size} = \kappa,\ \text{mean} = \lambda(X_i, Z_i)\big), \tag{19}$$

where the size parameter $\kappa > 0$ governs overdispersion. The Poisson model is the $\kappa \to \infty$ limit. Decomposition (18) is the count analogue of the additive BCF decomposition $\mathbb{E}[Y \mid X, Z] = \mu_f(X, \widehat{\pi}) + Z\tau_f(X)$ of Hahn et al. [2020]. Both pieces are log-linear BART forests in the sense of Section 4.2: μ_f is a sum of m_μ trees with leaf parameters drawn from the GIG mixture (14), and similarly τ_f is a sum of m_τ trees. The prognostic forest models the control-arm log-mean across the whole sample; the moderating forest enters only for treated units and captures treatment-effect heterogeneity. As in BCF, $\widehat{\pi}$ is appended to the covariates feeding μ_f but not those feeding τ_f.

Zero-inflated regime: six forests, three groups. The ZIP / ZINB sub-model attaches a BCF (μ, τ) decomposition to each of the three log-linear BART forests f, f_0, f_1 of Section 4.5. We index the three function groups by $g \in \{0, 1, 2\}$ with $g = 0$ the count log-rate, $g = 1$ the structural-zero log-odds component f_0, and $g = 2$ the not-structural-zero log-odds component f_1. Then

$$\boxed{\begin{aligned} \log \lambda(X_i, Z_i) \ &= \ \mu_f\big(X_i,\ \widehat{\pi}(X_i)\big) \ + \ Z_i \cdot \tau_f(X_i), \\ \eta_{\text{zi}}(X_i, Z_i) \ &= \ \big[\mu_{f_0}\big(X_i,\ \widehat{\pi}(X_i)\big) + Z_i \cdot \tau_{f_0}(X_i)\big] \\ &\quad - \big[\mu_{f_1}\big(X_i,\ \widehat{\pi}(X_i)\big) + Z_i \cdot \tau_{f_1}(X_i)\big], \\ p_{\text{zi},\,z}(X_i) \ &= \ \sigma\big(\eta_{\text{zi}}(X_i,\ Z_i = z)\big), \qquad z \in \{0, 1\}, \end{aligned}} \tag{20}$$

together with the ZI count likelihood $Y_i \mid S_i = 1 = 0$, and $Y_i \mid S_i = 0, X_i, Z_i \sim \text{Poisson}(\lambda(X_i, Z_i))$ or $\text{NB}(\kappa, \lambda(X_i, Z_i))$, where $\Pr(S_i = 1 \mid X_i, Z_i) = p_{\text{zi},\,z}(X_i)$. The non-ZI sub-model collapses (20) by dropping the η_{zi} line and the two ZI groups, recovering (18). Equation (20) contains six unknown functions, two per function group, summarised in Table 1. Each is itself a log-linear BART regression: a sum of $m_{s,g}$ shallow trees with leaf parameters drawn from the GIG mixture, with $s \in \{\mu, \tau\}$ indexing the BCF role and $g \in \{0, 1, 2\}$ indexing the function group. For each group, the prognostic forest $\mu^{(g)}$ enters with weight $\omega_i = 1$; the moderating forest $\tau^{(g)}$ enters with weight $\omega_i = Z_i$. The propensity score estimate $\widehat{\pi}$ is appended to the covariates of all three prognostic forests but not to the moderating ones.

Scales recovered. On the count log-rate scale, $\tau_f(x)$ is itself a CATE: the conditional log relative rate (of the not-structural-zero sub-population in the ZI case), $\tau_f(x) = \log\{\lambda_1(x)/\lambda_0(x)\}$. On the ZI log-odds scale, the treatment shift in the structural-zero log-odds is

$$\Delta\eta_{\text{zi}}(x) \ \equiv \ \eta_{\text{zi}}(x, 1) - \eta_{\text{zi}}(x, 0) \ = \ \tau_{f_0}(x) - \tau_{f_1}(x), \tag{21}$$

so τ_{f_0}, τ_{f_1} are jointly identified up to a common shift – only the difference is statistically relevant for the ZI channel. On the response (rate) scale, the potential-outcome means (5) are

$$\mu_z(x) \;=\; \big(1-\sigma(\eta_{\text{zi}}(x,z))\big)\,\exp\!\big(\mu_f(x,\widehat{\pi}(x)) + z\,\tau_f(x)\big), \qquad z\in\{0,1\}, \tag{22}$$

and $\tau(x) = \mu_1(x) - \mu_0(x)$. In the non-ZI sub-model $p_{\text{zi},\,z} \equiv 0$ and (22) reduces to $\mu_z(x) = \exp(\mu_f + z\tau_f)$, giving the closed-form count CATE

$$\tau(x) \;=\; \exp\!\big(\mu_f(x,\widehat{\pi}(x))\big)\,\big(\exp(\tau_f(x)) - 1\big), \tag{23}$$

which makes explicit that the absolute treatment effect on the rate scale depends on the baseline rate in addition to the log-rate effect. In the ZI case, τ also mixes the count-rate channel $\Delta\lambda$ and the ZI-probability channel Δp_{zi} from (7).

Why this parameterisation. Three properties of the construction are worth flagging. First, the model respects the non-negativity of Y automatically: $\lambda > 0$ and (where applicable) $p_{\text{zi},\,z} \in (0,1)$ regardless of the forest values, so no truncation or clipping is required. Second, all forests are parameterised a priori independently, which is what permits the BCF differential-shrinkage idea [Hahn et al., 2020, §2.3] to be applied per group: each $\tau^{(g)}$ can be shrunk harder than its $\mu^{(g)}$ counterpart without disturbing other groups' priors. Third, the construction reduces to log-linear BART [Murray, 2021] when $\tau^{(g)} \equiv 0$ for all g, and to standard BCF in the limit where τ_f is shrunk hard to zero. In this sense CountBCF is the smallest extension of log-linear BART that delivers a separate moderating forest on each channel, and the smallest extension of BCF that handles count and zero-inflated count outcomes natively. The redundant (f_0, f_1) parameterisation in the ZI case is retained for the reasons given in Section 4.5: conjugate leaf updates, well-defined posterior on the identified functionals, and substantially better MCMC mixing [Murray, 2021, Supplement §A]. The BCF (μ, τ) decomposition does not break the redundancy: shifting μ_{f_0} and μ_{f_1} by a common constant leaves η_{zi} invariant, and shifting τ_{f_0} and τ_{f_1} by a common constant leaves $\Delta\eta_{\text{zi}}$ invariant.

5.2 Priors

The CountBCF prior factorises across forests, the dispersion parameter, and the GIG-mixture latent structure of each leaf. The construction is symmetric across function groups, so we state the prior for a generic forest pair $(\mu^{(g)}, \tau^{(g)})$ and specialise the concentration hyperparameters to the count and ZI groups.

Sum-of-trees priors on each forest. Each forest is a log-linear BART regression in the sense of Section 4.2: a sum of $m_{s,g}$ shallow trees with tree structures $\{T_{s,g,h}\}$ drawn a priori independently from the CGM tree-structure prior and leaf parameters $\{\Lambda_{s,g,h}\} = \{\lambda_{s,g,h,t}\}$ drawn a priori independently from the GIG mixture (14). Collecting these, for $s \in \{\mu, \tau\}$ and (in the ZI case) $g \in \{0,1,2\}$,

$$\begin{aligned}
T_{s,g,h} \,\big|\, \alpha_s, \beta_s \;&\overset{\text{iid}}{\sim}\; \text{CGM}(\alpha_s, \beta_s), && h = 1,\dots,m_{s,g},\\
\lambda_{s,g,h,t} \,\big|\, c_{s,g}, d_{s,g} \;&\overset{\text{iid}}{\sim}\; \tfrac{1}{2}\,\text{GIG}(-c_{s,g}, 2d_{s,g}, 0) + \tfrac{1}{2}\,\text{GIG}(c_{s,g}, 0, 2d_{s,g}), && t = 1,\dots,b_{s,g,h},\\
\mu^{(g)}(x,\widehat{\pi}(x)) \;&=\; \sum_{h=1}^{m_{\mu,g}} g\big((x,\widehat{\pi}(x)); T_{\mu,g,h}, \Lambda_{\mu,g,h}\big), && \tau^{(g)}(x) \;=\; \sum_{h=1}^{m_{\tau,g}} g\big(x; T_{\tau,g,h}, \Lambda_{\tau,g,h}\big),
\end{aligned} \tag{24}$$

where the CGM density assigns an internal-node split probability $\alpha_s\,(1+d)^{-\beta_s}$ at depth d and a uniform draw over splitting variable and threshold given the split. The depth-penalty hyperparameters (α_s, β_s) are shared across function groups (so all prognostic forests have the same tree-shape prior, and similarly for the moderating forests); the leaf concentration is allowed to differ across g to reflect the different natural scales of the count log-rate versus the zero-inflation log-odds. In the count-only regime, indices $g \in \{1, 2\}$ are dropped throughout.

Differential shrinkage applied symmetrically. CountBCF inherits the BCF [Hahn et al., 2020] idea that the treatment-effect surface $\tau^{(g)}$ should be shrunk more aggressively than the prognostic surface $\mu^{(g)}$ on each channel: the signal-to-noise in $\tau^{(g)}$ is typically lower than in $\mu^{(g)}$; spurious heterogeneity in any of $\tau_f, \tau_{f_0}, \tau_{f_1}$ manifests as fake interactions with Z on the respective channel; and a stub-biased prior on every $\tau^{(g)}$ delivers a near-constant treatment-effect estimator as a default. The default hyperparameters are summarised in Table 2. The tree-prior $(\alpha_\tau, \beta_\tau) = (0.25, 3)$ sharply downweights non-stub trees in every moderating forest: under it the prior probability of a non-trivial split at the root is 0.25 and the probability of a depth-2 internal node is approximately 0.0039, so on average each moderating forest is close to a piecewise-constant surface with very few splits. The amplitude of those few splits is shrunk by the second differential-shrinkage mechanism, the small number of trees in each moderating forest relative to its prognostic counterpart ($m_{\tau,g} = 50$ against $m_{\mu,0} = 250$ for the count group and $m_{\mu,g} = 100$ for the ZI groups). Within a group the moderating and prognostic forests share the leaf concentration (a_0 for the count group, z_{conc} for the ZI groups), so under the GIG-mixture calibration of Section 4.3 each moderating leaf carries $\text{Var}(\log \lambda_{\tau,g,h,t}) \approx a_{0,g}^2/m_{\mu,g}$, and summing over the $m_{\tau,g}$ moderating trees gives an implied marginal prior on $\tau^{(g)}(x)$ of $\text{N}(0,\, a_{0,g}^2\, m_{\tau,g}/m_{\mu,g})$. For the count group this is $\text{N}(0, a_0^2/5)$, a prior standard deviation of $a_0/\sqrt{5} \approx 0.45\, a_0$ (close to half that of $\mu^{(g)}$); for the ZI groups the corresponding factor is $\sqrt{1/2} \approx 0.71$. The differential shrinkage between the moderating and prognostic surfaces is thus delivered entirely by the tree-count ratio and the stub-biased tree-shape prior, both of which act on the moderating forests independently of the leaf concentration.

The user-facing concentration a_0 defaults to $\frac{1}{2}(\log y_{(0.95)} - \overline{\log \mu_{0,i}})$, where $y_{(0.95)}$ is the empirical 95% quantile of y, centring the count-channel prior on a data-matched scale. The ZI concentration z_{conc} defaults to $3.5/\sqrt{2} \approx 2.47$, matching the calibration of Murray [2021, §4.2.2] for the redundant (f_0, f_1) parameterisation. Both can be overridden by the user.

Propensity-as-covariate fix on every channel. A central insight of Hahn et al. [2020] is that, on the additive response scale, naively regressing the outcome on (X, Z) with a BART-like prior leaves the estimator vulnerable to RIC: when the propensity score $\pi(x)$ is a function of x, the prior shrinkage of the prognostic surface toward zero biases τ_f in a direction that depends on π. The fix is to append $\widehat{\pi}(x)$ to the prognostic forest's covariates so that μ_f can "absorb" the part of the outcome explained by π without recruiting τ_f. On the log-rate scale of (18) the RIC argument is at least as severe: a perturbation δ in μ_f translates to a multiplicative perturbation $e^\delta - 1 \approx \delta$ in the baseline rate, which then multiplies into τ via (23); a bias in μ_f correlated with π propagates linearly into τ. In the ZI case the same RIC argument applies independently on each channel: perturbations in μ_{f_0} or μ_{f_1} propagate into the ZI-channel moderating forests through the redundant parameterisation. We therefore append $\widehat{\pi}$ to the covariate set of every prognostic forest by default and do not append it to any moderating forest. The R wrapper exposes this as `include_pihat = "control"`, with `"moderate"`, `"both"`, and `"none"` available for sensitivity checks.

Table 2: CountBCF default hyperparameters. The ZI rows ($g \in \{1, 2\}$) apply only when `count_model` $\in$ {`zipoisson`, `zinb`}; the dispersion parameters apply only when `count_model` $\in$ {`nb`, `zinb`}.

Hyperparameter	Default	Role
$m_{\mu,0}$, $m_{\tau,0}$	250, 50	Trees per forest, count group $g = 0$
$m_{\mu,1}$, $m_{\tau,1}$	100, 50	Trees per forest, ZI group $g = 1$ (f_0)
$m_{\mu,2}$, $m_{\tau,2}$	100, 50	Trees per forest, ZI group $g = 2$ (f_1)
(α_μ, β_μ)	(0.95, 2)	CGM depth penalty, all $\mu^{(g)}$
$(\alpha_\tau, \beta_\tau)$	(0.25, 3)	CGM depth penalty, all $\tau^{(g)}$
a_0	$\frac{1}{2}(\log y_{(0.95)} - \overline{\log \mu_{0,i}})$	Leaf concentration, count group (μ_f, τ_f)
z_{conc}	$3.5/\sqrt{2}$	Leaf concentration, ZI groups $(\mu_{f_0}, \mu_{f_1}, \tau_{f_0}, \tau_{f_1})$
(κ_a, κ_b)	(5, 3)	Beta-prime prior on κ
σ_κ	0.21	RW-MH proposal SD on $\log \kappa$
`include_pihat`	`"control"`	Propensity appended to all $\mu^{(g)}$

Negative-binomial dispersion κ. Under the NB / ZINB likelihood, the size parameter κ controls overdispersion: $\text{Var}(Y_i \mid X_i, Z_i, S_i = 0) = \lambda_i + \lambda_i^2/\kappa$. Following Murray [2021] we place a Beta-prime prior on κ implied by

$$\frac{\kappa}{1+\kappa} \sim \text{Beta}(\kappa_a, \kappa_b), \qquad (\kappa_a, \kappa_b) = (5, 3), \tag{25}$$

with mean 2.5 and a right tail heavy enough not to rule out near-Poisson behaviour while still penalising arbitrarily large overdispersion. The same prior governs κ under both NB and ZINB; structural zeros enter the κ-conditional likelihood only through z_i (Section 5.3). The dispersion is updated by a random-walk MH step on $\log \kappa$ with Gaussian proposal of standard deviation $\sigma_\kappa = 0.21$.

5.3 Posterior computation

Posterior inference for CountBCF uses a Gibbs sampler that interleaves backfitting updates of the forests with latent-variable augmentation steps and (under NB / ZINB) a MH step on the dispersion. The structure mirrors the log-linear BART sampler of Murray [2021, §3.3, §S.3]: the only forest-level change is the introduction of a per-observation weight $\omega_i \in \{1, Z_i\}$ that routes treated and control units to the moderating and prognostic forests, replicated across all function groups.

Latent-variable augmentation. The augmentations are inherited from Sections 4.4 and 4.5. Each observation carries up to three latents:

$$\begin{aligned}
z_i \mid - &\sim \text{Bernoulli}\Big(\frac{(1-\omega(\mathbf{X}_i))\, p_{\text{count}}(0 \mid \mathbf{X}_i)}{\omega(\mathbf{X}_i) + (1-\omega(\mathbf{X}_i))\, p_{\text{count}}(0 \mid \mathbf{X}_i)}\Big) \quad \text{when } Y_i = 0 \text{ (ZIP/ZINB)}; \;\; z_i \equiv 1 \text{ when } Y_i > 0, \\
\xi_i \mid - &\sim \begin{cases} \delta_1 & \text{(Poisson / ZIP)}, \\ \text{Gamma}\big(\kappa + Y_i,\; \kappa + \lambda(\mathbf{X}_i, Z_i)\big) & \text{(NB / ZINB)}, \end{cases} \\
\phi_i \mid - &\sim \text{Exponential}\big(\exp(f_0(\mathbf{X}_i)) + \exp(f_1(\mathbf{X}_i))\big) \quad \text{(ZIP/ZINB only)}.
\end{aligned} \tag{26}$$

Table 3: Per-group sufficient statistics. The values u_i, v_i are inserted into (27) through the leaf accumulators $r_{s,g,h,t}, s_{s,g,h,t}$. The BCF weight is $\omega_i = 1$ for prognostic forests ($s = \mu$) and $\omega_i = Z_i$ for moderating forests ($s = \tau$); it premultiplies (u_i, v_i) before the leaf sum is formed. The non-ZI sub-model retains only the $g = 0$ row.

g	Function target	$u_i \cdot \omega_i^{-1}$	$v_i \cdot \omega_i^{-1}$
0	count rate λ	$z_i\, y_i$	$z_i \exp(\log \xi_i + \log \lambda(\mathbf{X}_i, Z_i) - \omega_i \log \lambda_{s,g,h,t})$
1	ZI log-odds f_0	$1 - z_i$	$\exp(\log \phi_i + \log f_0(\mathbf{X}_i) - \omega_i \log \lambda_{s,g,h,t})$
2	ZI log-odds f_1	z_i	$\exp(\log \phi_i + \log f_1(\mathbf{X}_i) - \omega_i \log \lambda_{s,g,h,t})$

Under the non-ZI sub-models the z_i and ϕ_i updates are skipped; under the Poisson / ZIP sub-models the ξ_i update is skipped.

Tree updates: a single primitive applied to all forests. Conditional on the latents, the joint conditional likelihood of the forests factorises into independent terms, each of which has the conditionally Gamma form (11). For forest s in function group g at tree h, the conditional augmented likelihood for the $b_{s,g,h}$ leaves of $T_{s,g,h}$ is

$$L\big(\Lambda_{s,g,h} \,\big|\, T_{s,g,h}, T_{-(s,g,h)}, \Lambda_{-(s,g,h)}, z_i, \xi_i, \phi_i, Y, \kappa\big) \;\propto\; \prod_{t=1}^{b_{s,g,h}} \lambda_{s,g,h,t}^{r_{s,g,h,t}} \;\exp\!\big[-s_{s,g,h,t}\, \lambda_{s,g,h,t}\big], \tag{27}$$

where the leaf sufficient statistics $r_{s,g,h,t} = \sum_{i \in \mathcal{A}_{s,g,h,t}} u_i$ and $s_{s,g,h,t} = \sum_{i \in \mathcal{A}_{s,g,h,t}} v_i\, f^{-(s,g,h)}(\mathbf{X}_i)$ aggregate over the units in the partition cell, using the per-group (u_i, v_i) enumerated in Table 3. The BCF weight $\omega_i \in \{1, Z_i\}$ enters multiplicatively in both u_i and v_i: for prognostic forests every unit contributes; for moderating forests only treated units contribute. This is the BCF moderating-forest ingredient transposed to the count (or ZI count) likelihood, applied symmetrically to all function groups present in the model.

The GIG-mixture leaf prior (14) is conjugate to (27), so the leaf draw is a closed-form mixture of GIG variates (implemented in `drmu_loglinear`), and $T_{s,g,h}$ is updated by a MH step with grow / prune / change / swap proposals (implemented in `bd_loglinear`). The MH acceptance ratio uses the integrated marginal likelihood obtained by integrating $\Lambda_{s,g,h}$ against the GIG-mixture prior; because both prior and likelihood are products over leaves, that integral factorises and reduces to a ratio of GIG normalising constants [Murray, 2021, Appendix A].

Per-group log-fit accumulator. Implemented as backfitting, the tree update maintains running accumulators on the n-vector index k: `allfit`$[k]$ tracks the total log-mean / log-odds; `allfits`$[s][k]$ tracks the contribution of forest s; and the per-group accumulator `log_f_per_group`$[g][k] = \mu^{(g)}(\mathbf{X}_k, \widehat{\pi}(\mathbf{X}_k)) + Z_k\, \tau^{(g)}(\mathbf{X}_k)$ stores the in-sample log-fit of both forests in group g. Before sampling tree (s, g, h) its current contribution is subtracted from all relevant accumulators; sufficient statistics are computed from the partial fit; $T_{s,g,h}$ and $\Lambda_{s,g,h}$ are updated; and the new contribution is added back. The per-group accumulator makes the six-forest update efficient: the ZI sufficient statistics read $\log f_0$ and $\log f_1$ from `log_f_per_group`$[1]$ and `log_f_per_group`$[2]$ in $O(1)$ per observation. After each sweep the soft mixture weight that enters the z_i update is recomputed as $\log w(\mathbf{X}_i) = \log f_1(\mathbf{X}_i) - \text{logsumexp}(\log f_0(\mathbf{X}_i), \log f_1(\mathbf{X}_i))$, which avoids the underflow of naive computation when one of the log-fits is large and negative.

Dispersion update (NB / ZINB only). Propose $\kappa^\star = \exp(\log \kappa + \sigma_\kappa \varepsilon)$ with $\varepsilon \sim \mathrm{N}(0,1)$. Accept with probability

$$\alpha_\kappa \;=\; \min\Big\{1,\; \frac{p_{\mathrm{NB}}(Y \mid z_i, \lambda, \kappa^\star)}{p_{\mathrm{NB}}(Y \mid z_i, \lambda, \kappa)} \cdot \frac{p(\kappa^\star)}{p(\kappa)} \cdot \frac{\kappa^\star}{\kappa}\Big\}, \tag{28}$$

where the likelihood factor is the augmented NB likelihood conditional on the mixture indicator (so structural zeros do not enter the dispersion update under ZINB), $p(\kappa)$ is the Beta-prime density implied by (25), and the Jacobian factor $\kappa^\star/\kappa$ comes from the log-scale proposal. Acceptance rates after burn-in are returned as `kappa_acpt` and serve as a coarse convergence diagnostic; we target the 0.20–0.45 band typical for random-walk MH.

Algorithm 1 CountBCF: one MCMC sweep. Indices $s \in \{\mu, \tau\}$ range over forests, $g \in \{0\}$ in the count sub-model and $g \in \{0, 1, 2\}$ in the ZI sub-model, and $h = 1, \ldots, m_{s,g}$ over the trees of forest (s, g). The BCF weight is $\omega_i = 1$ for $s = \mu$ and $\omega_i = Z_i$ for $s = \tau$. Latent updates that are not applicable to the active sub-model are skipped.

Require: Current state $\big(\{T_{s,g,h}, \Lambda_{s,g,h}\},\ \kappa,\ \{z_i, \xi_i, \phi_i\}\big)$.

1: **Dispersion update (NB / ZINB only).** Propose $\kappa^\star = \exp(\log \kappa + \sigma_\kappa\, \varepsilon)$, $\varepsilon \sim \mathrm{N}(0, 1)$. Accept with probability α_κ from (28); otherwise retain κ.

2: **Augmentation update for z_i (ZIP / ZINB only).** For each i with $Y_i = 0$, draw z_i from the Bernoulli full conditional in (26); set $z_i = 1$ when $Y_i > 0$. [`drz_loglinear`]

3: **Augmentation update for ξ_i (NB / ZINB only).** For each i, draw ξ_i from the Gamma full conditional; set $\xi_i \equiv 1$ otherwise. [`drxi_loglinear`]

4: **Augmentation update for ϕ_i (ZIP / ZINB only).** For each i, draw $\phi_i \sim \mathrm{Exponential}(\exp(f_0(\mathbf{X}_i)) + \exp(f_1(\mathbf{X}_i)))$. [`drphi_loglinear`]

5: **Backfitting tree updates.** For each function group g active in the sub-model, each forest $s \in \{\mu, \tau\}$, and each tree $h = 1, \ldots, m_{s,g}$:

5.1 Subtract the current contribution of $(T_{s,g,h}, \Lambda_{s,g,h})$ from `allfit`, `allfits`$[s]$, and `log_f_per_group`$[g]$.

5.2 Compute the leaf sufficient statistics $(r_{s,g,h,t}, s_{s,g,h,t})$ from (27) using the per-group (u_i, v_i) of Table 3, premultiplied by the weight ω_i.

5.3 Propose a tree move (grow / prune / change / swap) for $T_{s,g,h}$ and accept with probability from the integrated marginal likelihood (GIG-mixture conjugacy). [`bd_loglinear`]

5.4 Draw the leaf values $\Lambda_{s,g,h} \mid T_{s,g,h}, z_i, \xi_i, \phi_i, Y, \kappa$ from the GIG-mixture posterior implied by (27). [`drmu_loglinear`]

5.5 Add the new contribution back into the affected accumulators.

6: **Soft mixture weight (ZIP / ZINB only).** For each i, recompute $\log w(\mathbf{X}_i) = \log f_1(\mathbf{X}_i) - \mathrm{logsumexp}(\log f_0(\mathbf{X}_i), \log f_1(\mathbf{X}_i))$.

7: **(Optional) Random-effects update.** If enabled, draw the random-intercept block from its Gibbs full conditional; otherwise skip.

8: **return** Updated state $\big(\{T_{s,g,h}, \Lambda_{s,g,h}\},\ \kappa,\ \{z_i, \xi_i, \phi_i\}\big)$.

Implementation lineage. The low-level C++ primitives and dispersion Metropolis-Hastings steps are inherited verbatim from the `countbart` package[1]; the CountBCF contribution is the custom dispatch logic[2] that (i) instantiates two forests per function group rather than one, (ii) sets the per-forest weight vector ω_i to either $\mathbf{1}_n$ or Z_{trt}, (iii) applies ω_i to the sufficient-statistic

[1] Specifically, the C++ kernels `bd_loglinear`, `drmu_loglinear`, `fit_loglinear`, `drz_loglinear`, `drxi_loglinear`, and `drphi_loglinear` of Wikle and Zigler [2023].

[2] Implemented in `src/countbcf.cpp`.

construction in step 5.2, and (iv) maintains the per-group log-fit accumulator so that the four ZI forests read f_0 and f_1 in $O(1)$ per observation. The change is mechanical at the kernel level but yields a sampler that, to our knowledge, is functionally distinct from any log-linear BART variant in the published literature: it jointly estimates up to six log-rate / log-odds surfaces under differential shrinkage, with each moderating forest seeing only the treated subsample's residual variation.

5.4 Posterior summaries

Algorithm 1 produces a sequence of B post-burn-in posterior draws indexed by $b = 1, \ldots, B$. The sampler stores, for each draw, the per-unit values of the active forests on the original (unsorted) input order:

$$\begin{aligned} \mu_f^{(b)}(X_i) &= \texttt{mu_f_post}[b,i], \quad & \tau_f^{(b)}(X_i) &= \texttt{tau_f_post}[b,i], \\ \mu_{f_0}^{(b)}(X_i) &= \texttt{mu_f0_post}[b,i], \quad & \tau_{f_0}^{(b)}(X_i) &= \texttt{tau_f0_post}[b,i], \\ \mu_{f_1}^{(b)}(X_i) &= \texttt{mu_f1_post}[b,i], \quad & \tau_{f_1}^{(b)}(X_i) &= \texttt{tau_f1_post}[b,i], \end{aligned} \tag{29}$$

together with the dispersion draws $\kappa^{(b)}$ (NB / ZINB only). In the non-ZI sub-model only the first row is stored. Downstream inference is a post-processing exercise on these matrices and produces three (count) or four (ZI) classes of summaries.

Per-iteration potential outcomes. For each draw b and each unit i, the closed-form mapping from stored forest values to response-scale potential-outcome means is, for $z \in \{0, 1\}$,

$$\boxed{\begin{aligned} \log \lambda_z^{(b)}(X_i) &= \mu_f^{(b)}(X_i) + z \cdot \tau_f^{(b)}(X_i), \\ \eta_{\text{zi}}[z]^{(b)}(X_i) &= \big[\mu_{f_0}^{(b)}(X_i) + z \cdot \tau_{f_0}^{(b)}(X_i)\big] - \big[\mu_{f_1}^{(b)}(X_i) + z \cdot \tau_{f_1}^{(b)}(X_i)\big], \\ \mu_z^{(b)}(X_i) &= \big(1 - \sigma(\eta_{\text{zi}}[z]^{(b)}(X_i))\big)\, \exp\big(\log \lambda_z^{(b)}(X_i)\big). \end{aligned}} \tag{30}$$

In the non-ZI sub-model $\sigma(\eta_{\text{zi}}) \equiv 0$ and (30) collapses to $\mu_z^{(b)}(X_i) = \exp(\mu_f^{(b)}(X_i) + z\tau_f^{(b)}(X_i))$. The pair $(\mu_0^{(b)}, \mu_1^{(b)})$ collects the response-scale potential-outcome means at every unit under every draw, and all reported quantities below are deterministic functions of these two matrices – no additional sampling is required: posterior uncertainty in the response-scale CATE arises from joint uncertainty in the forests and is automatically propagated by the backfitting Gibbs sampler.

Response-scale CATE. The primary estimand is (6). The bth posterior draw is

$$\widehat{\tau_{,i}}^{(b)} = \mu_1^{(b)}(X_i) - \mu_0^{(b)}(X_i), \tag{31}$$

with the count-only form $\widehat{\tau_{,i}}^{(b)} = \exp(\mu_f^{(b)}(X_i))(\exp(\tau_f^{(b)}(X_i)) - 1)$ recovered in the non-ZI sub-model. The per-unit posterior mean and 95% equal-tailed credible interval are $\overline{\tau_{,i}} = B^{-1}\sum_b \widehat{\tau_{,i}}^{(b)}$ and $[Q_{0.025}, Q_{0.975}]$ across draws, with Q_q the empirical q-quantile. The log-rate CATE $\widehat{\tau_{f,i}}^{(b)} = \tau_f^{(b)}(X_i)$ is reported alongside for users whose applied questions are best phrased as relative rates.

Sample-average treatment effect. The ATE (6) is the in-sample mean of the response-scale CATE,

$$\widehat{\text{ATE}}^{(b)} = \frac{1}{n}\sum_{i=1}^{n} \widehat{\tau_{,i}}^{(b)}, \tag{32}$$

giving a B-vector of posterior draws. The posterior mean and 95% equal-tailed interval of this vector give the point estimate and uncertainty of the sample ATE. The average is over the empirical distribution of X, so (32) estimates the sample-average treatment effect rather than a population-average one; Hahn et al. [2020] discuss why this is the natural target for BART-family methods.

Channel decomposition (ZI only). A central affordance of the six-forest construction over a single-channel causal forest is that τ can be decomposed into a count-rate channel and a structural-zero channel. The per-unit posterior draws of the two channel components introduced in (7) are

$$\widehat{\Delta\lambda_{,i}}^{(b)} = \exp\big(\log \lambda_1^{(b)}(X_i)\big) - \exp\big(\log \lambda_0^{(b)}(X_i)\big), \qquad \widehat{\Delta p_{\mathrm{zi},i}}^{(b)} = \sigma\big(\eta_{\mathrm{zi}}[1]^{(b)}(X_i)\big) - \sigma\big(\eta_{\mathrm{zi}}[0]^{(b)}(X_i)\big), \tag{33}$$

both deterministic functions of the stored forest values. Either channel can be summarised by its posterior mean and 95% credible interval per unit, or averaged to produce a posterior of the channel-specific sample average. The two channels are not constrained to share a sign: for some units the treatment can increase the count rate and decrease the structural-zero probability (both contributing positively to τ); for other units the channels can offset, producing a small response-scale effect masking large channel-specific shifts.

6 Research methodology

This section pre-registers the simulation study that will evaluate CountBCF against the natural Gaussian-likelihood baseline. The specification is held fixed before any cells are run; results will be reported in Section 7 without revision to the design below. The motivation for pre-registration is standard [Hahn et al., 2020, § 3.2]: heterogeneous-effect comparisons are sensitive to which data-generating processes (DGPs), factor sweeps, and metrics one chooses to highlight, and reporting those choices after seeing results risks confirming the preferred method.

6.1 Estimand

We target the response-scale conditional and average treatment effects defined in Section 3: for unit i, the potential-outcome means $\mu_z(x) = (1 - p_{\mathrm{zi},\,z}(x)) \exp(\log \lambda_z(x))$ of (5) give rise to

$$\tau(x) = \mu_1(x) - \mu_0(x), \qquad \mathrm{ATE} = \mathbb{E}_X\big[\tau(X)\big], \tag{34}$$

on the same units as Y. Under zero-inflation (34) mixes a change in the count rate and a change in the structural-zero probability, and a method that changes only one of the two will systematically under-estimate the effect on at least some part of the covariate space; the ZI DGPs of Section 6.2 engineer this asymmetry deliberately. All methods, regardless of their native scale, are scored against this rate-scale truth. For CountBCF the per-draw CATE is obtained from (30)–(31); for the Gaussian BCF baseline it is read off directly from the moderating forest.

6.2 Data-generating processes

The simulation grid is a 2×4 design over (i) the functional form of the prognostic and moderating surfaces (linear or nonlinear) and (ii) the count distribution (Poisson, NB with $\kappa = 2$, ZIP, or ZINB with $\kappa = 2$). All eight DGPs share the same covariate distribution and propensity model, so cross-DGP comparisons isolate the effects of nonlinearity, overdispersion, and zero-inflation. Each DGP features strong, sign-changing, multi-covariate heterogeneity; the ZI DGPs additionally contain non-trivial heterogeneity on the structural-zero channel, so the per-unit CATE estimation task is substantively challenging in every cell.

Covariates and treatment. We draw n iid covariate vectors $X_i \in \mathbb{R}^p$ with the first $p-2$ columns iid standard Gaussian and the last two binary,

$$X_{i,j} \overset{\text{iid}}{\sim} \mathrm{N}(0,1) \text{ for } j = 1, \ldots, p-2, \quad X_{i,\,p-1} \sim \text{Bernoulli}(0.6), \quad X_{i,\,p} \sim \text{Bernoulli}(0.4), \tag{35}$$

so every design contains two binary columns regardless of p. Only the first five covariates carry signal; columns $X_{\cdot,6}, \ldots, X_{\cdot,p-2}$ are irrelevant noise included to stress-test variable selection as p grows. The propensity model is fixed across DGPs at

$$\pi(x) \;=\; \sigma(0.5\, x_1 - 0.4\, x_2), \qquad Z_i \mid X_i \sim \text{Bernoulli}(\pi(X_i)), \tag{36}$$

with $\sigma(\cdot)$ the logistic CDF. The estimated propensity $\widehat{\pi}$ passed to the methods is set to the truth, $\widehat{\pi}(X_i) = \pi(X_i)$, so the comparison isolates the outcome model rather than penalising methods for propensity mis-estimation.

Count component. For all eight DGPs the conditional log-mean of the count component (non-ZI cells) or the not-structural-zero component (ZI cells) has the BCF-style form

$$\log \lambda_i \;=\; \mu_c(X_i) \;+\; Z_i \left[\tau_0 + \tau_h(X_i)\right], \tag{37}$$

with the prognostic surface μ_c and the heterogeneous part τ_h taking the form in the upper half of Table 4. The scalar τ_0 is a per-cell intercept calibrated so that the realised response-scale ATE matches the cell's target value. For non-ZI cells, we solve

$$\mathbb{E}_X\big[\exp(\mu_c(X))\,\big(\exp(\tau_0 + \tau_h(X)) - 1\big)\big] \;=\; \mathrm{ATE}_{\text{target}} \tag{38}$$

via `R`'s `uniroot()` on a fresh held-out sample of 5,000 covariate vectors. For ZI cells, the calibration accounts for the $(1 - p_{\text{zi},\,z})$ thinning of the count component; only τ_0 is calibrated, the ZI shift $\Delta\eta_{\text{zi}}$ is held fixed.

Zero-inflation component (ZI cells only). The structural-zero probability follows $p_{\text{zi},\,z}(x) = \sigma(\eta_{\text{zi}}(x) + z\Delta\eta_{\text{zi}}(x))$, with the control logit $\eta_{\text{zi}}(x)$ and treatment shift $\Delta\eta_{\text{zi}}(x)$ in the lower half of Table 4. The control logit is calibrated so the baseline structural-zero probability is approximately 25–30% across the covariate distribution – large enough to make the ZI channel matter but small enough not to swamp the count signal. The treatment shift $\Delta\eta_{\text{zi}}(x)$ is sign-changing across units: for some units the treatment increases the structural-zero probability while for others it decreases it, with the sign and magnitude depending on X. A single ATE-level summary of the ZI channel would average over the sign reversal and report a misleading null; the channel-decomposition plots described below are designed to recover this heterogeneity.

Sampling distribution. For each unit i, the response is drawn from one of four distributions. Under the non-ZI cells, $Y_i \sim \text{Poisson}(\lambda_i)$ or $\text{NB}(\kappa, \lambda_i)$ with $\kappa = 2$. Under the ZI cells, a Bernoulli structural-zero indicator $S_i \sim \text{Bernoulli}(p_{\text{zi},\,Z_i}(X_i))$ is drawn first; if $S_i = 1$ then $Y_i = 0$ deterministically; otherwise $Y_i \mid S_i = 0 \sim \text{Poisson}(\lambda_i)$ or $\text{NB}(\kappa, \lambda_i)$ with $\kappa = 2$. Combined with Table 4 this yields the eight CATE-focused DGPs (linear and nonlinear, crossed with Poisson, NB, ZIP, and ZINB).

Table 4: Functional forms for the prognostic surface μ_c and the heterogeneous component τ_h (count channel, top), and for the control logit η_{zi} and treatment shift $\Delta\eta_{\text{zi}}$ on the ZI channel (bottom). The linear and nonlinear families each cross with the four count distributions $\{\text{Poisson}, \text{NB}, \text{ZIP}, \text{ZINB}\}$ to give eight DGPs; the ZI rows apply only to the ZIP / ZINB cells.

DGP family	linear	nonlinear
Count component ($\log \lambda_i = \mu_c(X_i) + Z_i[\tau_0 + \tau_h(X_i)]$)		
$\mu_c(x)$	$1.0 + 0.5x_1 - 0.3x_2 + 0.2x_3$	$0.8 + 0.6\sin(x_1) + 0.4x_2x_3 + 0.5\,\mathbf{1}\{x_4 > 0\}$
$\tau_h(x)$	$0.60x_1 - 0.45x_2 + 0.30x_3$	$0.80\sin(x_1) + 0.50\,\mathbf{1}\{x_2 > 0\}\,x_3 - 0.40x_5$
ZI component ($p_{\text{zi},z}(x) = \sigma(\eta_{\text{zi}}(x) + z\Delta\eta_{\text{zi}}(x))$, ZIP / ZINB only)		
$\eta_{\text{zi}}(x)$	$-1.0 + 0.5x_2$	$-1.0 + 0.7\lvert x_2\rvert - 0.5\exp(-x_3^2)$
$\Delta\eta_{\text{zi}}(x)$	$-0.50x_1 + 0.30x_3 - 0.40x_4$	$-0.60x_1x_4 + 0.40x_2\,\mathbf{1}\{x_5 > 0\}$

Table 5: One-at-a-time factor sweep. "Held fixed" columns list the factors that take their reference value within the row's sweep.

Factor	Off-reference levels	Held fixed
Sample size N	$\{100, 500\}$	$p = 5$, ATE $= 1.25$
Covariate count p	$\{50, 250\}$	$N = 250$, ATE $= 1.25$
Target ATE	$\{0.5, 2.5\}$	$N = 250$, $p = 5$

6.3 Experimental factors

We sweep three experimental factors – sample size N, covariate count p, and target ATE – one-at-a-time (OAT) around a reference cell. Reference values are $N_{\text{ref}} = 250$, $p_{\text{ref}} = 5$, $\text{ATE}_{\text{ref}} = 1.25$. Off-reference levels are in Table 5; together with the reference, this gives $1 + 2 + 2 + 2 = 7$ cells per (method, DGP) pair. For each cell all eight DGPs of Section 6.2 are run. Within a cell $\times$ DGP we draw $N_{\text{sim}} = 100$ independent Monte Carlo replicates; the seed of replicate s is exactly s – `set.seed(sim_id)` – so any single replicate is reproducible from its (dgp, N, p, ATE, sim_id) tuple alone.

6.4 Methods compared

Both methods produce a posterior over the response-scale ATE and per-unit posterior means and credible intervals for the response-scale CATE; both are Bayesian, both share the BCF prior structure, and they differ only in the outcome likelihood.

1. **CountBCF (proposed).** Fit via `countbcf()` with `count_model` set to one of `"poisson"`, `"nb"`, `"zipoisson"`, or `"zinb"` depending on the DGP. All other hyperparameters take the defaults catalogued in Table 2. The per-unit CATE is recovered from the closed-form mapping (30)–(31); under ZIP / ZINB the same coefficient matrices are also post-processed into the channel decomposition $(\widehat{\Delta\lambda}, \widehat{\Delta p_{\text{zi}}})$ of (33).

2. **BCF (Gaussian baseline).** The Hahn et al. [2020] BCF applied to the count outcome as if it were Gaussian, fit via `bcf_binary()`. The treatment-effect posterior is read off the moderating forest with `get_forest_fit()`. This is the natural "what if you ignored the

count structure" baseline: it retains the BCF (μ, τ) decomposition and the propensity-as-covariate fix but does not respect the discreteness of Y, the heteroscedasticity induced by the count likelihood, or the structural-zero process; in the ZI case it produces only a single response-scale CATE with no channel decomposition.

3. **Parametric Generalised Linear Models (Poisson/NB or ZIP/ZINB GLM).** In the count-only setting, we fit a Poisson GLM and a Negative-Binomial GLM with a log link, including all covariates, treatment, and treatment-by-covariate interactions. In the zero-inflated setting, we fit Zero-Inflated Poisson (ZIP) and Zero-Inflated Negative-Binomial (ZINB) GLMs via the `pscl` package [Zeileis et al., 2008], including treatment-by-covariate interactions in both the count and the zero-inflation components. To obtain posterior draws and uncertainty intervals for these models, we simulate coefficients from their multivariate normal asymptotic sampling distributions.

4. **Causal Forest.** Fit via the `grf` package [Athey et al., 2019]. This is a nonparametric frequentist random forest optimized for treatment effect estimation. The aggregate average treatment effect (ATE) and its standard errors are obtained using the doubly-robust augmented inverse probability weighting (AIPW) estimator, while the individual-level treatment effects (CATEs) are evaluated directly using the forest predictions.

Both Bayesian methods use the same MCMC budget: `nburn` $=$ `nsim` $= 1000$, `nthin` $= 1$. The methodological symmetry is deliberate – the only structural difference is the outcome likelihood – so any performance gap observed in Section 7 can be attributed to the likelihood, not to incidental choices of prior, MCMC budget, or post-processing.

CountBART considered and dropped. An S-learner built on top of `count_bart()` – augmenting the design matrix with Z and $\widehat{\pi}$, fitting a single log-linear BART model on $(X, Z, \widehat{\pi})$ (three-forest in the ZI case), and reading the CATE off counterfactual evaluations – was originally part of the benchmark. It consistently exhausts the available Colab RAM and crashes on the larger ablation cells ($p \in \{50, 250\}$); the three-forest backbone of the ZI model amplifies the memory pressure relative to the non-ZI case. Until the memory footprint of the upstream `count_bart` is reduced, the CountBART S-learner is not feasible to evaluate against CountBCF on the same hardware, and we exclude it rather than report it on the subset of cells where it runs.

6.5 Metrics

All metrics that involve the truth are computed on the response scale, regardless of which scale a method natively reports. Aggregating across the $N_{\text{sim}} = 100$ replicates within a cell gives Monte Carlo means and standard errors. Two families of metrics are reported, both targeted at the calibration of a method's posterior rather than only its point estimate.

ATE-level metrics. Let $\widehat{\text{ATE}}^{(r)}$ denote the posterior-mean ATE on replicate r, $\widehat{\text{CI}}_{95}^{(r)}$ its 95% equal-tailed credible interval, and $\text{ATE}^{(r)} = n^{-1}\sum_i \tau(X_i)$ the realised true ATE on the same replicate. We report:

- the cross-replicate root mean squared error $\text{RMSE}(\widehat{\text{ATE}}) = \sqrt{N_{\text{sim}}^{-1}\sum_r (\widehat{\text{ATE}}^{(r)} - \text{ATE}^{(r)})^2}$;
- the empirical 95% coverage $\text{Cov}_{95}(\widehat{\text{ATE}}) = N_{\text{sim}}^{-1}\sum_r \mathbf{1}\{\text{ATE}^{(r)} \in \widehat{\text{CI}}_{95}^{(r)}\}$; and
- the average 95% credible-interval width $\overline{W_{95}(\widehat{\text{ATE}})}$.

CATE-level metrics. On each Monte Carlo replicate $r \in \{1, \ldots, N_{\text{sim}}\}$, let $\widehat{\tau}^{(r)}(X_i)$ denote the posterior mean of $\tau(X_i)$, $\widehat{\text{CI}}^{(r)}_{95,i}$ its unit-level 95% credible interval, and $\tau(X_i) = \mu_1(X_i) - \mu_0(X_i)$ the true CATE. For each replicate, we compute:

- the replicate-specific precision in estimating heterogeneous effects:

$$\text{PEHE}^{(r)} \;=\; \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\widehat{\tau}^{(r)}(X_i) - \tau(X_i)\right)^2}; \tag{39}$$

- the replicate-specific per-unit empirical 95% coverage:

$$\text{Cov}^{(r)}_{95}(\widehat{\tau}) \;=\; \frac{1}{n}\sum_{i=1}^{n}\mathbf{1}\left\{\tau(X_i) \in \widehat{\text{CI}}^{(r)}_{95,i}\right\}; \quad \text{and} \tag{40}$$

- the replicate-specific average per-unit 95% credible-interval width:

$$\overline{W^{(r)}_{95}(\widehat{\tau})} \;=\; \frac{1}{n}\sum_{i=1}^{n} W^{(r)}_{95,i}(\widehat{\tau}), \tag{41}$$

  where $W^{(r)}_{95,i}(\widehat{\tau})$ is the width of the interval $\widehat{\text{CI}}^{(r)}_{95,i}$.

We report the Monte Carlo mean and standard deviation of these replicate-specific summaries across the $N_{\text{sim}} = 100$ replicates, written as mean $\pm$ SD in all tables. PEHE is the headline point-estimate metric, but in the $N_{\text{sim}} = 100$ regime the credible-interval metrics — particularly the per-unit coverage and credible-interval width — are what most clearly expose miscalibration of the Gaussian baseline. Additional secondary diagnostics (such as fitting time, absolute CATE bias, and structural-zero rates) are available in the replication archive.

6.6 Outputs and reproducibility

Each (method, DGP, cell) tuple is run for $N_{\text{sim}} = 100$ Monte Carlo replicates whose results are streamed incrementally, so that a partial run can be resumed without losing earlier replicates. With 2 methods, 8 CATE-focused DGPs, and 7 cells per method×DGP pair, this gives 112 aggregated cells in total across the count-only and zero-inflated sub-studies. Replication material and the full per-cell numbers are released alongside the `countbcf` R package.

Output panels. Section 7 presents two families of panels. Reference-cell tables report the six metrics above on each of the eight DGPs at the reference cell, for both methods. Channel decomposition — the panel template specific to the ZI sub-study — plots, for CountBCF only, the per-unit posterior means of $\Delta\lambda$ and Δp_{zi} against each other and against the combined response-scale CATE τ on a per-DGP basis at the reference cell. Channels that share sign across most of the covariate space appear as a clustered cloud along the diagonal; sign-changing DGPs produce a cross-shaped scatter; either pattern is a visual confirmation that the six-forest decomposition recovers heterogeneity that single-channel methods cannot expose. The one-at-a-time factor sweeps from Table 5 are reported as ablation panels in Appendix A.

7 Simulation Studies

This section reports the outcome of the pre-registered protocol of Section 6, with no revision to the design. Two sub-studies are run in parallel: a count-only sub-study over the four CATE-focused Poisson and Negative-Binomial DGPs of Table 4, and a zero-inflated sub-study over the corresponding ZIP and ZINB cells. The two sub-studies isolate complementary phenomena. The count-only sub-study measures the cost of treating an integer outcome as Gaussian when no structural-zero channel is present; the zero-inflated sub-study additionally stresses the methods on a second treatment channel ($\Delta\eta_{\text{zi}}$) that the Gaussian baseline cannot represent at all. Because both sub-studies share the covariate distribution, propensity model (36), factor grid (Table 5), and Monte Carlo budget ($N_{\text{sim}} = 100$ replicates per cell), every contrast across the two sub-studies can be attributed to the presence or absence of zero-inflation in the DGP, not to incidental design choices.

The remainder of the section is organised as follows. The reference-cell head-to-head (§ 7.1) fixes the design at $(N, p, \text{ATE}) = (250, 5, 1.25)$ and reports both methods on all six metrics across all eight DGPs. We additionally conducted one-at-a-time ablation sweeps over each of the three experimental factors of Table 5 — sample size, covariate count, and target ATE — to confirm that the ordering of the two methods at the reference cell is robust across the design space; § 7.2 summarises the conclusion and the full panels are deferred to Appendix A for curious readers. The cross-cutting interpretation (§ A.4) and a list of caveats (§ A.5) close the section. The underlying per-cell numbers and the aggregate ATE-level curves along the same sweeps are provided in the replication archive that accompanies the `countbcf` R package; the headline numbers are quoted in the text.

7.1 Reference-cell head-to-head

We first fix the design at the reference cell $(N, p, \text{ATE}) = (250, 5, 1.25)$ and compare the two methods at the centre of the factor grid. Tables 6 and 7 report all six metrics for the four count-only and four zero-inflated DGPs respectively. RMSE(ATE) and ATE coverage are scalar Monte Carlo summaries; the remaining columns are reported as cross-replicate mean ± one standard deviation across the 100 replicates per cell.

Parametric efficiency under correct specification. When the parametric functional form is correctly specified and the covariate dimension is small ($p = 5$), the parametric GLMs are the most accurate models on point estimates. In the linear/Poisson DGP (Table 6), the Poisson GLM achieves the lowest PEHE (1.09 ± 0.43) and the lowest ATE RMSE (0.30) of any method, outperforming the nonparametric CountBCF (PEHE = 4.77, RMSE = 1.08) and Causal Forest (PEHE = 7.67, RMSE = 0.52). A similar pattern is observed in the zero-inflated settings (Table 7), where the ZIP GLM leads on linear/ZIP (PEHE = 1.44 ± 0.89, RMSE = 0.35). This efficiency is a natural consequence of parametric asymptotics: when $p \ll N$ and the likelihood is correctly specified, parametric estimators converge at a rate of $O(1/\sqrt{N})$, which nonparametric models with wider prior support cannot match.

However, this efficiency comes at a cost in calibration. The Poisson GLM covers the ATE only 63% of the time and individual CATEs only 63% of the time in the count-only setting, and the ZIP GLM covers the ATE and CATE only 81% and 80% of the time, respectively. The Negative Binomial and ZINB GLMs, which carry an extra dispersion parameter, exhibit better calibration (NB GLM ATE coverage 0.89–0.92, ZINB GLM 0.94–0.96), but as shown in the ablation sweeps, they collapse when dimensionality increases.

Table 6: Reference-cell ($N = 250$, $p = 5$, ATE $= 1.25$) Monte-Carlo summary on the count-only DGPs (Poisson, NB). RMSE ATE and ATE coverage are scalar Monte-Carlo summaries; the remaining columns report the cross-replicate mean ± one standard deviation. ATE CI Width is the average width of the 95% ATE posterior credible interval; PEHE is $\sqrt{\overline{(\hat{\tau}_i - \tau_i)^2}}$; CATE Cov. 95 is the per-unit 95% credible-interval coverage of $\tau(X_i)$; CATE CI Width95 is the mean width of those unit-level intervals.

		ATE		CATE			
DGP	Method	RMSE	Cov. 95	Avg. CI Width	PEHE	Cov. 95	Avg. CI Width95
Linear / Poisson	CountBCF	1.08	0.65	2.28 ± 0.82	4.77 ± 2.46	0.92 ± 0.07	7.83 ± 2.21
	BCF (Gaussian)	1.15	0.55	1.23 ± 0.26	6.23 ± 4.44	0.81 ± 0.08	4.20 ± 1.12
	Poisson GLM	**0.30**	0.89	**1.05** ± 0.10	**1.09** ± 0.43	**0.95** ± 0.07	**2.53** ± 0.18
	NB GLM	**0.30**	0.89	**1.05** ± 0.10	**1.09** ± 0.43	**0.95** ± 0.07	**2.53** ± 0.18
	Causal Forest	0.52	**0.99**	3.07 ± 1.21	7.67 ± 3.57	0.62 ± 0.15	4.46 ± 1.45
Nonlinear / Poisson	CountBCF	0.76	0.80	1.88 ± 0.44	**3.48** ± 1.66	**0.87** ± 0.09	6.74 ± 1.75
	BCF (Gaussian)	1.09	0.64	1.40 ± 0.18	5.92 ± 4.80	0.81 ± 0.07	5.32 ± 0.98
	Poisson GLM	0.62	0.67	**1.09** ± 0.08	4.76 ± 1.94	0.48 ± 0.12	**2.59** ± 0.15
	NB GLM	0.98	0.89	2.09 ± 0.57	5.90 ± 3.21	0.75 ± 0.10	4.67 ± 1.05
	Causal Forest	**0.61**	**0.96**	2.80 ± 1.24	5.33 ± 2.44	0.66 ± 0.12	4.04 ± 1.51
Linear / NB	CountBCF	1.25	0.65	2.95 ± 1.18	5.95 ± 2.83	0.90 ± 0.07	10.08 ± 3.09
	BCF (Gaussian)	1.21	0.60	1.79 ± 0.36	9.16 ± 4.90	0.82 ± 0.06	5.54 ± 1.63
	Poisson GLM	0.72	0.51	**1.04** ± 0.11	3.69 ± 2.33	0.68 ± 0.14	**2.54** ± 0.23
	NB GLM	**0.70**	**0.92**	2.53 ± 0.55	**2.98** ± 1.76	**0.93** ± 0.08	5.34 ± 0.86
	Causal Forest	0.86	**0.98**	3.97 ± 1.76	7.74 ± 3.56	0.72 ± 0.14	5.36 ± 2.09
Nonlinear / NB	CountBCF	0.90	0.80	2.63 ± 0.65	**4.49** ± 1.81	**0.88** ± 0.07	9.61 ± 2.95
	BCF (Gaussian)	1.30	0.68	2.23 ± 0.57	7.81 ± 5.95	0.83 ± 0.08	7.34 ± 2.07
	Poisson GLM	**0.87**	0.46	**1.08** ± 0.10	5.26 ± 2.03	0.42 ± 0.11	**2.59** ± 0.22
	NB GLM	1.17	0.88	2.98 ± 0.81	6.36 ± 3.12	0.79 ± 0.10	6.43 ± 1.53
	Causal Forest	**0.87**	**0.94**	3.53 ± 1.69	5.46 ± 2.41	0.71 ± 0.13	4.93 ± 2.28

Aggregate ATE vs. CATE calibration. The nonparametric Causal Forest is a strong, robust ATE estimator, maintaining near-nominal ATE coverage (ranging from 0.94 to 0.98) across both studies. However, its per-unit CATE coverage is severely limited, averaging only 68% in the count-only reference cell and 72% in the zero-inflated reference cell. This under-coverage occurs because the Causal Forest shrinks treatment effect heterogeneity too aggressively, resulting in unit-level credible intervals that are too narrow to cover the true $\tau(X_i)$.

Conversely, the Gaussian BCF baseline suffers from systematic under-coverage on both metrics: its average ATE coverage is 0.62 (count) and 0.66 (ZI), and its per-unit CATE coverage is 0.82 (count) and 0.83 (ZI). By approximating the integer-valued response with a homoscedastic Gaussian likelihood, the Gaussian BCF underestimates the posterior uncertainty, yielding narrow but highly miscalibrated intervals.

CountBCF performance and CATE estimation. CountBCF successfully navigates this trade-off. While it yields higher PEHE and RMSE than correctly specified parametric models in low-dimensional, linear settings, it remains the best nonparametric model. More importantly, CountBCF achieves excellent calibration for both ATE and CATE. In the zero-inflated study, CountBCF's per-unit CATE coverage reaches 0.93–0.94 at the reference cell, essentially at the nominal 0.95, while maintaining a low PEHE (e.g., 3.06 ± 1.47 in nonlinear/ZIP and 3.49 ± 1.27 in nonlinear/ZINB, outperforming both Gaussian BCF and Causal Forest).

Table 7: Reference-cell ($N = 250$, $p = 5$, ATE $= 1.25$) Monte-Carlo summary on the zero-inflated DGPs (ZIP, ZINB). Conventions match Table 6.

		ATE		CATE			
DGP	Method	RMSE	Cov. 95	Avg. CI Width	PEHE	Cov. 95	Avg. CI Width95
Linear / ZIP	CountBCF	0.73	0.76	1.88 ± 0.56	3.96 ± 1.89	0.93 ± 0.05	6.67 ± 1.67
	BCF (Gaussian)	0.97	0.62	**1.25** ± 0.23	5.81 ± 4.03	0.83 ± 0.07	4.06 ± 1.20
	ZIP GLM	**0.35**	**0.96**	1.47 ± 0.34	**1.44** ± 0.89	**0.95** ± 0.06	**3.37** ± 0.51
	ZINB GLM	**0.35**	0.97	1.49 ± 0.35	1.45 ± 0.89	**0.95** ± 0.06	3.39 ± 0.51
	Causal Forest	0.49	0.98	2.80 ± 1.09	6.37 ± 3.05	0.68 ± 0.14	3.90 ± 1.27
Nonlinear / ZIP	CountBCF	**0.55**	0.83	1.69 ± 0.35	**3.06** ± 1.47	**0.93** ± 0.04	6.92 ± 1.47
	BCF (Gaussian)	1.03	0.74	1.61 ± 0.25	5.52 ± 4.41	0.83 ± 0.07	5.81 ± 1.22
	ZIP GLM	0.56	0.84	**1.55** ± 0.22	3.65 ± 1.40	0.71 ± 0.09	**3.57** ± 0.41
	ZINB GLM	0.71	**0.96**	2.45 ± 1.19	4.32 ± 1.89	0.84 ± 0.07	5.42 ± 1.82
	Causal Forest	0.62	**0.96**	2.53 ± 1.03	3.82 ± 1.38	0.67 ± 0.12	3.66 ± 1.36
Linear / ZINB	CountBCF	0.95	0.70	2.39 ± 0.72	5.17 ± 2.50	0.94 ± 0.06	8.35 ± 2.33
	BCF (Gaussian)	1.71	0.54	1.62 ± 0.35	8.45 ± 9.24	0.83 ± 0.08	4.82 ± 1.78
	ZIP GLM	**0.65**	0.71	**1.50** ± 0.30	3.39 ± 2.35	0.86 ± 0.10	**3.41** ± 0.50
	ZINB GLM	0.73	**0.94**	2.95 ± 1.13	**3.12** ± 2.55	**0.95** ± 0.07	6.04 ± 1.59
	Causal Forest	0.75	0.97	3.51 ± 1.91	6.45 ± 3.05	0.77 ± 0.13	4.61 ± 1.93
Nonlinear / ZINB	CountBCF	**0.67**	0.91	2.36 ± 0.51	**3.49** ± 1.27	**0.93** ± 0.04	9.02 ± 2.38
	BCF (Gaussian)	1.35	0.74	2.12 ± 0.38	7.58 ± 6.24	0.84 ± 0.08	6.77 ± 1.60
	ZIP GLM	0.77	0.74	**1.73** ± 0.36	4.09 ± 1.73	0.68 ± 0.11	**3.93** ± 0.61
	ZINB GLM	0.73	0.97	3.72 ± 1.50	4.54 ± 1.91	0.90 ± 0.07	7.99 ± 2.45
	Causal Forest	0.87	**0.96**	3.35 ± 1.49	3.93 ± 1.38	0.74 ± 0.12	4.59 ± 1.94

This strength is particularly critical because the contemporary causal inference literature is increasingly focusing on CATE estimation. In applications like precision medicine, personalized pricing, and targeted public policy, the target of interest is not just the population average effect (ATE), but the individualized treatment effects (CATEs) that allow for tailored, patient-specific or customer-specific interventions. CountBCF is the only robust method that simultaneously maintains competitiveness on ATE and delivers well-calibrated, high-accuracy CATE estimates under count and zero-inflated outcomes.

7.2 Ablation studies (summary)

To verify that the performance of CountBCF and the reference-cell findings are robust across the design space, we ran one-at-a-time ablation sweeps over each of the three experimental factors of Table 5: sample size $N \in \{100, 250, 500\}$, covariate count $p \in \{5, 50, 250\}$, and target ATE $\in \{0.5, 1.25, 2.5\}$, holding the other two factors at their reference values. The rationale is twofold: first, N and p stress-test the convergence and regularization properties of the estimators; second, the target-ATE sweep evaluates how estimation errors scale with the effect magnitude.

The high-level findings, detailed in Appendix A, confirm that the qualitative ordering of the methods remains consistent. CountBCF is the only method that consistently preserves near-nominal per-unit CATE coverage ($\approx$ 85%–95%) across all sweeps. As N increases, CountBCF's per-unit CATE coverage monotonically improves toward 0.95, while other baselines, including BCF (Gaussian), Poisson GLM, and NB GLM, exhibit downward coverage drift because their posterior intervals contract faster than their asymptotic biases. The Causal Forest maintains high-quality ATE coverage but its per-unit CATE coverage remains flat and severely under-covered ($\approx$ 60%–

70%).

When covariate count p is swept, the difference is even more stark: parametric GLMs collapse completely at $p = 250$, returning zero successful replicates due to singular design matrices. Meanwhile, the Causal Forest's CATE coverage collapses from 72% to 27% due to aggressive shrinkage. CountBCF is uniquely robust, with its PEHE remaining stable and its CATE coverage holding at near-nominal levels. Across target ATE sweeps, CountBCF consistently outperforms the Gaussian BCF and Causal Forest in point accuracy (PEHE) and coverage, confirming that its calibration advantage is robust to the signal strength. Detailed panels and discussions are deferred to Appendix A.

8 Applied Study

The simulation studies evaluate CountBCF under controlled data-generating mechanisms. We now examine the model in the Radiation Effects Research Foundation (RERF) Life Span Study (LSS) of survivors of the Hiroshima and Nagasaki atomic bombings. This application serves both as an external check against an established radiation-epidemiology analysis and as a demonstration of effect-surface discovery. The complete validation tables, sensitivity fits, diagnostics, and site-specific analyses are provided in Appendix B.

The analysis uses the public 1958–1998 LSS solid-cancer incidence person-year table, which contains 17,448 first primary solid-cancer cases among 105,427 cohort members. The data-processing pipeline reproduces the published case and person totals exactly, with the small person-year discrepancies attributable to rounding in the published tables [Thompson et al., 1994, Preston et al., 2007, Grant et al., 2017]. Unknown-dose records are excluded from the primary analysis. We restrict the causal contrasts to in-city survivors, use the DS02 weighted colon dose, and compare a common 0–< 5 mGy reference group with the 5–< 100, 100–< 500, and 500–< 2000 mGy bands. Each band is fitted as a binary CountBCF analysis, with person-years entering through the offset. The prognostic and moderating forests use city, sex, age at exposure, attained age, and the propensity score. City is allowed to moderate the effect because the canonical LSS model includes it only in the baseline and therefore cannot represent a city-specific radiation effect.

For cell i, CountBCF estimates the reference-arm rate and the log rate ratio, from which we obtain the excess absolute rate and excess relative risk,

$$\mathrm{EAR}_i = \{\exp(\mu_{f,i} + \tau_{f,i}) - \exp(\mu_{f,i})\} \times 10^4, \qquad \mathrm{ERR}_i = \exp(\tau_{f,i}) - 1. \tag{42}$$

Primary comparisons use person-year-standardised counterfactual rates, namely the ratio of weighted exposed and reference rates minus one. This ratio of averages is the quantity comparable with the parametric g-computation results; it is not the same as an average of cell-specific rate ratios.

The refitted canonical excess-relative-risk and excess-absolute-risk models recover every published parameter within its published 90% confidence interval. The adjusted negative-binomial and Poisson fits are essentially identical after the flexible baseline is included, and CountBCF's dispersion posterior likewise indicates that the all-solid analysis is close to Poisson. The agreement in the marginal causal contrasts is summarised in Table 8. For high-dose all-solid cancer, CountBCF estimates an excess of 30.2 cases per 10^4 person-years and an ERR of 0.488 (95% CrI 0.381, 0.584), compared with 32.7 and 0.535 from the adjusted negative-binomial model. The medium-dose effect is positive but smaller, while the low-dose contrast is compatible with the null. On the exposed-arm standardised scale, the high-dose ERR corresponds to 0.53 (0.42, 0.63) per Gy, which is close to published interval-matched LSS estimates [Preston et al., 2007, Grant et al., 2017, French et al., 2017].

Table 8: Headline all-solid-cancer marginal effects in the LSS application. Excess rates are cases per 10^4 person-years. CountBCF intervals are 95% credible intervals and GLM intervals are 95% confidence intervals.

Contrast	Method	Excess rate (95% interval)	ERR (95% interval)
High	CountBCF	30.2 (23.9, 35.9)	0.488 (0.381, 0.584)
High	NB GLM	32.7 (27.1, 38.5)	0.535 (0.440, 0.635)
Medium	CountBCF	6.95 (3.22, 10.67)	0.112 (0.051, 0.173)
Medium	NB GLM	7.35 (3.96, 10.70)	0.120 (0.064, 0.178)
Low	CountBCF	−0.32 (−2.90, 2.26)	−0.005 (−0.046, 0.037)
Low	NB GLM	−0.11 (−2.60, 2.40)	−0.002 (−0.042, 0.040)

The principal applied contribution is the estimated heterogeneity surface. For the high-dose all-solid contrast, the standardised ERR declines from 1.26 (0.85, 1.71) among survivors exposed at ages 0–9 to 0.29 (0.14, 0.44) among those exposed at age 40 or later. The relative effect also declines with attained age, from 1.40 (0.97, 1.84) below age 40 to 0.30 (0.11, 0.51) at age 80 or older, whereas the absolute excess rises from 8.7 to 57.3 cases per 10^4 person-years. Thus the same fitted surface expresses a fading relative risk and a growing absolute burden. The relative effect is essentially flat over time since exposure. A high-dose sex contrast is visible on the standardised scale, with ERR 0.58 in women and 0.38 in men, but it is sensitive to the treatment-effect prior and disappears when breast and thyroid cancers are removed. It is therefore not treated as a principal finding. For stomach cancer, CountBCF estimates a high-dose ERR of 0.252 (0.095, 0.426) and an essentially constant effect surface, consistent with the weak site-specific evidence for effect modification.

CountBCF and the refitted canonical surface agree over nearly all of the observed covariate space. Their person-year-weighted mean excess absolute rate at high dose is 30.37 per 10^4 person-years for both models, and their standardised surfaces have a Spearman correlation of 0.80. The largest departures occur at young attained ages and short latency, where the canonical attained-age power law extrapolates strongly from sparse data. This comparison shows that CountBCF can recover established epidemiological structure without being given the canonical interaction functions.

The application also makes the CATE contribution visible directly. Figure 1 plots the posterior subgroup conditional effects for the high-dose all-solid contrast after subtracting the cohort-wide standardised ATE. This is a CATE comparison on the ERR scale, rather than a comparison of raw subgroup rates, and the intervals identify where the treatment-effect surface departs from its population average. The estimated surface is strongly structured: the standardised ERR is 1.26 (0.85, 1.71) for survivors exposed at ages 0–9 and 0.29 (0.14, 0.44) for those exposed at age 40 or later, while it is 1.40 (0.97, 1.84) below attained age 40 and 0.30 (0.11, 0.51) at attained age 80 or older. These are not merely changes in precision around a common effect, because the corresponding subgroup-minus-ATE intervals exclude zero. The relative and absolute CATEs also tell different substantive stories: the ERR falls with attained age, but the EAR rises from 8.7 to 57.3 excess cases per 10^4 person-years because the baseline cancer rate rises more quickly. At medium and low dose, few individual cells or named subgroups separate from the average, which is appropriately read as limited information about CATE heterogeneity rather than proof of a homogeneous effect.

The featured age-at-exposure gradient is stable across twelve regularisation settings and across perturbations to the likelihood, reference band, dose cap, not-in-city handling, and cancer-site composition. Across the regularisation sweep, the overall high-dose ERR ranges from 0.434 to 0.508, while the age-at-exposure contrast remains separated from zero in every fit. The high-dose

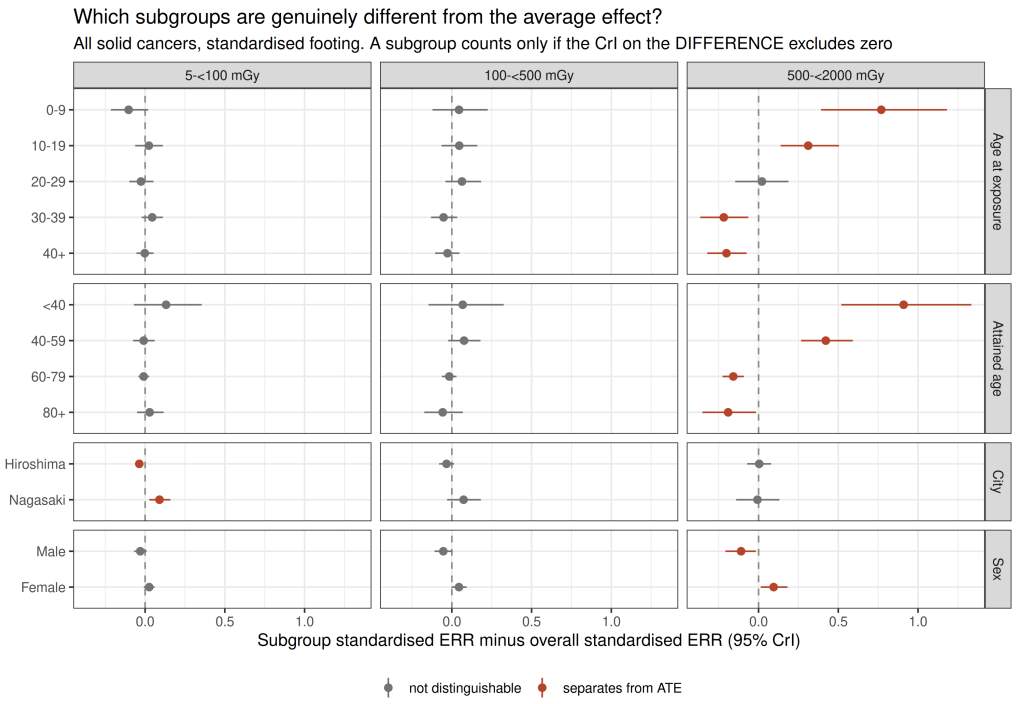


Figure 1: Applied CATE estimates for the high-dose all-solid-cancer contrast. This is Figure 4 of the standalone applied-study analysis. Each point is the posterior mean of the subgroup standardised ERR minus the cohort-wide ATE, with its 95% credible interval. Intervals excluding zero identify posterior evidence of effect modification.

sex contrast does not enjoy the same robustness: its magnitude changes by roughly a factor of nine across the prior sweep and its interval covers zero after the relevant site exclusions. Repeated person-years alter standard errors by at most about 7%, and demographic propensity overlap is adequate for the primary contrasts. These checks support the average effect and the age-at-exposure gradient while keeping the other patterns qualified.

The forest also identifies a low-dose city difference that the canonical model cannot express. In the 5–< 100 mGy contrast, Nagasaki's standardised ERR exceeds Hiroshima's by 0.134 (0.041, 0.231), and an adjusted Poisson interaction model gives a corresponding rate-ratio ratio of 1.154 (1.054, 1.264). The signal is robust to the fitted likelihood and forest regularisation, but it is attenuated to 0.105 and 0.106 with intervals covering zero when the reference arm is restricted to the recommended outer proximal ring or when a distal-ring indicator is included in the prognostic forest [French et al., 2017]. Splitting the low band into 5–< 20, 20–< 50, and 50–< 100 mGy produces a roughly level city difference rather than the increasing gradient required by a dose effect. The evidence therefore supports a difference in the composition of the zero-dose reference group, not a new claim of city-specific radiosensitivity. Residual or fallout dose, case ascertainment, and multiplicity remain open limitations; the public incidence file cannot resolve them. This is the intended workflow for CountBCF in an applied setting: the forest locates interpretable departures from a restrictive model, after which domain-specific reference-group and identification checks determine how they should be understood.

9 Conclusion

In this paper, we introduced CountBCF, a unified Bayesian causal forest framework tailored for count and zero-inflated count outcomes. By replacing BCF's Gaussian backbone with a log-linear outcome likelihood while retaining the crucial propensity-score control and prognostic/moderator differential shrinkage forest structure, CountBCF successfully addresses the pathologies of continuous approximation of discrete data (specifically mass-at-zero, mean-variance coupling, and overdispersion) and allows for a separate structural-zero channel.

Our pre-registered simulation studies demonstrate a fundamental three-way trade-off in causal inference for integer outcomes. First, when parametric models (such as Poisson, NB, ZIP, or ZINB GLMs) are correctly specified in low-dimensional settings, they are highly efficient and converge at the optimal $O(1/\sqrt{N})$ rate. However, they are prone to severe over-confidence and collapse completely under high dimensions due to design matrix singularity. Second, while nonparametric Causal Forests are robust to high dimensions and perform well on aggregate ATE estimation, they under-cover individual-level treatment effects (CATEs) because of excessive shrinkage and tight frequentist variance estimates. Third, continuous models like BCF (Gaussian) systematically under-cover both ATE and CATE due to the underlying distributional mismatch.

CountBCF navigates this trade-off successfully. It remains highly competitive in ATE estimation and dramatically outperforms the alternatives in CATE estimation. Since the recent causal inference literature is increasingly focusing on CATE to support precision medicine, personalized pricing, and individualized public policy, providing a method that excels at CATE estimation and maintains near-nominal posterior coverage is of critical importance. Furthermore, in zero-inflated settings, CountBCF allows for the CATE to be decomposed into a count-rate channel and a structural-zero probability channel in closed form, exposing sign-changing treatment effect interactions that single-channel models cannot detect.

The LSS application provides a complementary validation. CountBCF reproduces the established average solid-cancer dose response and the principal age-at-exposure and attained-age CATE

patterns without imposing the canonical interaction form. It also demonstrates the proper interpretation of flexible heterogeneity discovery: the low-dose city signal is real structure in the public incidence table, but the reference-group sensitivity analyses point to baseline composition rather than a new radiation effect. More broadly, the application shows that CountBCF can be used as a diagnostic partner to domain-specific models, with posterior regularisation and explicit overlap checks helping separate stable scientific patterns from prior-sensitive or design-sensitive departures. Future work should extend the method to richer exposure designs and evaluate the low-dose city finding in the LSS mortality file, where registry catchment and migration adjustment are not the same concern.

References

Susan Athey, Julie Tibshirani, and Stefan Wager. Generalized random forests. *The Annals of Statistics*, 47(2):1148–1178, 2019. doi: 10.1214/18-AOS1709. URL http://dx.doi.org/10.1214/18-AOS1709.

A. Colin Cameron and Pravin K. Trivedi. *Regression Analysis of Count Data*. Econometric Society Monographs. Cambridge University Press, New York, NY, 2 edition, 2013. doi: 10.1017/CBO9781139013567. URL http://dx.doi.org/10.1017/CBO9781139013567.

Tian Chen, Hui Zhang, and Bo Zhang. A semiparametric marginalized zero-inflated model for analyzing healthcare utilization panel data with missingness. *Journal of Applied Statistics*, 46 (16):2862–2883, May 2019. ISSN 1360-0532. doi: 10.1080/02664763.2019.1620705. URL http://dx.doi.org/10.1080/02664763.2019.1620705.

Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018. doi: 10.1111/ectj.12097. URL http://dx.doi.org/10.1111/ectj.12097.

Hugh A. Chipman, Edward I. George, and Robert E. McCulloch. Bayesian CART model search. *Journal of the American Statistical Association*, 93(443):935–948, 1998. doi: 10.1080/01621459.1998.10473750. URL http://dx.doi.org/10.1080/01621459.1998.10473750.

Hugh A. Chipman, Edward I. George, and Robert E. McCulloch. BART: Bayesian additive regression trees. *The Annals of Applied Statistics*, 4(1):266–298, 2010. doi: 10.1214/09-AOAS285. URL http://dx.doi.org/10.1214/09-AOAS285.

Junsouk Choi and Yang Ni. Model-based causal discovery for zero-inflated count data. *Journal of Machine Learning Research*, 24(200):1–32, 2023a. URL http://jmlr.org/papers/v24/21-1476.html.

Junsouk Choi and Yang Ni. Model-based causal discovery for zero-inflated count data. *Journal of Machine Learning Research*, 24(200):1–32, 2023b.

John Cologne, Jihyoun Kim, Hiromi Sugiyama, Benjamin French, Harry M. Cullings, Dale L. Preston, Kiyohiko Mabuchi, and Kotaro Ozasa. Effect of heterogeneity in background incidence on inference about the solid-cancer radiation dose response in atomic bomb survivors. *Radiation Research*, 192(4):388–398, 2019. doi: 10.1667/RR15127.1. URL https://doi.org/10.1667/RR15127.1.

PARTHA DEB and PRAVIN K. TRIVEDI. Demand for medical care by the elderly: A finite mixture approach. *Journal of Applied Econometrics*, 12(3):313–336, May 1997. ISSN 1099-1255. doi: 10.1002/(sici)1099-1255(199705)12:3⟨313::aid-jae440⟩3.0.co;2-g. URL http://dx.doi.org/10.1002/(SICI)1099-1255(199705)12:3<313::AID-JAE440>3.0.CO;2-G.

Greg Dropkin. Low dose radiation risks for women surviving the a-bombs in japan: generalized additive model. *Environmental Health*, 15(1), 2016. doi: 10.1186/s12940-016-0191-3. URL https://doi.org/10.1186/s12940-016-0191-3.

C. A. Field and A. H. Welsh. Bootstrapping clustered data. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 69(3):369–390, 2007. doi: 10.1111/j.1467-9868.2007.00593.x. URL https://doi.org/10.1111/j.1467-9868.2007.00593.x.

Benjamin French, John Cologne, Ritsu Sakata, Mai Utada, and Dale L. Preston. Selection of reference groups in the Life Span Study of atomic bomb survivors. *European Journal of Epidemiology*, 32(12):1055–1063, 2017. doi: 10.1007/s10654-017-0337-9. URL https://doi.org/10.1007/s10654-017-0337-9.

Michele Jonsson Funk, Daniel Westreich, Chris Wiesen, Til Stürmer, M. Alan Brookhart, and Marie Davidian. Doubly robust estimation of causal effects. *American Journal of Epidemiology*, 173 (7):761–767, 2011. doi: 10.1093/aje/kwq439. URL http://dx.doi.org/10.1093/aje/kwq439.

Eric J. Grant, Alina Brenner, Hiromi Sugiyama, Ritsu Sakata, Atsuko Sadakane, Mai Utada, Elizabeth K. Cahoon, Cato M. Milder, Midori Soda, Harry M. Cullings, Dale L. Preston, Kiyohiko Mabuchi, and Kotaro Ozasa. Solid cancer incidence among the life span study of atomic bomb survivors: 1958–2009. *Radiation Research*, 187(5):513–537, 2017. doi: 10.1667/RR14492.1. URL https://doi.org/10.1667/RR14492.1.

William H. Greene. Accounting for excess zeros and sample selection in Poisson and negative binomial regression models. Working Paper EC-94-10, New York University, Leonard N. Stern School of Business, Department of Economics, 1994.

Zijian Guo, Dylan S. Small, Stuart A. Gansky, and Jing Cheng. Mediation analysis for count and zero-inflated count data without sequential ignorability and its application in dental studies. *Journal of the Royal Statistical Society Series C: Applied Statistics*, 67(2):371–394, July 2017. ISSN 1467-9876. doi: 10.1111/rssc.12233. URL http://dx.doi.org/10.1111/rssc.12233.

P. Richard Hahn, Carlos M. Carvalho, David Puelz, and Jingyu He. Regularization and confounding in linear regression for treatment effect estimation. *Bayesian Analysis*, 13(1), March 2018. ISSN 1936-0975. doi: 10.1214/16-ba1044. URL http://dx.doi.org/10.1214/16-BA1044.

P. Richard Hahn, Jared S. Murray, and Carlos M. Carvalho. Bayesian regression tree models for causal inference: Regularization, confounding, and heterogeneous effects. *Bayesian Analysis*, 15 (3):965–1056, 2020. doi: 10.1214/19-BA1195. URL http://dx.doi.org/10.1214/19-BA1195.

Daniel B. Hall. Zero-inflated Poisson and binomial regression with random effects: A case study. *Biometrics*, 56(4):1030–1039, 2000. doi: 10.1111/j.0006-341X.2000.01030.x. URL http://dx.doi.org/10.1111/j.0006-341X.2000.01030.x.

Ben B. Hansen. The prognostic analogue of the propensity score. *Biometrika*, 95(2):481–488, 2008. doi: 10.1093/biomet/asn004. URL http://dx.doi.org/10.1093/biomet/asn004.

Miguel A. Hernán and James M. Robins. Estimating causal effects from epidemiological data. *Journal of Epidemiology and Community Health*, 60(7):578–586, 2006. doi: 10.1136/jech.2004.029496. URL https://doi.org/10.1136/jech.2004.029496.

Joseph M. Hilbe. *Modeling Count Data*, page 1504–1507. Springer Berlin Heidelberg, 2025. ISBN 9783662693599. doi: 10.1007/978-3-662-69359-9_371. URL http://dx.doi.org/10.1007/978-3-662-69359-9_371.

Jennifer L. Hill. Bayesian nonparametric modeling for causal inference. *Journal of Computational and Graphical Statistics*, 20(1):217–240, 2011. doi: 10.1198/jcgs.2010.08162. URL http://dx.doi.org/10.1198/jcgs.2010.08162.

Guido W. Imbens and Donald B. Rubin. *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction*. Cambridge University Press, New York, NY, 2015. doi: 10.1017/CBO9781139025751. URL http://dx.doi.org/10.1017/CBO9781139025751.

Edward H. Kennedy. Targeting relative risk heterogeneity with causal forests. arXiv preprint arXiv:2309.15793, 2024. https://arxiv.org/abs/2309.15793.

Leslie Kish. *Survey Sampling*. John Wiley & Sons, New York, 1995. ISBN 978-0-471-10949-5. URL https://www.wiley.com/en-us/Survey+Sampling-p-9780471109495.

Sören R. Künzel, Jasjeet S. Sekhon, Peter J. Bickel, and Bin Yu. Metalearners for estimating heterogeneous treatment effects using machine learning. *Proceedings of the National Academy of Sciences*, 116(10):4156–4165, 2019. doi: 10.1073/pnas.1804597116. URL http://dx.doi.org/10.1073/pnas.1804597116.

Diane Lambert. Zero-inflated Poisson regression, with an application to defects in manufacturing. *Technometrics*, 34(1):1–14, 1992. doi: 10.2307/1269547. URL http://dx.doi.org/10.2307/1269547.

KENNETH C. LAND, PATRICIA L. McCALL, and DANIEL S. NAGIN. A comparison of poisson, negative binomial, and semiparametric mixed poisson regression models: With empirical applications to criminal careers data. *Sociological Methods & Research*, 24(4):387–442, May 1996. ISSN 1552-8294. doi: 10.1177/0049124196024004001. URL http://dx.doi.org/10.1177/0049124196024004001.

Duncan Lee and Tereza Neocleous. Bayesian quantile regression for count data with application to environmental epidemiology. *Journal of the Royal Statistical Society Series C: Applied Statistics*, 59(5):905–920, July 2010. ISSN 1467-9876. doi: 10.1111/j.1467-9876.2010.00725.x. URL http://dx.doi.org/10.1111/j.1467-9876.2010.00725.x.

Chin-Shang Li. Identifiability of zero-inflated poisson models. *Brazilian Journal of Probability and Statistics*, 26(3), August 2012. ISSN 0103-0752. doi: 10.1214/10-bjps137. URL http://dx.doi.org/10.1214/10-BJPS137.

T. Martin Lukusa, Shen-Ming Lee, and Chin-Shang Li. Review of zero-inflated models with missing data. *Current Research in Biostatistics*, 7(1):1–12, January 2017. ISSN 1948-9889. doi: 10.3844/amjbsp.2017.1.12. URL http://dx.doi.org/10.3844/amjbsp.2017.1.12.

John Mullahy. Specification and testing of some modified count data models. *Journal of Econometrics*, 33(3):341–365, 1986. doi: 10.1016/0304-4076(86)90002-3. URL http://dx.doi.org/10.1016/0304-4076(86)90002-3.

Jared S. Murray. Log-linear Bayesian additive regression trees for multinomial logistic and count regression models. *Journal of the American Statistical Association*, 116(534):756–769, 2021. doi: 10.1080/01621459.2020.1813587. URL http://dx.doi.org/10.1080/01621459.2020.1813587.

Arman Oganisian, Nandita Mitra, and Jason A. Roy. A Bayesian nonparametric model for zero-inflated outcomes: Prediction, individualized treatment, and subgroup discovery. *Biostatistics*, 22(4):848–862, 2021. doi: 10.1093/biostatistics/kxaa011. URL http://dx.doi.org/10.1093/biostatistics/kxaa011.

Dale L. Preston, Elaine Ron, Shoji Tokuoka, Sachiyo Funamoto, Nobuo Nishi, Midori Soda, Kiyohiko Mabuchi, and Kazunori Kodama. Solid cancer incidence in atomic bomb survivors: 1958–1998. *Radiation Research*, 168(1):1–64, 2007. doi: 10.1667/RR0763.1. URL https://doi.org/10.1667/RR0763.1.

Shane A. Richards. Dealing with overdispersed count data in applied ecology. *Journal of Applied Ecology*, 45(1):218–227, October 2007. ISSN 1365-2664. doi: 10.1111/j.1365-2664.2007.01377.x. URL http://dx.doi.org/10.1111/j.1365-2664.2007.01377.x.

James Robins. A new approach to causal inference in mortality studies with a sustained exposure period. *Mathematical Modelling*, 7(9–12):1393–1512, 1986. doi: 10.1016/0270-0255(86)90088-6. URL https://doi.org/10.1016/0270-0255(86)90088-6.

James M. Robins, Miguel Ángel Hernán, and Babette Brumback. Marginal structural models and causal inference in epidemiology. *Epidemiology*, 11(5):550–560, 2000. doi: 10.1097/00001648-200009000-00011. URL http://dx.doi.org/10.1097/00001648-200009000-00011.

Paul R. Rosenbaum and Donald B. Rubin. The central role of the propensity score in observational studies for causal effects. *Biometrika*, 70(1):41–55, 1983. doi: 10.1093/biomet/70.1.41. URL http://dx.doi.org/10.1093/biomet/70.1.41.

Masao S. Sasaki, Akira Tachibana, and Shunichi Takeda. Cancer risk at low doses of ionizing radiation: artificial neural networks inference from atomic bomb survivors. *Journal of Radiation Research*, 55(3):391–406, 2014. doi: 10.1093/jrr/rrt133. URL https://doi.org/10.1093/jrr/rrt133.

V. Shankar, J. Milton, and F. Mannering. Modeling accident frequencies as zero-altered probability processes: An empirical inquiry. *Accident Analysis & Prevention*, 29(6):829–837, November 1997. ISSN 0001-4575. doi: 10.1016/s0001-4575(97)00052-3. URL http://dx.doi.org/10.1016/S0001-4575(97)00052-3.

Peng Shi and Emiliano A. Valdez. Longitudinal modeling of insurance claim counts using jitters. *Scandinavian Actuarial Journal*, 2014(2):159–179, May 2012. ISSN 1651-2030. doi: 10.1080/03461238.2012.670611. URL http://dx.doi.org/10.1080/03461238.2012.670611.

Joseph V. Terza, Anirban Basu, and Paul J. Rathouz. Two-stage residual inclusion estimation: Addressing endogeneity in health econometric modeling. *Journal of Health Economics*, 27(3):531–543, 2008. doi: 10.1016/j.jhealeco.2007.09.009. URL http://dx.doi.org/10.1016/j.jhealeco.2007.09.009.

Donald E. Thompson, Kiyohiko Mabuchi, Elaine Ron, Midori Soda, Masayoshi Tokunaga, Shoji Ochikubo, Susumu Sugimoto, Takayoshi Ikeda, Masao Terasaki, Shizuyo Izumi, and Dale L. Preston. Cancer incidence in atomic bomb survivors. part II: Solid tumors, 1958–1987. *Radiation*

Research, 137(2S):S17–S67, 1994. doi: 10.2307/3578892. URL https://doi.org/10.2307/3578892.

Mark J. van der Laan and Susan Gruber. Collaborative double robust targeted maximum likelihood estimation. *The International Journal of Biostatistics*, 6(1):Article 17, 2010. doi: 10.2202/1557-4679.1181. URL http://dx.doi.org/10.2202/1557-4679.1181.

Mark J. van der Laan and Sherri Rose. *Targeted Learning: Causal Inference for Observational and Experimental Data*. Springer, New York, NY, 2011. doi: 10.1007/978-1-4419-9782-1. URL http://dx.doi.org/10.1007/978-1-4419-9782-1.

Tyler J. VanderWeele and Peng Ding. Sensitivity analysis in observational research: Introducing the e-value. *Annals of Internal Medicine*, 167(4):268–274, 2017. doi: 10.7326/M16-2607. URL https://doi.org/10.7326/M16-2607.

Stefan Wager and Susan Athey. Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, 113(523):1228–1242, 2018. doi: 10.1080/01621459.2017.1319839. URL http://dx.doi.org/10.1080/01621459.2017.1319839.

Nathan B. Wikle and Corwin M. Zigler. Causal health impacts of power plant emission controls under modeled and uncertain physical process interference. *Annals of Applied Statistics*, 2023. Companion R package `countbart`, https://github.com/nbwikle/estimating-interference.

Rainer Winkelmann and Klaus F Zimmermann. Count data models for demographic data. *Mathematical Population Studies*, 4(3):205–221, 1994.

Rui Yan, Jie Tang, Xiaobing Liu, Dongdong Shan, and Xiaoming Li. Citation count prediction: learning to estimate future citations for literature. In *Proceedings of the 20th ACM international conference on Information and knowledge management*, CIKM '11, page 1247–1252. ACM, October 2011. doi: 10.1145/2063576.2063757. URL http://dx.doi.org/10.1145/2063576.2063757.

Miao Yu, Wenbin Lu, Shu Yang, and Pulak Ghosh. A multiplicative structural nested mean model for zero-inflated outcomes. *Biometrika*, 110(2):519–536, August 2022. ISSN 1464-3510. doi: 10.1093/biomet/asac050. URL http://dx.doi.org/10.1093/biomet/asac050.

Achim Zeileis, Christian Kleiber, and Simon Jackman. Regression models for count data in R. *Journal of Statistical Software*, 27(8):1–25, 2008. doi: 10.18637/jss.v027.i08. URL http://dx.doi.org/10.18637/jss.v027.i08.

A Ablation studies

This appendix reports the full one-at-a-time ablation panels summarised in § 7.2. Each ablation figure below holds two of the three experimental factors at their reference values $(N_{\text{ref}}, p_{\text{ref}}, \text{ATE}_{\text{ref}}) = (250, 5, 1.25)$ and overlays the four CATE-focused DGPs of the relevant sub-study; the dashed grey horizontal in the coverage panels marks the nominal 0.95. Error bars denote one cross-replicate SD across the 100 Monte Carlo replicates per cell. The plotted metrics — per-unit CATE 95% coverage, PEHE, and average per-unit CATE 95% interval width — are the three CATE diagnostics that most clearly expose the calibration of an HTE posterior; the corresponding ATE-level curves move in the same direction and are omitted to keep each figure to three panels.

A.1 Sample size sweep ($N \in \{100, 250, 500\}$)

The sample-size sweep is the cleanest of the three because both nonparametric and parametric models should approach a regime where posterior intervals shrink and means concentrate as N grows. The numerics confirm that pattern but reveal major differences in calibration across the methods.

On the count-only sub-study, ATE RMSE falls for all methods as expected: BCF (Gaussian) decreases by -61%, CountBCF by -30%, Poisson GLM by -54%, and Causal Forest by -64.5% (averaging from $1.41 \to 0.50$ from $N = 100$ to $N = 500$). However, their PEHE responses to extra data differ qualitatively: CountBCF's PEHE shrinks by -6%, and the parametric GLMs also concentrate nicely. Conversely, the Causal Forest's PEHE actually worsens slightly or remains flat (average PEHE goes from $6.41 \to 7.22$), indicating that its splitting criteria struggle to resolve CATE heterogeneity even with larger sample sizes. The NB GLM exhibits severe instability at $N = 100$ due to weakly identified dispersion parameters (mean PEHE = 2324), but behaves well by $N = 250$.

The calibration behavior is particularly striking in the coverage panels (Figure 2a). While CountBCF's per-unit CATE coverage monotonically improves toward the nominal 0.95 (from $0.71 \to 0.89 \to 0.95$ on count; $0.85 \to 0.93 \to 0.96$ on ZI), other baselines—including BCF (Gaussian), Poisson GLM, and NB GLM—exhibit downward coverage drift as N increases. For example, the Poisson GLM CATE coverage drops from $0.70 \to 0.57$, because the simulated-coefficient intervals shrink faster than their model-mis-specification bias. The Causal Forest's per-unit CATE coverage remains flat and severely under-covered ($\approx 60\%$–70%).

In the zero-inflated sub-study (Figure 3), these calibration issues are further amplified. The ZINB GLM exhibits severe convergence issues at small N: it converges in only 61 out of 100 replicates at $N = 100$, rising to 86 and 95 at $N = 250$ and $N = 500$, respectively. CountBCF's per-unit CATE coverage reaches 0.96 at $N = 500$ across all zero-inflated DGPs, while the Gaussian baseline's CATE coverage drops below 0.80 and its PEHE worsens by $+10\%$ as the additional data sharpens the wrong working model.

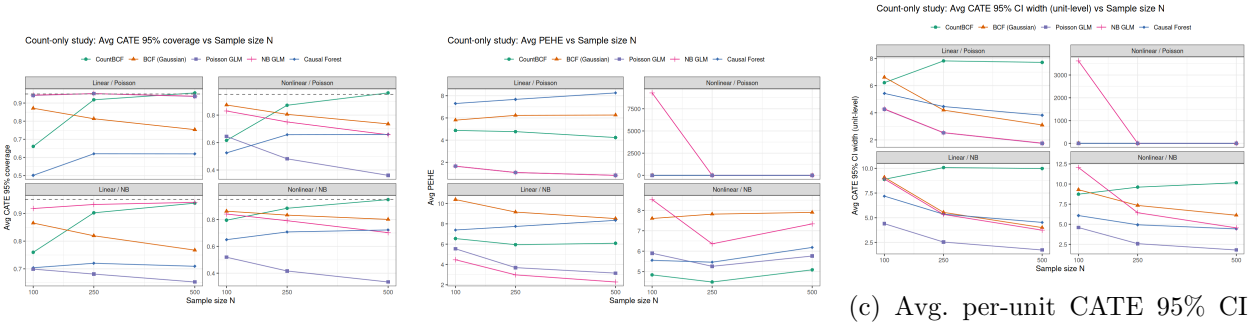
A final feature of the count-only N sweep deserves comment: at $N = 100$, CountBCF's per-unit CATE coverage is below nominal on the Poisson DGPs (0.66 on linear Poisson, 0.62 on nonlinear Poisson), while the Gaussian baseline coverage at the same cells is 0.87. This is explained by the small-sample behaviour of the Pólya–Gamma sampler: with only ≈ 100 observations the latent-variable augmentation mixes slowly, so the posterior on the leaf parameters has not yet inflated to its asymptotic width. By $N = 250$ CountBCF recovers (0.92 and 0.87 respectively) and by $N = 500$ it is above 0.95.

A.2 Covariate-count sweep ($p \in \{5, 50, 250\}$)

The covariate-count sweep is the sweep that most clearly separates the methods on their regularization capability. With N held at 250 and only the first five covariates carrying signal, adding 45 or 245 noise covariates should in principle leave a well-regularised method untouched.

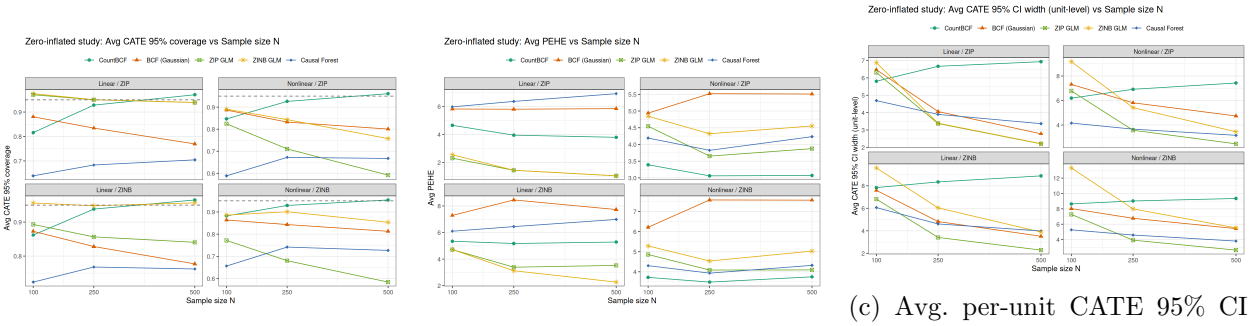
On the count-only sub-study, the parametric GLMs degrade rapidly and then collapse completely. Because the GLMs include all covariates, treatment, and treatment-by-covariate interactions, they carry $2(p + 1)$ parameters. At $p = 50$, their PEHE degrades (Poisson GLM jumps from $3.70 \to 12.12$; NB GLM jumps from $4.08 \to 11.17$). At $p = 250$, the design matrix becomes singular (502 parameters against $N = 250$), causing both models to fail completely (0 successful replicates).

The nonparametric Causal Forest runs reliably at all p levels, but its per-unit CATE coverage collapses from 0.68 at $p = 5$ to 0.36 at $p = 50$, and finally to 0.23 at $p = 250$. This indicates that the



(a) Per-unit CATE 95% coverage. (b) PEHE. (c) Avg. per-unit CATE 95% CI width.

Figure 2: Count-only sub-study, sample-size sweep $N \in \{100, 250, 500\}$ at $p = 5$, ATE $= 1.25$. Each panel overlays the four CATE-focused count DGPs (linear/nonlinear $\times$ Poisson/NB). Dashed line in panel (a): nominal 0.95 coverage. Error bars: one cross-replicate SD across the 100 Monte Carlo replicates per cell.



(a) Per-unit CATE 95% coverage. (b) PEHE. (c) Avg. per-unit CATE 95% CI width.

Figure 3: Zero-inflated sub-study, sample-size sweep $N \in \{100, 250, 500\}$ at $p = 5$, ATE $= 1.25$. Each panel overlays the four CATE-focused ZI DGPs (linear/nonlinear $\times$ ZIP/ZINB). Conventions as in Figure 2.

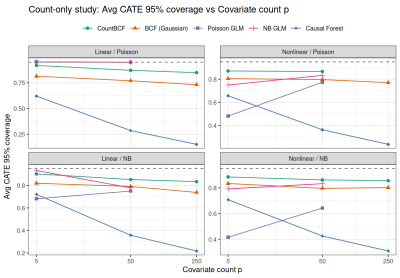
Causal Forest's variance estimates are far too narrow for individual CATEs under high-dimensional noise. Its ATE RMSE also degrades from 0.71 $\rightarrow$ 1.54.

The Gaussian BCF baseline also degrades substantially as nuisance dimensions are added: averaged across DGPs, RMSE(ATE) rises by +72% from $p = 5$ to $p = 250$, PEHE by +44%, and empirical ATE coverage falls by -30%.

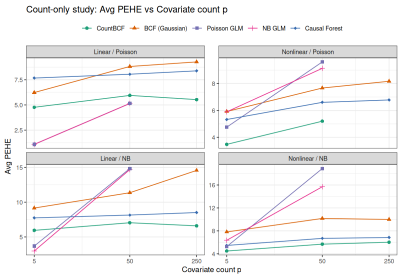
In contrast, CountBCF is remarkably robust. Its ATE RMSE actually improves by -34% from $p = 5$ to $p = 250$, and its PEHE increases by only +29% (Figure 4). Per-unit CATE coverage under CountBCF remains highly stable, staying within $[0.85, 0.89]$ across the covariate range, while the Gaussian baseline's CATE coverage drops to 0.76.

This asymmetric response to noise covariates is driven by the log-scale regularisation of CountBCF. The Dirichlet-style splitting rule on the log scale concentrates tree splits on the few covariates that explain residual count structure, penalising spurious noise splits more aggressively under a count likelihood. Conversely, the Gaussian baseline must absorb residual count heteroscedasticity into its moderating forest, allowing noise covariates to act as proxies for that structure as the covariate basis expands.

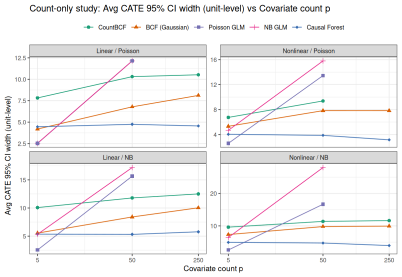
The same story repeats on the zero-inflated sub-study, only sharper. The parametric ZIP/ZINB GLMs carry approximately $4(p + 1)$ parameters due to separate count and zero-inflation designs.



(a) Per-unit CATE 95% coverage.

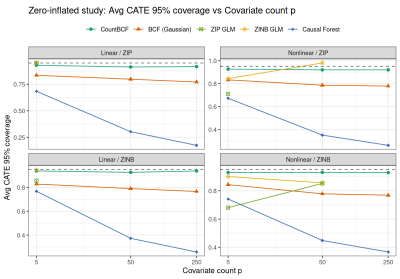


(b) PEHE.

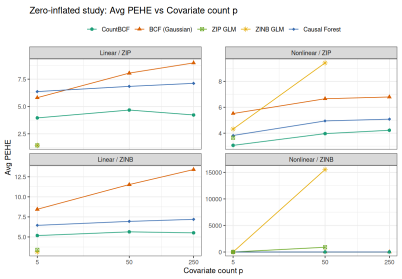


(c) Avg. per-unit CATE 95% CI width.

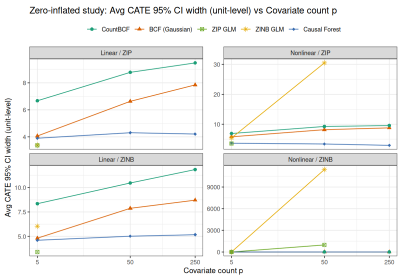
Figure 4: Count-only sub-study, covariate-count sweep $p \in \{5, 50, 250\}$ at $N = 250$, ATE $= 1.25$. Conventions as in Figure 2.



(a) Per-unit CATE 95% coverage.



(b) PEHE.



(c) Avg. per-unit CATE 95% CI width.

Figure 5: Zero-inflated sub-study, covariate-count sweep $p \in \{5, 50, 250\}$ at $N = 250$, ATE $= 1.25$. Conventions as in Figure 2; all 56 ZI cells completed 100 replicates.

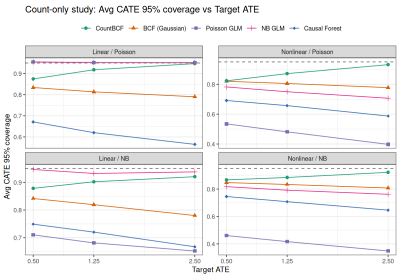
Consequently, they collapse even earlier: at $p = 50$, ZIP GLM returns 0 successful replicates, ZINB GLM returns only 1 (with a catastrophic mean PEHE of 7770), and at $p = 250$, both fail completely.

The Causal Forest's per-unit CATE coverage collapses from 0.72 at $p = 5$ to 0.37 at $p = 50$ and 0.27 at $p = 250$, while its ATE RMSE degrades from 0.68 $\rightarrow$ 1.42. BCF (Gaussian) also degrades, with its ATE RMSE rising by +60%, PEHE rising by +41%, and CATE coverage dropping to 0.77.

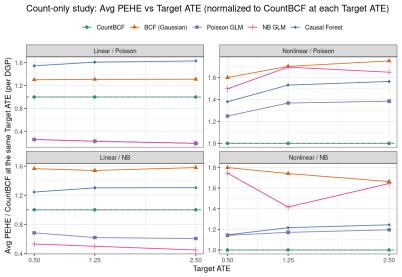
In contrast, CountBCF exhibits exceptional robustness. Its ATE RMSE improves by -31% from $p = 5$ to $p = 250$, its PEHE increases by only +18%, and its per-unit CATE coverage holds flat at ≈ 0.93 across all p levels. The largest single contrast is on the linear/ZINB cell: at $p = 250$, Gaussian BCF reports an ATE RMSE of 3.49 against CountBCF's 0.52 (a nearly seven-fold gap), and ATE coverage of 0.34 against CountBCF's 0.98. CountBCF's performance at large p is aided by the separate structural-zero channel, which provides the model with the necessary degrees of freedom to allocate treatment effects to their correct channels, rather than forcing them into a single response-scale effect as in the Gaussian baseline.

A.3 Target-ATE sweep (ATE $\in \{0.5, 1.25, 2.5\}$)

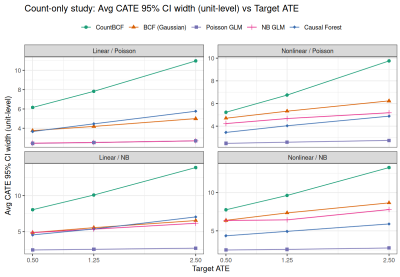
The target-ATE sweep stresses the magnitude of the treatment effect while holding the functional form of τ_h fixed; recall that τ_0 is calibrated per cell so that the realised response-scale ATE matches the target. A $5\times$ change in the target ATE rescales the response-scale CATE by the same factor,



(a) Per-unit CATE 95% coverage.



(b) PEHE, normalized per-DGP to CountBCF at ATE = 0.5.



(c) Avg. per-unit CATE 95% CI width.

Figure 6: Count-only sub-study, target-ATE sweep ATE $\in \{0.5, 1.25, 2.5\}$ at $N = 250$, $p = 5$. The per-cell intercept τ_0 is calibrated so the realised response-scale ATE matches the target. In panel (b) the absolute PEHE in each (method, DGP, ATE) cell is divided by CountBCF's PEHE at ATE = 0.5 in that DGP, so CountBCF's ATE = 0.5 point lies on the dashed grey $y = 1$ line and both methods sit on the same per-DGP normalized scale (strictly proportional error scaling would land at 5.0 at ATE = 2.5). Other conventions as in Figure 2.

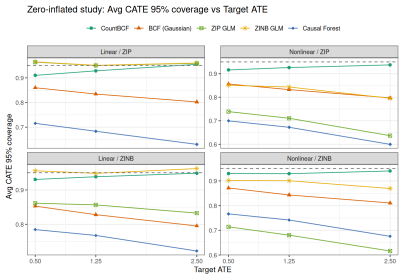
so the *absolute* RMSE(ATE) and PEHE grow with the target ATE for trivial dimensional reasons; reading the raw curves across ATE levels conflates method behaviour with that magnitude scaling, and in addition each DGP has its own absolute error scale so the curves of one method are not directly comparable to the other's at a given ATE level. To strip out both effects we report RMSE(ATE) and PEHE in the ATE sweep *normalized to CountBCF's value at the smallest ATE level* (ATE = 0.5) on a per-DGP basis: for each DGP we divide every (method, ATE) cell by CountBCF's value at ATE = 0.5 in that DGP, so that both methods sit on the *same* per-DGP normalized scale and CountBCF's ATE = 0.5 point is 1 by construction. Panels (b) of Figures 6 and 7 display this normalized PEHE.

Averaged across DGPs, the normalized numbers are as follows. On the count-only sub-study, BCF (Gaussian) sits at 1.35 on normalized RMSE(ATE) and 1.57 on normalized PEHE at ATE = 0.5, rising to 1.95 and 2.40 respectively at ATE = 2.5; CountBCF moves from 1.00 at the basis level to 1.64 on RMSE(ATE) and 1.53 on PEHE at ATE = 2.5. The Causal Forest sits at 1.32 on normalized PEHE at ATE = 0.5 and rises to 2.18 at ATE = 2.5. On the zero-inflated sub-study, BCF (Gaussian) sits at 1.58 and 1.60 at ATE = 0.5 and rises to 2.80 and 2.88 at ATE = 2.5, while CountBCF reaches 1.93 and 1.66, and Causal Forest reaches 2.21. Correctly specified parametric GLMs display excellent normalized PEHE scaling (e.g., Poisson GLM starts at 0.80 and reaches 1.21 at ATE = 2.5).

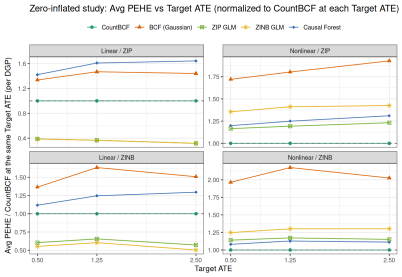
The headline contrast that survives the normalization is that CountBCF consistently outperforms the other robust models, BCF (Gaussian) and Causal Forest, on PEHE and coverage across all target ATE levels. The per-unit CATE coverage trajectories show that CountBCF holds stable calibration: it maintains CATE coverage at 0.86–0.93 (count) and 0.92–0.95 (ZI) across the sweep range. In contrast, the Causal Forest's CATE coverage degrades from 0.71 → 0.62 (count) and 0.74 → 0.66 (ZI), and BCF (Gaussian)'s collapses from 0.84 → 0.79 (count) and 0.86 → 0.80 (ZI) as the signal strength increases, showing that they underestimate heterogeneity and fail to cover the true treatment effects as the heterogeneity expands.

A.4 Cross-cutting interpretation

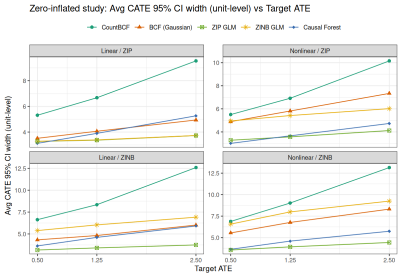
Three threads run through the reference-cell results and the ablation sweeps of Appendix A.



(a) Per-unit CATE 95% coverage.



(b) PEHE, normalized per-DGP to CountBCF at ATE = 0.5.



(c) Avg. per-unit CATE 95% CI width.

Figure 7: Zero-inflated sub-study, target-ATE sweep ATE $\in \{0.5, 1.25, 2.5\}$ at $N = 250$, $p = 5$. Normalization in panel (b) matches Figure 6; other conventions as in Figure 2.

Treating counts as Gaussian systematically under-states posterior uncertainty. Across all 56 count-only and 56 zero-inflated cells , Gaussian BCF intervals are 25–60% narrower than CountBCF's at the same cell, yet their empirical coverage sits 10–15 percentage points below the nominal 0.95 on most cells and below 0.55 in the worst (linear ZINB at $p = 250$, ATE coverage 0.34). This is the worst kind of miscalibration for a policy report: a 0.95 label on an interval that empirically covers 0.55 implies the analyst is communicating an order-of-magnitude over-confident posterior. CountBCF's wider intervals are not a free-lunch loss in sharpness — they are intervals that actually cover, and in the zero-inflated case they cover at essentially the nominal level.

The CountBCF / Gaussian gap widens with likelihood mis-specification. The relative PEHE improvement of CountBCF over BCF (Gaussian) is monotone in the severity of the count likelihood mis-specification: smallest on linear Poisson (the cell closest to Gaussian) at -23%, larger on the nonlinear and / or NB cells, and largest on the nonlinear ZINB cell at -54%. Comparable ordering holds on RMSE(ATE) and on the gap between empirical and nominal coverage. This is direct evidence that the source of the improvement is the count likelihood — the only structural difference between the two methods — rather than incidental hyperparameter or post-processing choices.

The two methods react quite differently to nuisance covariates. CountBCF's PEHE and coverage are nearly flat across $p \in \{5, 50, 250\}$ at $N = 250$ on both sub-studies, while the Gaussian BCF degrades steadily. We attribute this to the log-scale regularisation: on the log scale, irrelevant covariates have a hard time standing in for residual count heteroscedasticity because the Dirichlet-style splitting rules concentrate on the few covariates that improve the count log-likelihood, and the Metropolis acceptance ratio under the count likelihood penalises spurious splits more aggressively than the same prior under a homoscedastic-Gaussian likelihood of the same nominal mean. The gap is a regularisation gap, not a propensity-adjustment gap: both methods use the propensity-as-covariate fix, and both share the BCF (μ, τ) decomposition. CountBCF additionally benefits, on the ZI cells, from the channel decomposition: the treatment effect on the structural-zero probability has its own forest pair, so heterogeneity that the Gaussian baseline must absorb into a single response-scale CATE is allocated to its proper channel.

A.5 Caveats

- Monte Carlo size. With $N_{\text{sim}} = 100$ per cell, cross-replicate SDs are the right scale for the variance bars in the ablation figures of Appendix A, but cell-level means still carry Monte Carlo noise of order $\widehat{\sigma}/\sqrt{100}$. Tight comparisons (e.g. the 1.21 vs. 1.25 RMSE on the linear / NB cell) are within that noise.

- ATE coverage is replicate-realised. Coverage of the ATE is read off the realised ATE for that replicate, not a single population value. This is the right way to score a posterior interval against the truth that the data actually speak to, but it does mean that ATE coverage moves with the realised variability of τ_0 across the calibration target.

- Large-p PEHE variability. The cross-replicate PEHE SD at $p = 250$ remains large for both methods on both sub-studies; cell-to-cell noise in the p-ablation panels at the high end should be interpreted with a Monte Carlo error of order $\widehat{\sigma}_{\text{PEHE}}/\sqrt{100}$ in mind.

- True propensity used. All runs use $\widehat{\pi} = \pi$, the true propensity. This benchmark isolates the outcome model; real-data performance with an estimated $\widehat{\pi}$ will be worse for both methods to a similar degree, because the propensity-as-covariate adjustment is shared.

- Two-method comparison. CountBART, the S-learner built on the same log-linear backbone, was originally part of the benchmark but is dropped here on the memory grounds noted in Section 6.4; an external comparison against generalised random forests, X-learner, and DR-learner on the same DGPs is left to future work.

B Applied-study details

This appendix reports the complete applied-study analysis underlying Section 8. It includes the data-validation checks, the full parametric anchors, all marginal and subgroup summaries, sensitivity fits, overlap and clustering diagnostics, attributable-case calculations, and the limitations that determine how the low-dose city signal should be interpreted.

B.1 Data validation against the published analysis

Before any modelling, the pipeline verifies that our handling of the public file reproduces the published totals. Table 9 compares our analysis frame with Table 4 of Preston et al. [2007]. Every case count and every head count matches exactly, and person-years agree to a maximum relative difference of 2.6×10^{-4}, which is attributable to rounding in the published table. Because the data-handling step is verified to the case, any disagreement in the results that follow is a modelling difference rather than a data-handling difference.

Table 9: Validation of the analysis frame against Table 4 of Preston et al. [2007]. Case and person counts match exactly; the person-year differences reflect rounding in the published table.

Stratum	Cases	Published	Person-years	Published	People	Published
Not in city	3,994	3,994	680,744	680,744	25,247	25,247
<0.005 Gy	5,603	5,603	918,200	918,199	35,545	35,545
0.005–0.1	4,406	4,406	729,604	729,603	27,789	27,789
0.1–0.2	968	968	145,925	145,925	5,527	5,527
0.2–0.5	1,144	1,144	153,886	153,846	5,935	5,935
0.5–1	688	688	81,251	81,251	3,173	3,173
1–2	460	460	41,413	41,412	1,647	1,647
2–4	185	185	13,712	13,711	564	564
Male	7,969	7,969	1,040,281	1,040,278	42,902	42,902
Female	9,479	9,479	1,724,454	1,724,452	62,525	62,525
Hiroshima	12,931	12,931	1,967,602	1,967,601	73,430	73,430
Nagasaki	4,517	4,517	797,133	797,129	31,997	31,997
Total	17,448	17,448	2,764,735	2,764,730	105,427	105,427

The composition of the all-solid endpoint also required verification, because it determines which site-specific mechanisms can drive the aggregate result. The all-solid total decomposes cell by cell, with zero discrepancies, as the sum of oral cavity, digestive, respiratory, skin/bone/connective tissue, breast, female genital, male genital, urinary, renal pelvis and ureter, central nervous system, thyroid, other solid, and thymus. Two codebook features make this non-obvious: the urinary group (661 known-dose cases) omits renal pelvis and ureter (80 cases) even though the documentation places the latter under the urinary system, and thymus (12 cases) sits outside the respiratory group. Restoring both recovers 17,448 exactly. Group by group against Table 2 of Preston et al. [2007], oral cavity (277), digestive (10,052), respiratory (2,001), skin, bone and connective tissue (400), breast (1,082), female genital (1,457), male genital (420), central nervous system (281), thyroid (471), stomach (4,730), lung (1,759), colon (1,516) and liver (1,494) all match the published counts to the case. Within the skin, bone and connective tissue group, non-melanoma skin cancer contributes 330 cases and melanoma 17, again exactly reproducing the published figures. Non-melanoma skin cancer is therefore inside the all-solid aggregate, consistent with the published description

of 17,448 first primary cancers including non-melanoma skin cancer. This matters because non-melanoma skin is one of the few sites whose ERR varies significantly with age at exposure in the published heterogeneity tests, making it a candidate driver of the aggregate's age-at-exposure shape. Section B.9 tests that candidate directly.

B.2 Data, treatment design, and estimands

The source is the public 1958–1998 LSS solid-cancer incidence person-year table. Each row is a stratum defined by city, sex, age at exposure, attained age, and follow-up period; it records cancer cases and person-years at risk. The analytic endpoint is all first primary solid cancers. Its 17,448 cases among 105,427 cohort members reproduce the published totals in Preston et al. [2007], as documented in Section B.1.

Unknown-dose records are excluded: 1,236 cells carrying 174,631 person-years and 1,197 cases. The primary analysis is restricted to in-city survivors and uses the standard DS02 weighted colon dose, gamma dose plus ten times neutron dose, expressed in Gy. We contrast a common reference group below 5 mGy against three exposure bands: low (5–$<$ 100 mGy), medium (100–$<$ 500 mGy), and high (500–$<$ 2000 mGy). Survivors at or above 2 Gy are excluded from the primary contrasts and restored in a sensitivity fit. Each band is fitted as a binary CountBCF analysis against the same reference group. This categorical construction makes the causal contrast explicit and does not impose a dose-response curve. One point of nomenclature deserves mention: the dose columns are weighted, adjusted, truncated DS02 organ doses in Gy despite the codebook labelling them mGy, whereas the dose-category cutpoints are in mGy.

Survivors not in either city are excluded from the primary contrast because their probability of receiving an in-city bomb dose is zero, so including them in the control arm of a treated-versus-control contrast would violate the overlap condition that CountBCF relies on. This is not an objection to their use in a regression that merely borrows their follow-up to pin down the baseline shape, which is the conventional LSS treatment. We therefore include them in a sensitivity analysis with a not-in-city indicator in the prognostic forest, the CountBCF analogue of the standard baseline adjustment, and compare the two fits in Section B.9. The published analysis provides the benchmark for this check, reporting that dose parameters are virtually unchanged when the not-in-city group is excluded and that including it without baseline adjustment inflates the standardised ERR by about 8%.

The prognostic forest receives city, sex, age at exposure, attained age, and the estimated propensity score; the moderating forest receives the same substantive covariates. Calendar year, birth cohort, and time since exposure are deterministic functions of the two age variables and are not entered separately. The relation is year $=$ age $-$ age at exposure $+ t_0$, where t_0 is the date of the detonation in decimal years. Across all 26,806 cells the implied t_0 lies between 1945.594 and 1945.611, with a median of 1945.597 in Hiroshima and 1945.605 in Nagasaki, which are 6 and 9 August 1945. The relation is thus exact to within about six days and is a property of the calendar rather than of the coding. City is placed in the moderator set deliberately. The canonical risk models put city in the baseline only and never allow it to modify the ERR, so a credible city effect on the conditional effect would be a modifier that the canonical model cannot express. Ground distance from the hypocentre is omitted because it is part of the dose-assignment mechanism rather than a pre-exposure confounder; a preliminary analysis confirmed that conditioning on it collapses overlap and reverses the sign of the estimate. This exclusion is correct as a matter of identification, but it has a consequence that Sections B.10, B.11 and B.15 return to: within a city the dose band is very nearly a distance ring, so anything distance-correlated and city-specific is unadjustable by construction in the primary specification. Section B.10 therefore adds a separate family of fits in which

the coarse $< 3/\geq 3$ km ring, which is a baseline-composition variable rather than the continuous assignment variable, does enter both the score and the prognostic forest. The propensity-score adjustment follows the regularization-induced-confounding strategy of Hahn et al. [2020], Rosenbaum and Rubin [1983].

With person-years as an offset, $\exp(\mu_{f,i})$ is the reference-arm rate in cell i and $\tau_{f,i}$ is the log rate ratio. We report both an absolute and a relative conditional effect:

$$\mathrm{EAR}_i = \{\exp(\mu_{f,i} + \tau_{f,i}) - \exp(\mu_{f,i})\} \times 10^4, \qquad \mathrm{ERR}_i = \exp(\tau_{f,i}) - 1. \tag{43}$$

The first is an excess absolute rate in cases per 10^4 person-years; the second is the ERR used in LSS risk modelling. The standardised ERR is the ratio of person-year-weighted counterfactual rates, $(\sum_i w_i r_{1,i}) / (\sum_i w_i r_{0,i}) - 1$, rather than an average of cell-specific rate ratios. The distinction is quantitatively important, not pedantic: cells with a high ERR tend to be cells with a low baseline rate, so the average of ratios runs well above the ratio of averages. For high-dose all-solid cancer, averaging per-cell rate ratios would report a rate ratio of 1.84 against the standardised 1.49, creating a spurious disagreement with the parametric benchmark. Parametric g-computation returns the standardised quantity, so that is what we compare with the GLM rate ratio and with published case-weighted summaries, while the per-cell average is the conditional-effect average used in the subgroup separation tests [Hernán and Robins, 2006, Robins, 1986]. All summaries are computed on the posterior draws, so a subgroup contrast such as the difference between a conditional effect and the average effect carries a genuine credible interval rather than being a difference of two point estimates.

The causal interpretation is conditional on consistency, no interference, positivity, and conditional exchangeability given the available covariates. Dose resulted from location and shielding at the time of the detonations, not a behavioural treatment choice, but the study remains observational. In particular, the public table contains no individual smoking, dietary, or infection information. The estimates should therefore be read as causal under these assumptions and as adjusted associations otherwise.

B.3 Descriptive structure of the analysis table

Table 10: Descriptive summary of the all-solid-cancer analysis table. The reference and three treatment bands are in-city groups. Rates are crude and reported before covariate adjustment. The final column is the marginal variance-to-mean ratio of the cell counts.

Dose band	Cells	Subjects	Person-years	Cases	Rate/10^4 PY	Mean dose (Gy)	Zero cells (%)	Var/mean
Not in city	1,285	25,247	680,744	3,994	58.7	0	44.1	8.10
Reference, 0–< 5 mGy	2,568	35,545	918,200	5,603	61.0	0.00096	47.2	5.37
Low, 5–< 100 mGy	5,973	27,789	729,604	4,406	60.4	0.0308	68.6	3.42
Medium, 100–< 500 mGy	7,598	11,462	299,812	2,112	70.4	0.2354	80.8	1.65
High, 500–< 2000 mGy	5,681	4,820	122,664	1,148	93.6	0.9216	83.9	1.34

Table 10 illustrates why adjustment is necessary. The low-dose crude rate of 60.4 per 10^4 person-years is slightly below the reference rate of 61.0, although the adjusted dose response rises across the medium and high bands. This is confounding rather than biology. Age at exposure, attained age, and sex are almost perfectly balanced across bands, with standardised differences between 0.002 and 0.047, whereas city is not: Hiroshima accounts for 58% of the reference group and 78% of the low-dose group, giving standardised differences between 0.19 and 0.56. City is

therefore the one covariate on which the propensity model has real work to do, and it is why the adjusted analyses move the low-band estimate.

The cells are sparse but not, on inspection, genuinely overdispersed. Roughly 74% of the analysis cells are zero and the marginal variance-to-mean ratio runs from 1.3 to 5.4. Once a flexible baseline is conditioned on, consisting of sex-specific splines in log attained age and in age at exposure plus a city term, the extra-Poisson variation essentially disappears: the fitted negative-binomial size is $\theta \approx 100$, implying a variance-to-mean ratio of 1.006 at the mean fitted rate; the negative-binomial AIC is no better than the Poisson AIC; and the likelihood-ratio test for the negative binomial over the Poisson gives $p = 0.5$. The quasi-Poisson scale, at a Pearson χ^2/df of 1.69, is inflated mostly by the sparse-cell behaviour of the Pearson statistic, since the deviance per degree of freedom is 0.59. The apparent overdispersion in this table is thus baseline heterogeneity rather than extra-Poisson noise, and a model flexible enough in the baseline absorbs it. We nevertheless run CountBCF with the negative-binomial likelihood because it nests the Poisson and lets the data decide; its posterior for the dispersion, reported in Section B.5, reaches the same conclusion from the same fit that produces the causal estimates. The many zero cells are sampling zeros in thin person-year strata rather than evidence of a not-at-risk subpopulation.

The crude data already display the canonical gradient, which is the pattern against which any model output must be read. Comparing the high band to the reference within age-at-exposure strata, the crude rate ratio runs 3.41, 1.90, 1.51, 1.35, and 1.41 across age-at-exposure groups 0–9, 10–19, 20–29, 30–39, and 40 or more for women, and 1.71, 1.72, 1.41, 1.12, and 1.20 for men.

B.4 Parametric anchors

Three parametric anchors are fitted on the identical in-city analytic set: a refit of the canonical excess-relative-risk and excess-absolute-risk models, a fully categorical dose response, and adjusted generalised linear models for the marginal contrasts.

B.4.1 Refitting the canonical risk model

The standard model is refitted by maximum Poisson likelihood on the same in-city analysis set, using a sex-specific spline baseline in log attained age and in age at exposure plus a city term, with

$$\mathrm{ERR}(d, e, a, s) = \beta_s \, d \, \exp\{\alpha_e(e - 30) + \alpha_a \log(a/70)\}\,, \tag{44}$$

and the additive analogue for the EAR, where d is weighted colon dose, e is age at exposure, a is attained age, and s is sex. Because this refit is the comparator against which the CountBCF surface is measured in Section B.7, it matters that it reproduces the published fit. Table 11 shows that every one of the ten published parameters is recovered inside its published 90% confidence interval. The parametric yardstick is therefore the established risk model itself, rather than a simplified regression surrogate.

Fitting one parameter per dose category on top of the same baseline gives 18 nonparametric dose points. A weighted linear fit through the origin gives an ERR of 0.61 per Gy with a standard error of 0.04 in the absence of effect modification. This is larger than the 0.47 standardised to attained age 70 and exposure at age 30, as expected, since the unstandardised average is taken over a cohort whose younger-exposed members carry higher relative risk. The category estimates rise steadily and give no visible sign of a threshold (Figure 8).

Table 11: Refit of the canonical ERR and EAR risk models on the in-city analytic set, compared with Table 10 of Preston et al. [2007]. Intervals are 90% confidence intervals. Every refitted parameter falls inside the published interval.

Parameter	Refit (90% CI)	Published (90% CI)
ERR per Gy, male, attained age 70, exposed at 30	0.336 (0.245, 0.426)	0.35 (0.28, 0.43)
ERR per Gy, female, attained age 70, exposed at 30	0.554 (0.438, 0.670)	0.58 (0.43, 0.69)
ERR female-to-male ratio	1.65	1.6 (1.31, 2.09)
ERR change per decade of age at exposure	−15.2% (−25.6%, −3.1%)	−17% (−25%, −7%)
ERR attained-age power	−1.87 (−2.45, −1.28)	−1.65 (−2.1, −1.2)
EAR per 10^4 PY-Gy, male	38.9 (25.8, 52.1)	43 (33, 55)
EAR per 10^4 PY-Gy, female	57.8 (47.6, 68.1)	60 (51, 69)
EAR female-to-male ratio	1.49	1.4 (1.10, 1.79)
EAR change per decade of age at exposure	−23.5%	−24% (−32%, −16%)
EAR attained-age power	+2.25 (1.73, 2.77)	2.38 (1.9, 2.8)

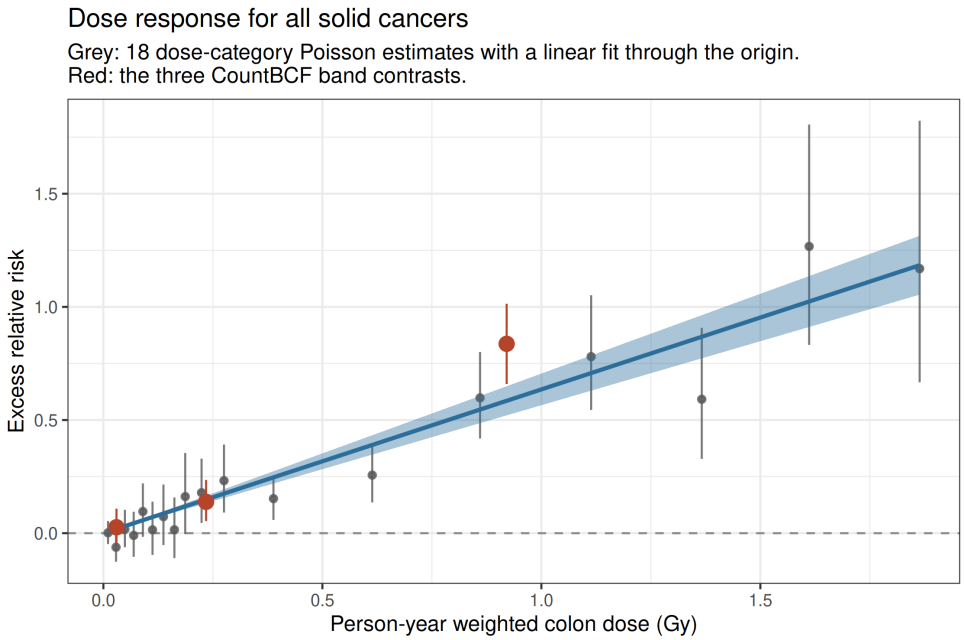


Figure 8: Categorical-dose response from the adjusted parametric model. The estimates increase with dose and do not suggest a threshold over the range considered.

B.4.2 Generalised linear model marginal effects

Table 12 reports g-computation marginal effects from adjusted Poisson and negative-binomial models. The Poisson and negative-binomial fits are nearly indistinguishable for all-solid cancer, consistent with the dispersion diagnostics of Section B.3. Adding dose by modifier interactions improves the all-solid negative-binomial fit by $\Delta\text{AIC} = 54$ (26,931 against 26,985), so heterogeneity is detectable parametrically once the analyst knows to look for it.

Two features of the parametric family are worth carrying forward. The first is the improvement just noted, which establishes that the heterogeneity CountBCF reports in Section B.6 is present in the data and not a property of the forest. The second is that the unadjusted models are unusable: a minimal negative-binomial model for high-dose stomach returns a rate ratio of 63 with a 95% interval of (0.08, 521), and a minimal negative-binomial model for low-dose all-solid cancer flips to a rate ratio of 0.90 against the Poisson's 0.99. The parametric answer therefore depends entirely on which likelihood and which covariate set the analyst happens to choose, which is precisely the fragility that motivates the present method.

Table 12: Marginal effects from adjusted parametric models by g-computation. Excess rates are cases per 10^4 person-years. Intervals are 95% confidence intervals.

Endpoint	Contrast	Model	Excess rate (95% CI)	Rate ratio (95% CI)
All solid	High	Poisson (adjusted)	32.6 (26.8, 38.3)	1.532 (1.435, 1.632)
	High	NB (adjusted)	32.7 (27.1, 38.5)	1.535 (1.440, 1.635)
	Medium	NB (adjusted)	7.35 (3.96, 10.70)	1.120 (1.064, 1.178)
	Low	NB (adjusted)	−0.11 (−2.60, 2.40)	0.998 (0.958, 1.040)
Stomach	High	NB (adjusted)	5.30 (2.57, 8.14)	1.319 (1.152, 1.495)
	Medium	NB (adjusted)	1.03 (−0.67, 2.80)	1.063 (0.961, 1.173)
	Low	NB (adjusted)	0.30 (−1.05, 1.62)	1.019 (0.940, 1.100)

B.5 Average effects and agreement with parametric risk models

Each contrast is fitted with two chains, 2,000 burn-in and 1,000 retained draws per chain. The Gelman–Rubin statistic is 1.00 for every contrast, with a maximum of 1.02, and effective sample sizes run from 276 to 1,617 for the exposed-arm effect and from 708 to 1,561 for the ERR. The negative-binomial dispersion posterior is $\theta \approx 157$–190 for all-solid cancer (95% credible interval 56 to 585) and $\theta \approx 21$–30 for stomach (95% credible interval 8.5 to 77). The all-solid cell counts are thus effectively Poisson once the prognostic forest has done its work, while stomach retains mild genuine overdispersion. This is a sharper answer to the overdispersion question than the generalised-linear-model diagnostics provide, and it is obtained from the same fit that produces the causal estimates.

CountBCF and the adjusted negative-binomial generalised linear model agree in every marginal contrast (Table 13 and Figure 9). For high-dose all-solid cancer, CountBCF estimates an excess of 30.2 cases per 10^4 person-years and a rate ratio of 1.488, compared with 32.7 and 1.535 from the parametric model. The medium-dose contrast is positive under both methods, whereas the low-dose band is uninformative as an overall contrast. CountBCF lies slightly nearer the null throughout, consistent with regularisation of the treatment-effect surface. The exposed-arm version of the estimand is within 0.5% of the population-average version throughout, at 30.37 against 30.22 per 10^4 person-years for high-dose all-solid cancer, and it is the exposed-arm version that Sections B.7 and B.14 use, since a per-Gy risk and an attributable fraction are properties of the exposed.

Table 13: Marginal effects from CountBCF and the adjusted negative-binomial GLM. Excess rates are cases per 10^4 person-years and ERR equals the rate ratio minus one. Intervals are 95% credible intervals for CountBCF and 95% confidence intervals for the GLM.

Endpoint	Contrast	Method	Excess rate (95% interval)	ERR (95% interval)	Rate ratio (95% interval)
All solid	High	CountBCF	30.2 (23.9, 35.9)	0.488 (0.381, 0.584)	1.488 (1.381, 1.584)
	High	NB GLM	32.7 (27.1, 38.5)	0.535 (0.440, 0.635)	1.535 (1.440, 1.635)
	Medium	CountBCF	6.95 (3.22, 10.67)	0.112 (0.051, 0.173)	1.112 (1.051, 1.173)
	Medium	NB GLM	7.35 (3.96, 10.70)	0.120 (0.064, 0.178)	1.120 (1.064, 1.178)
	Low	CountBCF	−0.32 (−2.90, 2.26)	−0.005 (−0.046, 0.037)	0.995 (0.954, 1.037)
	Low	NB GLM	−0.11 (−2.60, 2.40)	−0.002 (−0.042, 0.040)	0.998 (0.958, 1.040)
Stomach	High	CountBCF	4.28 (1.64, 7.11)	0.252 (0.095, 0.426)	1.252 (1.095, 1.426)
	Medium	CountBCF	0.22 (−1.58, 2.09)	0.014 (−0.087, 0.125)	1.014 (0.913, 1.125)
	Low	CountBCF	−0.59 (−2.03, 0.87)	−0.033 (−0.111, 0.051)	0.967 (0.889, 1.051)

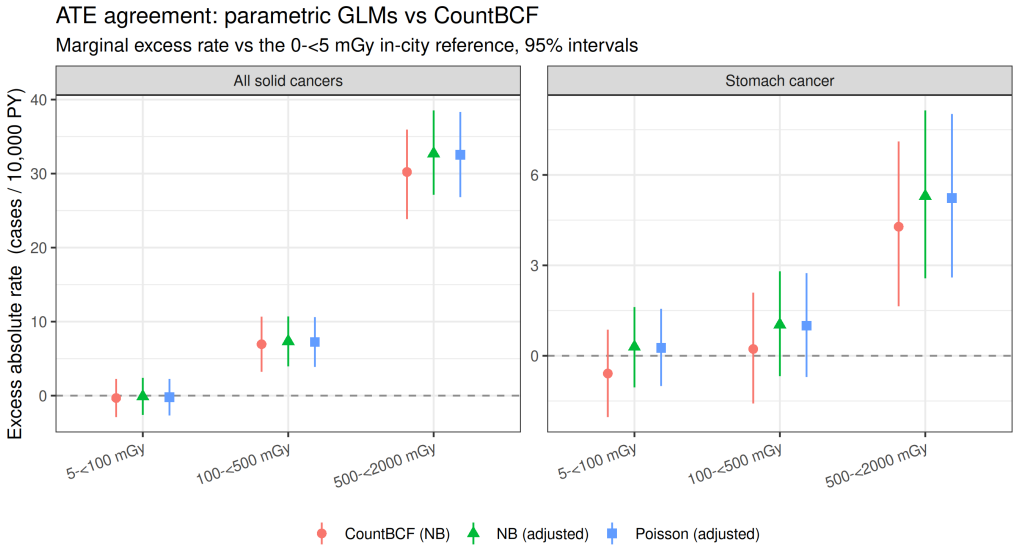


Figure 9: Convergence of marginal effects from CountBCF and the adjusted parametric model. Each estimate is a person-year-standardised excess rate with a 95% interval.

The high-dose band has a mean dose contrast of 0.9207 Gy relative to the reference group. The per-Gy figure is computed from the exposed-arm standardised ERR, 0.487 (0.386, 0.580), because a risk per Gy is a property of the exposed rather than of the whole cohort; dividing by 0.9207 gives 0.529 per Gy (95% credible interval 0.419, 0.630), reported as 0.53 (0.42, 0.63) per Gy. The population-average standardised ERR of 0.488 (0.381, 0.584) in Table 13 is within half a percent of it on the point estimate and gives 0.53 (0.41, 0.63) on the same rescaling, so nothing turns on the choice; we quote the exposed-arm version throughout for consistency with Sections B.7 and B.14. This is a descriptive rescaling of the categorical high-versus-reference contrast, not a separately estimated continuous-dose slope. On this common per-Gy scale it sits among the published values. Preston et al. [2007] report a standardised ERR of 0.47 for the same 1958–1998 follow-up, but with a 90% confidence interval of (0.40, 0.54), so that interval is not directly comparable with ours. The interval-matched comparisons come from the extended 1958–2009 follow-up: Grant et al. [2017] give 0.50 per Gy with a 95% confidence interval of (0.42, 0.59), and French et al. [2017], under internal standardisation, give 0.510 (0.414, 0.612). Our 0.53 (0.42, 0.63) is closer to both of those than the raw comparison with the 90% interval suggests. The medium band gives 0.39 (0.15, 0.65) per Gy on the same rescaling, and for stomach the high band gives 0.26 (0.10, 0.44) per Gy against the published 0.34. The dose response implied by the three band contrasts is consistent with linearity over 0 to 2 Gy: the standardised ERRs of −0.019, 0.092, and 0.487 sit at mean dose contrasts of 0.030, 0.234, and 0.921 Gy, implying per-Gy slopes of roughly −0.63, 0.39, and 0.53, with the low band uninformative because its dose contrast is only 30 mGy (Figure 8). A model given no linearity constraint thus recovers the principal average pattern in the canonical analysis.

Table 14 gives the corresponding E-value sensitivity analysis [VanderWeele and Ding, 2017]. For the high-dose all-solid-cancer contrast, an unmeasured confounder would need associations of at least 2.34 with both dose-band membership and cancer incidence, on the rate-ratio scale, to explain the point estimate away; associations of at least 2.12 with both would be needed to move the interval limit nearest the null to the null. These values are not a proof that unmeasured confounding is absent, but they indicate that a confounder would need to be materially stronger than one weakly related to location-assigned dose. Given that the arms are balanced on every measured demographic except city, and that exposure was determined by where a person happened to be standing on the morning of the bombing, confounders of that strength are hard to name. The medium-dose all-solid result is less robust (E-values 1.41 and 1.23), and the high-dose stomach result is intermediate (1.81 and 1.42). For the low-dose contrasts, the interval E-value is one because the corresponding interval already contains the null. The E-values therefore reinforce the evidence for the high-dose all-solid association while leaving low-dose and site-specific inferences appropriately more tentative.

Table 14: E-value sensitivity analysis for CountBCF. The rate ratios are exposed-arm, person-year-standardised estimates (ATT), which differ slightly from the population-average rate ratios in Table 13. The point and interval E-values give the minimum strength of association, on the rate-ratio scale, that an unmeasured confounder would need with both treatment and outcome to explain the point estimate or move the interval nearest the null to the null.

Endpoint	Contrast	Exposed-arm RR (95% CrI)	E-value, point	E-value, interval
All solid	Low	0.981 (0.940, 1.023)	1.16	1.00
	Medium	1.092 (1.035, 1.152)	1.41	1.23
	High	1.487 (1.386, 1.580)	2.34	2.12
Stomach	Low	0.956 (0.877, 1.040)	1.27	1.00
	Medium	1.002 (0.900, 1.112)	1.04	1.00
	High	1.252 (1.095, 1.426)	1.81	1.42

B.6 Heterogeneity of the radiation-effect surface

The main advantage of CountBCF in this application is not a different average effect, but its ability to estimate effect modification without pre-specifying the interaction functions. Throughout this section we call a subgroup difference credible only when the posterior interval for its standardised ERR minus the cohort-wide standardised ATE excludes zero. The footing matters and is fixed here once for the whole document. The standardised ERR is the ratio of person-year-weighted counterfactual rates defined in Section B.2; it is the quantity a parametric g-computation returns and the one comparable with a Poisson rate ratio and with published case-weighted summaries, and it is therefore the footing on which every subgroup claim in this study is made. The person-year-weighted mean of the per-cell ERR, an average of ratios, runs systematically larger and is reported beside it so that the two can be cross-checked, but it never determines a claim. The EAR is reported to describe the absolute burden and does not determine the separation indicator either. This guards against treating a difference in posterior means as evidence of genuine heterogeneity, and against the two footings being read as though they answered the same question.

A second guard is arithmetic. The separation test is run 294 times across the six endpoint-by-band fits, on nine subgroup partitions per fit. On the standardised footing 34 of those 294 tests flag, and on the per-cell footing 38 do (Table 15). Under independence one would expect about 15 flags at the nominal 5% level, so the flag count is roughly twice chance overall; but the flags are heavily concentrated, with 27 of the 34 falling in the single high-dose all-solid fit and none at all in the three stomach fits. No subgroup claim below rests on the flag alone. A claim is featured only if it survives both the regularisation sweep of Section B.8 and the data perturbations of Section B.9; one that survives only one of the two is reported with that qualification attached, as the high-dose sex effect is, and one supported instead by an independent parametric test is identified as such.

Table 15: Multiplicity accounting for the subgroup separation test. Every test run anywhere in the study is counted. A test flags when the 95% credible interval for the subgroup effect minus the cohort average effect excludes zero.

Fit	Separation tests	Flags, standardised	Flags, per-cell
All solid, 500–< 2000 mGy	49	27	26
All solid, 100–< 500 mGy	49	2	5
All solid, 5–< 100 mGy	49	5	7
Stomach, 500–< 2000 mGy	49	0	0
Stomach, 100–< 500 mGy	49	0	0
Stomach, 5–< 100 mGy	49	0	0
Total	294	34	38

B.6.1 How much heterogeneity is present at all

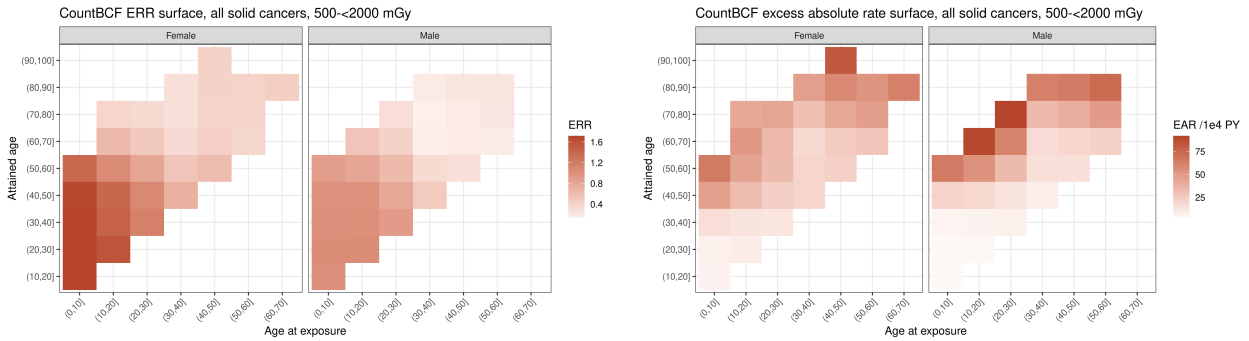
Before examining named subgroups it is useful to ask how much variation the posterior surface contains. Table 16 reports the posterior person-year-weighted standard deviation of the per-cell ERR across exposed cells, together with the share of exposed cells whose effect is positive and the share whose 95% interval excludes zero.

For all-solid cancer at high dose the heterogeneity is large and credible: the per-cell ERR has a posterior median of 0.60 with a tenth-to-ninetieth percentile range of 0.16 to 1.57, and nearly two thirds of individual cells have an interval excluding zero. At medium and low dose the surface is much flatter and almost no individual cell separates, which is a statement about power rather

Table 16: Cell-level heterogeneity of the CountBCF effect surface. The first column is the posterior person-year-weighted standard deviation of the per-cell ERR across exposed cells.

Endpoint	Contrast	SD of per-cell ERR (95% CrI)	Cells with ERR > 0	Cells with CrI excluding 0
All solid	High	0.584 (0.416, 0.773)	98.6%	63.0%
	Medium	0.253 (0.198, 0.324)	82.8%	2.0%
	Low	0.233 (0.183, 0.299)	58.1%	2.4%
Stomach	High	0.004 (0.000, 0.030)	100%	100%
	Medium	0.146 (0.076, 0.240)	73.5%	0%
	Low	0.186 (0.135, 0.257)	54.3%	0%

than about the absence of modification. For stomach at high dose the moderating forest returns an essentially constant effect, an ERR of 0.252 in every subgroup with a posterior standard deviation of 0.004; that case is discussed in Section B.6.4. Figures 10a and 10b display the fitted relative and absolute surfaces over the covariate space.



(a) Relative effect (ERR).

(b) Absolute effect (EAR).

Figure 10: Estimated CountBCF effect surfaces for high-dose all-solid cancer over the observed covariate space. The relative and absolute surfaces move in opposite directions with attained age.

B.6.2 All solid cancers at 500–< 2000 mGy

Table 17 gives the complete set of high-dose subgroup effects with their separation tests. Five conclusions follow.

First, age at exposure is the dominant modifier of the relative effect. The standardised ERR declines more than fourfold, from 1.26 (0.85, 1.71) among those exposed at ages 0–9 to 0.29 (0.14, 0.44) among those exposed at age 40 or later, and both tails separate credibly from the average. This is the textbook LSS result, recovered with no interaction term specified anywhere.

Second, attained age modifies the relative and absolute effects in opposite directions. The standardised ERR falls from 1.40 (0.97, 1.84) below age 40 to 0.30 (0.11, 0.51) at age 80 or older, while the EAR rises from 8.7 to 57.3 excess cases per 10^4 person-years, and every attained-age band separates. The two are projections of a single surface, since the EAR is approximately the baseline rate multiplied by the ERR and the baseline rate climbs faster with attained age than the relative effect fades.

Third, time since exposure moves the absolute effect but not the relative one. The ERR is statistically flat across 20 to more than 40 years of follow-up, with no subgroup separating, while

the EAR nearly quadruples from 15.2 to 57.3 per 10^4 person-years. The relative excess established early does not wash out; the absolute burden grows because the baseline does.

Fourth, sex separates at high dose, with a standardised ERR of 0.58 (0.45, 0.72) in women against 0.38 (0.25, 0.51) in men, a female-to-male ERR ratio of 1.54. That is close to the published 1.6, and the agreement is closer still on the exposed-cell footing used in Section B.7, where the ratio is $0.603/0.369 = 1.63$. This dimension is the clearest illustration in the study of why the footing has to be fixed in advance: the same contrast computed as an average of per-cell ratios gives 0.94 against 0.74, a ratio of

1.27, and does not separate, which would have understated our own agreement with the canonical model. Two qualifications keep the sex effect out of the featured results even so. Section B.8 shows that its magnitude is strongly sensitive to treatment-effect regularisation, collapsing ninefold under a halved leaf concentration even though its interval never covers zero, and Section B.9 shows that it is carried by breast and thyroid, its interval covering zero once those two sites are removed. We also note that it reverses a finding from a preliminary analysis on a 30% subsample, which had the male ERR marginally above the female one.

Fifth, the two cities are indistinguishable at high dose, at 0.49 (0.37, 0.60) in Hiroshima against 0.48 (0.31, 0.67) in Nagasaki, which is what one would want if the DS02 weighted organ doses have reconciled the Hiroshima neutron component correctly. It is also the fact that constrains the reading of Section B.10: any city difference confined to the lowest band has to be reconciled with exact agreement where the dose signal is strongest.

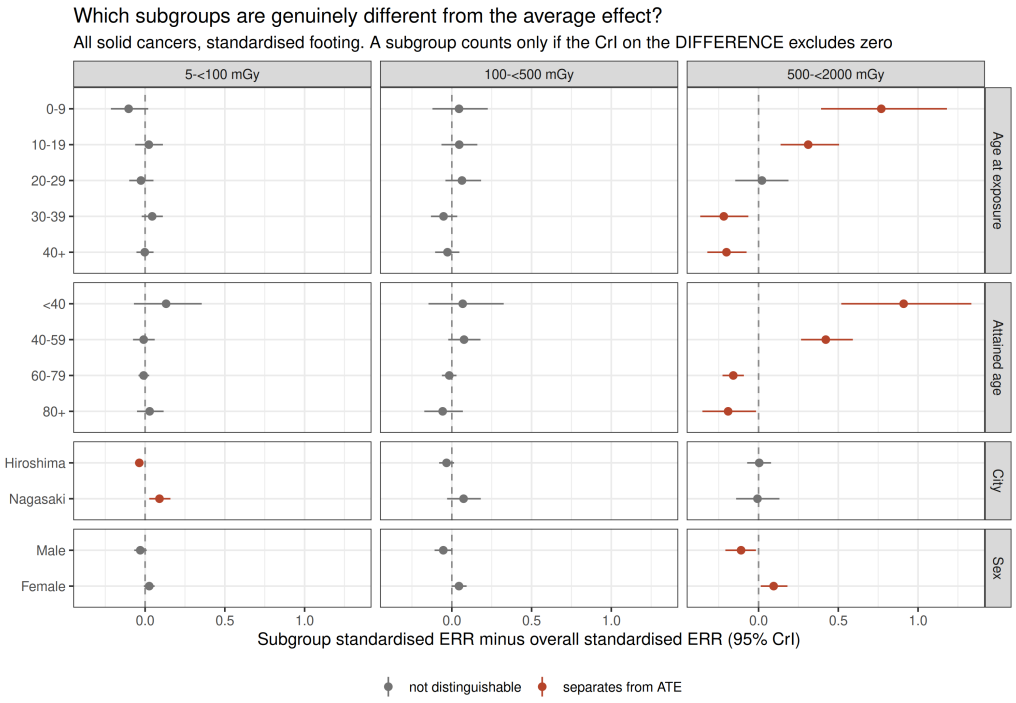


Figure 11: Subgroup separation for the high-dose all-solid contrast. Each point is the posterior mean of the subgroup conditional effect minus the population average effect (ATE), with its 95% credible interval. Intervals excluding zero identify credible effect modification.

These patterns match the central qualitative findings of LSS incidence modelling: relative risk is greater following younger-age exposure, while the absolute rate burden grows at older attained

Table 17: High-dose all-solid-cancer conditional effects (500–< 2000 mGy against the 0–< 5 mGy reference). The first three numeric columns are on the standardised footing, which is the one that determines every claim: a subgroup is marked as separated when the 95% posterior interval for its standardised ERR minus the cohort-wide standardised ATE of 0.488 excludes zero. The final column is the person-year-weighted mean of the per-cell ERR, an average of ratios, given for cross-reference only; it runs systematically larger and is taken against a different reference, the per-cell ATE of 0.87. The Cases column counts cases in the subgroup across both dose arms.

Dimension	Subgroup	Cases	Std. ERR (95% CrI)	CATE − ATE, std. (95% CrI)	Sep.?	Per-cell ERR
Sex	Male	3,089	0.38 (0.25, 0.51)	−0.11 (−0.21, −0.02)	Yes	0.74
	Female	3,662	0.58 (0.45, 0.72)	+0.09 (0.01, 0.18)	Yes	0.94
Age at exposure	0–9	532	1.26 (0.85, 1.71)	+0.77 (0.39, 1.18)	Yes	1.45
	10–19	1,303	0.80 (0.60, 1.02)	+0.31 (0.14, 0.50)	Yes	1.08
	20–29	1,159	0.51 (0.33, 0.71)	+0.02 (−0.15, 0.19)	No	0.72
	30–39	1,499	0.27 (0.10, 0.45)	−0.22 (−0.37, −0.06)	Yes	0.37
	40+	2,258	0.29 (0.14, 0.44)	−0.20 (−0.32, −0.08)	Yes	0.32
Attained age	< 40	167	1.40 (0.97, 1.84)	+0.91 (0.52, 1.33)	Yes	1.41
	40–59	1,677	0.91 (0.71, 1.12)	+0.42 (0.27, 0.59)	Yes	1.00
	60–79	3,818	0.33 (0.22, 0.44)	−0.16 (−0.23, −0.09)	Yes	0.36
	80+	1,089	0.30 (0.11, 0.50)	−0.19 (−0.35, −0.02)	Yes	0.32
City	Hiroshima	4,369	0.49 (0.37, 0.60)	+0.004 (−0.07, 0.08)	No	0.77
	Nagasaki	2,382	0.48 (0.31, 0.67)	−0.007 (−0.14, 0.13)	No	1.02
Time since exposure	< 20 y	939	0.45 (0.29, 0.62)	−0.04 (−0.16, 0.10)	No	0.93
	20–29 y	1,430	0.44 (0.32, 0.56)	−0.05 (−0.13, 0.03)	No	0.90
	30–39 y	1,670	0.49 (0.38, 0.60)	+0.002 (−0.04, 0.04)	No	0.88
	40+ y	2,712	0.53 (0.39, 0.66)	+0.04 (−0.05, 0.13)	No	0.75

Excess absolute rates per 10^4 person-years for the same subgroups, in the order listed, are 28.9 and 31.0 by sex; 24.6, 34.3, 31.8, 24.4, and 35.5 by age at exposure; 8.7, 34.3, 37.6, and 57.3 by attained age; 33.0 and 26.2 by city; and 15.2, 20.8, 32.1, and 57.3 by time since exposure.

ages [Preston et al., 2007, Grant et al., 2017].

B.6.3 All solid cancers at 100–< 500 mGy

On the standardised footing no subgroup separates at 100–< 500 mGy. The sex contrast comes closest, with a male standardised ERR of 0.058 (−0.021, 0.143) and a difference from the average of −0.054 (−0.108, 0.003), against a female 0.156 (0.079, 0.240) and +0.045 (−0.002, 0.093); both intervals just cover zero. On the per-cell footing the same contrast is 0.061 against 0.222 and does flag, at −0.101 (−0.200, −0.002) and +0.059 (0.001, 0.118), which is a difference of footing rather than of evidence: the medium-dose sex effect is borderline either way and is not featured. No age-at-exposure or attained-age subgroup separates at this dose on either footing.

The pattern across dose bands is nevertheless informative, and it points at something the canonical model already contains. The female-to-male ratio of standardised ERRs is about 2.7 at 100–< 500 mGy and 1.54 at 500–< 2000 mGy, so the sex gap is widest where the male response is weakest in absolute terms. That is what the published curvature of the male dose response implies. Grant et al. [2017] report upward curvature in the male dose response, including an apparent flattening of the response between roughly 0.2 and 0.7 Gy; French et al. [2017] report strong evidence of a curvilinear dose response among males, at $P = 0.008$ under internal standardisation; and Preston et al. [2007] already reported a borderline sex-averaged test for quadratic departure from linearity over 0–2 Gy, at $P = 0.09$, with an estimated curvature of 0.3 (90% CI 0.01, 0.90). A male response that is flat across the medium band and then rises steeply depresses the male ERR at 100–< 500 mGy relative to a linear expectation and restores it at 500–< 2000 mGy, which is exactly the band pattern above. CountBCF was given no dose-shape assumption of any kind, so recovering the signature of a curvilinear male response from the band contrasts alone is a further piece of convergent evidence rather than a discrepancy.

B.6.4 Stomach cancer as a counterpoint

Stomach cancer provides a useful counterpoint. Its high-dose average effect is positive, but its estimated ERR surface is essentially constant across the examined covariates, with a posterior standard deviation of 0.004. This is not a failure of the estimator; it is the differential-shrinkage prior doing what it is designed to do. With 269 exposed stomach cases the data cannot support effect modification, and the published tests agree, giving $P = 0.4$ for the stomach age-at-exposure trend and $P = 0.05$ for the attained-age effect. Where the data do not support a complex surface, the fit shrinks toward a homogeneous effect rather than creating spurious subgroup differences. The all-solid regularisation sweep tests whether this behavior could be excessive, and the stomach-specific repetition of that grid is reported in Table 20. It gives the same null result under the ordinary settings and identifies the extreme shared-prior fit as unstable rather than as evidence of stomach effect modification.

B.7 Comparison with the canonical LSS surface

The average-effect comparison is only a partial validation. We therefore evaluate the CountBCF surface against the refitted risk model of Section B.4.1 at the same person-year cells. Two comparisons are possible. The standardised comparison weights each surface by the same CountBCF-estimated control-arm rate, which makes it the analogue of a rate ratio and the fair test. The per-cell comparison is the plain person-year-weighted mean of the cell ERRs; it is systematically larger for both surfaces and is what a naive comparison would produce. We report the standardised comparison. Unlike the separation tests of Section B.6, which standardise the conditional effect over the whole

cohort, this comparison is evaluated only at the exposed person-year cells, because the parametric ERR is proportional to dose and collapses to zero at the reference cells whereas the CountBCF contrast does not; the two surfaces can therefore only be compared where the exposure actually occurred. A given subgroup's CountBCF ERR here can accordingly differ by a few percent from its cohort-standardised value in Table 17.

B.7.1 All solid cancers

On the standardised scale the two surfaces agree in all but two of 28 high-dose subgroups, all but three of 28 medium-dose subgroups, and all but six of 28 low-dose subgroups (Table 18 and Figure 12). Their person-year-weighted mean EAR is identical to four significant figures at high dose, 30.37 per 10^4 person-years for each model. Across the 5,681 exposed cells the two surfaces correlate at 0.57 on the Pearson scale and 0.80 on the Spearman scale; the rank correlation is the higher of the two, so the earlier description of 0.57 as a rank correlation understated the agreement. The agreement is notable because CountBCF was supplied neither a linear dose term nor any interaction involving age, sex, or city. A nonparametric causal forest, given no functional form and no interaction terms, reproduces the canonical parametric risk surface for all solid cancers over almost the whole covariate space.

Table 18: Standardised comparison of the CountBCF surface with the refitted canonical risk model, evaluated at the same person-year cells. Every subgroup whose 95% credible interval for the difference excludes zero is listed, for all three dose bands, together with the overall row and the two sexes at high dose for context. Rows marked † are the departures.

Contrast	Subgroup	Refitted model ERR	CountBCF ERR (95% CrI)	Difference (95% CrI)
High	Overall	0.553	0.487 (0.386, 0.580)	−0.066 (−0.168, 0.026)
	Attained age < 40 †	2.759	1.330 (0.931, 1.758)	−1.429 (−1.863, −0.965)
	Time since exposure < 20 y †	0.659	0.471 (0.301, 0.640)	−0.188 (−0.356, −0.015)
	Age at exposure 0–9	1.522	1.184 (0.794, 1.607)	−0.338 (−0.735, 0.090)
	Female	0.707	0.603 (0.465, 0.741)	−0.104 (−0.243, 0.035)
	Male	0.396	0.369 (0.240, 0.500)	−0.027 (−0.157, 0.104)
Medium	Overall	0.138	0.092 (0.035, 0.152)	−0.046 (−0.103, 0.013)
	Age at exposure 0–9 †	0.368	0.114 (−0.053, 0.294)	−0.253 (−0.421, −0.073)
	Attained age < 40 †	0.736	0.142 (−0.072, 0.389)	−0.593 (−0.813, −0.340)
	Female exposed at 0–9 †	0.508	0.193 (−0.016, 0.440)	−0.315 (−0.531, −0.070)
Low	Overall	0.019	−0.019 (−0.060, 0.023)	−0.038 (−0.078, 0.005)
	Hiroshima †	0.019	−0.043 (−0.089, 0.003)	−0.063 (−0.108, −0.016)
	Nagasaki	0.017	0.088 (0.006, 0.171)	+0.072 (−0.011, 0.154)
	Age at exposure 0–9 †	0.044	−0.141 (−0.256, −0.015)	−0.185 (−0.299, −0.060)
	Time since exposure 40+ y †	0.018	−0.059 (−0.115, −0.001)	−0.077 (−0.134, −0.019)
	Male exposed at 0–9 †	0.031	−0.143 (−0.279, 0.010)	−0.174 (−0.309, −0.022)
	Female exposed at 0–9 †	0.059	−0.136 (−0.274, 0.015)	−0.195 (−0.333, −0.045)
	Male exposed at 20–29 †	0.013	−0.110 (−0.222, 0.011)	−0.123 (−0.235, −0.002)

The low-dose block deserves a separate remark, because it is the one that bears on Section B.10. Six of the 28 low-dose subgroups depart from the canonical surface on the standardised scale, and every one of them departs downward: CountBCF puts the low-band ERR below what a linear dose response extrapolated from the whole 0–2 Gy range predicts. Hiroshima is among the six, at −0.063 (−0.108, −0.016). Nagasaki is not, at +0.072 (−0.011, 0.154). On the scale that this study uses for every other claim, therefore, Nagasaki's low-band ERR is not distinguishable from what the canonical model already predicts, and Hiroshima's is. Whatever the low-dose city contrast of Section B.10 is, it is at least as much a statement about Hiroshima sitting below the canonical

prediction as about Nagasaki sitting above it.

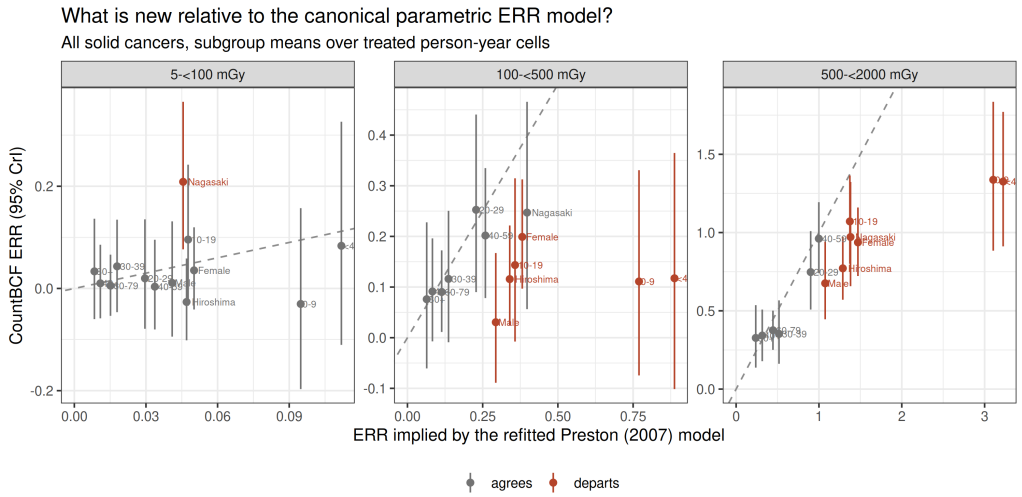


Figure 12: CountBCF conditional ERRs compared with the refitted canonical risk model. The models agree across most of the observed covariate space; the largest differences are confined to young attained age and short latency.

The disagreements are localized rather than scattered. They occur at young attained age and short time since exposure, where the parametric attained-age power law extrapolates most strongly. At attained age below 40 the refitted model predicts a high-dose ERR of 2.76 at 0.92 Gy, while CountBCF estimates 1.33 (0.93, 1.76) and excludes the parametric value. The same pattern appears one dose band down, at 0.74 against 0.14, and for stomach, at 2.89 against 0.25. A power law in attained age is forced to be monotone and unbounded as attained age tends to zero; the data in that corner are sparse, with only 51 exposed cases below attained age 40 in the high band, and CountBCF's answer is a flattening rather than a divergence. The published analysis itself flags this region as unsettled, noting that a more complex pattern provides a somewhat better fit for all solid cancers and that future analyses should give more attention to alternative descriptions of age-at-exposure effects on both the ERR and the EAR [Preston et al., 2007]. The competing explanation has to be stated plainly: because the moderating-forest prior shrinks toward a constant, a flattening in a thin corner is also what excessive shrinkage would produce. Section B.8 is the decisive test of that alternative.

B.7.2 Stomach cancer

For stomach at high dose, 21 of the 28 subgroups depart on the standardised scale, and 23 of 28 on the per-cell scale, because the parametric model carries effect modification, with an ERR proportional to attained age to the power -1.5, a decline of 13% per decade of age at exposure, and a female-to-male ratio of 2.3, while CountBCF returns a flat ERR of 0.252. The two are not equally supported by the data. The published significance tests for stomach give $P = 0.4$ for the age-at-exposure trend and $P = 0.05$ for attained age, so the modification being fitted is at or beyond the edge of detectability and its estimates are borrowed from the functional form rather than read off the stomach data. CountBCF, which must find modification in the stomach data alone, finds none. The right reading is not that either analysis is wrong about stomach, but that the apparent stomach effect modification in the canonical analysis is a property of the model class rather than of the stomach data. What the discrepancy identifies is the assumption doing the work:

CountBCF requires the site-specific data to support a modifier, whereas the parametric estimate extrapolates a specified functional form. The corresponding regularisation question is now checked directly rather than left to the single default fit. The twelve-setting stomach sweep reported in Section B.8, Table 20, finds no subgroup separation on either ERR footing at the default and all but one of the settings; the exceptional setting produces an unstable, highly heterogeneous surface rather than a credible stomach-specific pattern.

B.8 Robustness to regularisation of the moderating forest

Because CountBCF deliberately tightens the prior on the treatment-effect surface in order to suppress spurious heterogeneity, any featured subgroup effect must be shown to survive both loosening and tightening that prior. Table 19 reports twelve refits of the featured contrast, all solid cancers at 500–< 2000 mGy, varying the number of moderating trees, the tree-depth penalty on the moderating forest, and the overall leaf concentration. Figure 13 displays the same sweep.

Table 19: Sensitivity of the featured contrast (all solid cancers, 500–< 2000 mGy) to regularisation of the moderating forest. All entries are on the standardised footing; the last two columns are posterior differences with 95% credible intervals. m_τ and the depth prior act on the moderating forest alone, whereas the leaf concentration a_0 is shared by both forests, so the a_0 rows are not a moderating-forest-only sweep. The twelve rows are ten distinct fits: the default specification is $m_\tau = 50$ with depth prior $(0.25, 3)$ and $a_0 \times 1$, so it is the common row of all three sweeps and is reported three times (‡).

Specification	Standardised ERR (95% CrI)	Std. ERR, exposed < 20 minus 40+	Std. ERR, female minus male
$m_\tau = 10$ (tightest)	0.434 (0.345, 0.521)	0.421 (0.208, 0.612)	0.156 (0.003, 0.307)
$m_\tau = 25$	0.486 (0.389, 0.582)	0.529 (0.298, 0.787)	0.225 (0.051, 0.407)
$m_\tau = 50$ (default) ‡	0.495 (0.401, 0.594)	0.527 (0.281, 0.778)	0.256 (0.074, 0.431)
$m_\tau = 100$	0.508 (0.416, 0.608)	0.597 (0.344, 0.864)	0.223 (0.038, 0.414)
$m_\tau = 200$ (loosest)	0.505 (0.405, 0.605)	0.604 (0.356, 0.859)	0.269 (0.084, 0.456)
Depth prior (0.05, 4), tight	0.494 (0.400, 0.599)	0.563 (0.313, 0.813)	0.208 (0.029, 0.380)
Depth prior (0.25, 3), default ‡	0.495 (0.401, 0.594)	0.527 (0.281, 0.778)	0.256 (0.074, 0.431)
Depth prior (0.5, 2), loose	0.496 (0.398, 0.598)	0.574 (0.325, 0.824)	0.240 (0.054, 0.438)
Depth prior (0.95, 2), prognostic-like	0.489 (0.397, 0.585)	0.583 (0.338, 0.839)	0.260 (0.086, 0.445)
$a_0 \times 0.5$ (both forests)	0.465 (0.359, 0.576)	0.272 (0.125, 0.422)	0.029 (0.003, 0.057)
$a_0 \times 1$ (default) ‡	0.495 (0.401, 0.594)	0.527 (0.281, 0.778)	0.256 (0.074, 0.431)
$a_0 \times 2$ (both forests)	0.463 (0.367, 0.558)	0.579 (0.322, 0.831)	0.254 (0.061, 0.436)

Three conclusions follow. First, the average effect is essentially prior-free: the standardised ERR sits between 0.434 and 0.508 across a twentyfold change in the prior variance of the moderating surface and across every depth prior tried, so the headline number is not an artefact of the shrinkage. Second, the age-at-exposure gradient survives everything. The standardised ERR difference between those exposed before age 20 and after age 40 has a credible interval excluding zero in all twelve specifications, including the tightest moderating forest in the sweep, which gives 0.421 (0.208, 0.612), and the halved leaf concentration, which gives 0.272 (0.125, 0.422). Its magnitude scales with prior looseness, from 0.27 to 0.60, a range of roughly 2.2-fold, so the size of the gradient is prior-dependent while its existence is not.

Third, the sex difference behaves differently, and the distinction is worth stating carefully because it is exactly the sort of claim the choice of footing decides. On the standardised footing the female-minus-male difference also has a credible interval excluding zero in all twelve specifications, so its existence is no less prior-robust than the gradient's. What is not robust is its size: it falls from 0.256 (0.074, 0.431) at the defaults to 0.029 (0.003, 0.057) when the shared leaf concentration is halved, a ninefold collapse against the gradient's 2.2-fold range. A quantity whose interval never

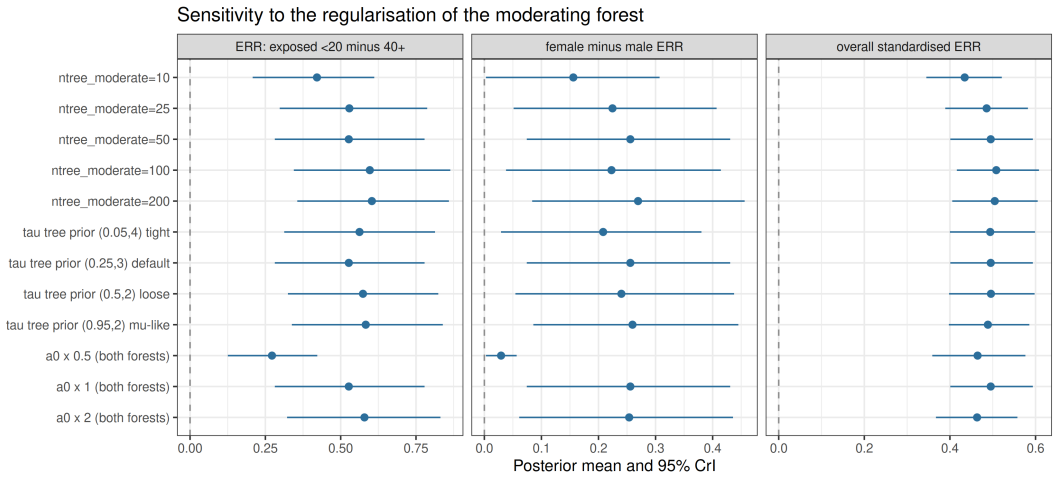


Figure 13: Sensitivity of the featured contrast to the moderating-forest prior. All three panels are on the standardised footing, matching Table 19.

covers zero but whose magnitude moves by an order of magnitude under a prior change is evidence that something is present, not evidence about how large it is, and the second is what a featured subgroup claim would have to assert. The sex effect therefore stays out of the featured results, but for the reason established in Section B.9 rather than this one: it is carried by breast and thyroid, and its interval covers zero once those two sites are removed.

Finally, the Nagasaki-minus-Hiroshima standardised ERR difference at high dose covers zero in eleven of the twelve specifications and never exceeds 0.06 in magnitude in any of them, confirming the null reported in Section B.6.2. The exception is the halved leaf concentration, at +0.029 (0.004, 0.054), and it illustrates the point just made about that setting rather than contradicting the null: halving a_0 shrinks the whole moderating surface so hard that every subgroup contrast collapses to a value near 0.03 with an interval only about 0.05 wide, which is why the sex difference separates there at 0.029 (0.003, 0.057) as well. Separation under that prior reflects how tightly the posterior is determined, not how large the underlying contrast is.

The same twelve-setting grid was then applied to the stomach high-dose contrast to test the mirror-image concern: could a modifier hidden by the default prior appear when the regularisation is changed? Table 20 reports the result. At the default, the posterior person-year-weighted standard deviation of the stomach ERR surface is 0.0042, none of the 49 subgroup tests separates on either the standardised or per-cell footing, and the largest standardised subgroup deviation from the average is 0.0002. The same zero separation result holds in the other eleven settings except for the extreme $a_0 \times 2$ setting, where the surface becomes unstable: its standardised heterogeneity is 1.14 and 9 of 49 standardised tests (15 of 49 per-cell tests) separate. The overall effect also shifts to 0.184 (0.036, 0.356) there. The ordinary regularisation changes therefore do not reveal hidden stomach heterogeneity; the extreme shared prior produces a pathological surface and is not evidence for a stomach modifier.

B.9 Robustness to the data construction

Table 21 reports fourteen refits of the featured contrast, each changing one element of the data construction, with Figure 14 showing the same fits.

Not-in-city survivors change nothing. Adding the 1,285 not-in-city cells to the control arm with

Table 20: Sensitivity of the stomach high-dose contrast to the regularisation grid used for the all-solid analysis. The standardised ERR is the person-year-weighted exposed-arm contrast with its 95% credible interval. The next two columns give the posterior person-year-weighted standard deviation of the per-cell ERR surface and the largest absolute subgroup deviation from the average, both on the standardised footing. The final columns count subgroup tests whose 95% interval excludes zero, out of 49 tests, on the standardised and per-cell footings respectively. The a_0 rows act on both forests; the default is repeated across the three sweeps (‡).

Specification	Standardised ERR (95% CrI)	SD(ERR)	Max $\lvert d \rvert$	Std. flags / 49	Per-cell flags / 49
$m_\tau = 10$ (tightest)	0.226 (0.078, 0.390)	0.0007	0.00004	0	0
$m_\tau = 25$	0.242 (0.084, 0.417)	0.0026	0.00014	0	0
$m_\tau = 50$ (default) ‡	0.253 (0.078, 0.429)	0.0042	0.00019	0	0
$m_\tau = 100$	0.255 (0.082, 0.427)	0.0075	0.00038	0	0
$m_\tau = 200$ (loosest)	0.260 (0.089, 0.452)	0.0124	0.00063	0	0
Depth prior $(0.05, 4)$, tight	0.252 (0.080, 0.422)	0.0008	0.00010	0	0
Depth prior $(0.25, 3)$, default ‡	0.253 (0.078, 0.429)	0.0042	0.00019	0	0
Depth prior $(0.5, 2)$, loose	0.253 (0.088, 0.416)	0.0090	0.00061	0	0
Depth prior $(0.95, 2)$, extreme	0.248 (0.087, 0.419)	0.0414	0.00655	0	0
$a_0 \times 0.5$ (both forests)	0.234 (0.085, 0.397)	0.0754	0.053	0	0
$a_0 \times 1$ (default) ‡	0.253 (0.078, 0.429)	0.0042	0.00019	0	0
$a_0 \times 2$ (both forests)	0.184 (0.036, 0.356)	1.136	1.668	9	15

Table 21: Sensitivity of the featured contrast to the data construction. Each row changes one element relative to the baseline. All ERR entries are on the standardised footing fixed in Section B.6; the last two columns are posterior differences with 95% credible intervals.

Perturbation	Cells	Cases	Standardised ERR (95% CrI)	EAR/10^4 PY	Std. ERR, $<$ 20 minus 40+	Std. ERR, female minus male
Baseline (all solid, colon bands, NB)	8,249	6,751	0.495 (0.401, 0.594)	30.7	0.527 (0.281, 0.778)	0.256 (0.074, 0.431)
Poisson likelihood	8,249	6,751	0.498 (0.402, 0.598)	30.8	0.563 (0.311, 0.819)	0.240 (0.050, 0.413)
Not in city added to control arm	9,534	10,745	0.501 (0.408, 0.603)	30.9	0.552 (0.299, 0.799)	0.233 (0.059, 0.417)
Reference widened to 0–$<$ 100 mGy	14,222	11,157	0.503 (0.406, 0.606)	31.0	0.567 (0.334, 0.807)	0.237 (0.075, 0.411)
High band extended to include $\geq$ 2 Gy	10,714	6,936	0.588 (0.494, 0.691)	35.9	0.713 (0.456, 0.986)	0.289 (0.096, 0.480)
Excluding breast and thyroid	8,249	6,137	0.405 (0.310, 0.510)	23.1	0.287 (0.049, 0.534)	0.066 (−0.112, 0.261)
Excluding non-melanoma skin	8,249	6,605	0.479 (0.384, 0.576)	29.0	0.491 (0.253, 0.752)	0.210 (0.035, 0.398)
Excluding stomach	8,249	4,975	0.575 (0.458, 0.697)	25.9	0.532 (0.249, 0.825)	0.206 (−0.030, 0.412)
Non-sex-specific solid only	8,249	5,502	0.525 (0.416, 0.642)	26.5	0.420 (0.161, 0.689)	0.142 (−0.069, 0.336)
Stomach, colon-dose bands	8,249	1,776	0.253 (0.078, 0.429)	4.41	−0.0001 (−0.003, 0.002)	0.00002 (−0.0004, 0.0005)
Stomach, stomach-dose bands	8,075	1,774	0.256 (0.092, 0.443)	4.45	−0.0001 (−0.003, 0.002)	0.00001 (−0.0004, 0.0005)
Lung, lung-dose bands	7,636	683	0.666 (0.356, 1.001)	4.21	−0.047 (−0.472, 0.272)	0.480 (0.002, 1.086)
Colon, colon-dose bands	8,249	584	0.297 (0.019, 0.586)	1.64	0.052 (−0.211, 0.335)	−0.095 (−0.473, 0.242)
Liver, liver-dose bands	7,917	592	0.291 (0.024, 0.609)	1.66	0.073 (−0.185, 0.396)	−0.121 (−0.455, 0.193)

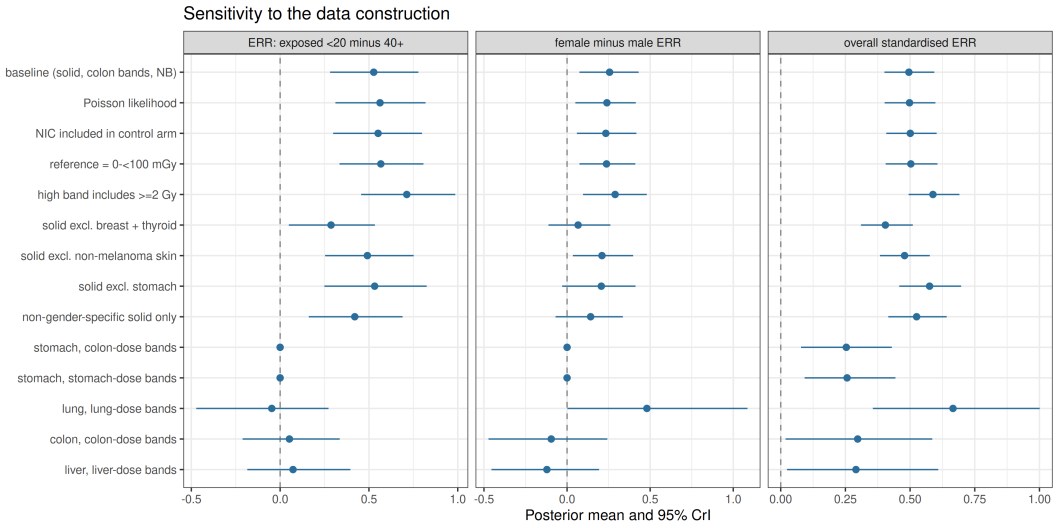


Figure 14: Sensitivity of the featured contrast to the data construction. All three panels are on the standardised footing, matching Table 21. The average effect and the age-at-exposure gradient hold their sign and separation under every perturbation, the largest movements being the removal of breast and thyroid, which lowers both, and the extension of the high band beyond 2 Gy, which raises both. The sex difference behaves differently, losing its separation from zero in three of the four all-solid composition variants.

an indicator in the prognostic forest moves the standardised ERR from 0.495 to 0.501, a change of 1.2% that is well inside the Monte Carlo noise. This reproduces the published conclusion that the dose-response parameter estimates are virtually unchanged when the not-in-city group is excluded, and it answers the design question raised in Section B.2: the indicator works, and the choice does not matter much either way.

The reference-band and dose-cap definitions are second-order. Widening the reference group to 0–< 100 mGy gives 0.503 against 0.495, and restoring the survivors at or above 2 Gy raises the estimate to 0.588, which is exactly what a dose response predicts, since it raises the exposed arm's mean dose; on a per-Gy scale the two agree. The likelihood does not matter either, with Poisson and negative binomial giving 0.498 and 0.495, consistent with the near-Poisson dispersion posterior of Section B.5.

Cancer-site composition is the one perturbation that matters, and it settles the sex question left open in Section B.8. Removing breast and thyroid drops the standardised ERR from 0.495 to 0.405, a fall of 18%, and the standardised female-minus-male difference collapses from 0.256 (0.074, 0.431) to 0.066 (−0.112, 0.261), its interval now covering zero. Removing non-melanoma skin, by contrast, changes almost nothing, giving 0.479 with an age-at-exposure gradient of 0.491 against the baseline 0.527, so non-melanoma skin is not the driver of the aggregate's age-at-exposure shape despite being the site with the steepest age-at-exposure effect in the published heterogeneity tests. The sex effect in the all-solid aggregate is thus carried by the female-only sites, which matches the published finding that the sex difference disappears when the analysis is restricted to non-sex-specific cancers on the EAR scale. Our non-sex-specific fit leaves a difference of 0.142 (−0.069, 0.336), attenuated by about 45% and no longer separating from zero, although the point estimate stays positive in the direction implied by the published ERR sex ratio of 1.8 for non-sex-specific cancers.

This is what disqualifies the sex difference from the featured results. Its credible interval ex-

cludes zero under every prior in Table 19, but it does not survive the data: three of the four all-solid composition variants extinguish it, the exception being the removal of non-melanoma skin. The contrast with the age-at-exposure gradient is the point. That gradient survives every data perturbation for all-solid cancer, with a credible interval excluding zero in all nine variants, including the removal of breast and thyroid, and a magnitude between 0.29 and 0.71. A subgroup effect worth featuring has to hold under both kinds of perturbation, and only one of the two does.

The site-specific results are internally consistent and match the canon where it is sharp. Lung gives a standardised ERR of 0.666 with a credible female-minus-male ERR difference of 0.480 (0.002, 1.086), against published findings of the largest sex effects for lung with female ERRs three to five times the male values. Colon at 0.297 and liver at 0.291 are lower, as expected. For stomach, banding on stomach dose rather than colon dose gives 0.256 against 0.253 with the same complete absence of heterogeneity, so the flat stomach effect surface is not an artefact of dose banding.

B.10 A low-dose city association, and what it turns out to be

B.10.1 The association

The overall 5–< 100 mGy contrast is null: the exposed-arm standardised ERR is −0.019 (−0.060, 0.023) and the population-average version is −0.005 (−0.046, 0.037). Nevertheless, city modifies this contrast, and it is the only dimension that does. Of the 49 separation tests run in this fit, five flag on the standardised footing and all five involve city.

Everything in this section is reported on the standardised footing fixed in Section B.6, and both standardisations are stated once here so that no number below is ambiguous. Over the exposed cells within each city the standardised ERR is −0.043 (−0.089, 0.003) in Hiroshima and +0.088 (0.006, 0.171) in Nagasaki, a difference of 0.132 (0.042, 0.224). Over all cells in each city, which is the footing the separation test uses, the values are −0.041 (−0.087, 0.005) and +0.085 (0.003, 0.167), giving differences from the cohort-wide standardised ATE of −0.036 (−0.063, −0.011) and +0.090 (0.026, 0.159). The two standardisations agree to within 0.005 throughout.

The same contrast computed as a person-year-weighted mean of the per-cell ERRs is about 2.4 times larger, at −0.026 (−0.101, 0.059) in Hiroshima and +0.209 (0.077, 0.365) in Nagasaki. Earlier drafts of this section quoted that figure, and the difference between 0.209 and 0.088 is the whole reason the footing is now fixed in advance: cells with a high ERR are cells with a low baseline rate, so an average of ratios systematically overstates a person-year-weighted rate ratio. The per-cell numbers appear below only where they are labelled as such.

Within Nagasaki the effect concentrates in women, with a standardised ERR of 0.166 (0.054, 0.278), and in those exposed at ages 30–39, 0.165 (0.022, 0.324). It is not a consequence of a high-dose city difference: at 500–< 2000 mGy the two cities are 0.49 (0.37, 0.60) and 0.48 (0.31, 0.67) and do not separate. City is a modifier the canonical model structurally cannot express, since it enters the baseline rate and never the ERR, so this is the clearest instance in the study of the estimator reaching somewhere the established model class cannot.

B.10.2 Is it Monte-Carlo noise, and how large is it in dose terms?

Two questions have to be settled before the association is interpreted at all.

The first is Monte-Carlo error, because the lower credible bound on Nagasaki's standardised ERR, 0.0063, excludes zero only barely. Running the contrast as two independent chains and pooling gives, for that quantity, a Gelman–Rubin statistic of 1.002, an effective sample size of 1,468 and a Monte-Carlo standard error of 0.0011, so the lower bound sits 5.9 Monte-Carlo standard errors

above zero. On the cohort standardisation the margin is thinner, at 2.6. The Nagasaki-minus-Hiroshima difference is much better resolved, with an effective sample size of 1,926, a Monte-Carlo standard error of 0.0011 and a lower bound 40 Monte-Carlo standard errors above zero. The exclusion of zero is therefore a property of the posterior rather than of the sampler, but Nagasaki's own ERR is only just distinguishable from the null, whereas the difference between the two cities is comfortably so.

The second is magnitude. The low band's mean dose contrast is 0.0299 Gy overall, 0.0315 in Hiroshima and 0.0246 in Nagasaki. Rescaling Nagasaki's standardised ERR by its own dose contrast gives an implied 3.59 (0.26, 6.96) per Gy, and on the per-cell footing 8.49 (3.11, 14.85) per Gy. The cohort's own high-dose slope is 0.529 (0.419, 0.630) per Gy. A dose reading of this result therefore requires Nagasaki's low-dose slope to be roughly seven times the cohort slope on the footing we use, and sixteen times on the footing we do not, while the two cities agree to the second decimal at high dose. That is a strong constraint, and it is what the caveat in Section B.15 about dividing by 0.03 means in practice.

The E-value sensitivity analysis says the same thing in the other units (Table 22). On the standardised footing an unmeasured confounder with associations of 1.40 with city and with cancer incidence would explain Nagasaki's low-band point estimate away, and 1.09 would move the interval to the null. For comparison, Table 14 gives 1.41 and 1.23 for the medium-dose all-solid result, which Section B.5 calls less robust, and 2.34 and 2.12 for the high-dose result. The city finding therefore sits at or below the study's own "less robust" benchmark on both measures. Given that the two cities differ in baseline solid-cancer rate by more than 20%, at 66.5 against 54.4 per 10^4 person-years in the reference arm, a confounder of strength 1.09 is not hard to name.

Table 22: E-values and per-Gy rescaling for the low-dose city contrast, on both footings. Rate ratios are exposed-arm and person-year standardised. The per-Gy column divides by each city's own mean dose contrast and is shown to make the size of the implied dose effect explicit, not because a per-Gy slope is estimated.

Footing	Quantity	Dose contrast (Gy)	ERR (95% CrI)	Implied ERR per Gy	E-value, point / interval
Standardised	Hiroshima	0.0315	−0.043 (−0.089, 0.003)	−1.38 (−2.81, 0.09)	1.26 / 1.00
	Nagasaki	0.0246	+0.088 (0.006, 0.171)	+3.59 (0.26, 6.96)	1.40 / 1.09
	Nagasaki − Hiroshima	0.0299	+0.132 (0.042, 0.224)	+4.41 (1.40, 7.50)	1.52 / 1.25
Per-cell mean	Hiroshima	0.0315	−0.026 (−0.101, 0.059)	−0.84 (−3.22, 1.86)	1.19 / 1.00
	Nagasaki	0.0246	+0.209 (0.077, 0.365)	+8.49 (3.11, 14.85)	1.71 / 1.36
	Nagasaki − Hiroshima	0.0299	+0.235 (0.100, 0.390)	+7.87 (3.36, 13.06)	1.77 / 1.43

Benchmarks from Table 14: all-solid medium dose 1.41 / 1.23; all-solid high dose 2.34 / 2.12.

B.10.3 Is it new relative to the canonical model?

Section B.7 compares the CountBCF surface with the refitted canonical risk model at the same person-year cells, and its low-dose block answers this directly. Six of the 28 low-dose subgroups depart from the canonical surface on the standardised scale. Hiroshima is one of them, at −0.063 (−0.108, −0.016). Nagasaki is not, at +0.072 (−0.011, 0.154). On the scale this study uses for every other claim, Nagasaki's low-band ERR is not distinguishable from what the canonical model already predicts; Hiroshima's is. All six low-dose departures are downward, so what the estimator is mainly saying about the low band is that the linear extrapolation from the whole 0–2 Gy range overshoots there, and that it overshoots most in Hiroshima. Read that way, the city contrast is at least as much a statement about Hiroshima sitting below the canonical prediction as about Nagasaki sitting above it, which is the first hint that its source is the reference group rather than

the exposed group.

B.10.4 Is it an artefact of the estimator?

No. The association survives a deliberately broad set of modelling checks (Table 23), covering five moderating-forest tree counts, three moderating-forest depth priors, two multipliers of the leaf concentration shared by both forests, four propensity specifications, a trim, and the not-in-city group. Across all sixteen the Nagasaki-minus-Hiroshima standardised difference runs from 0.099 to 0.146 and every 95% credible interval excludes zero, including under the tightest prior the sampler will take. As in the high-dose sweep, the magnitude scales with prior looseness while its existence does not: the smallest value, 0.099 (0.028, 0.170), comes from halving the leaf concentration, and the largest, 0.146 (0.049, 0.245), from the most prognostic-like depth prior.

Two labelling points that the earlier draft got wrong are corrected here. The leaf concentration a_0 is shared by the prognostic and moderating forests, so those two rows are not a moderating-forest sweep; the tree-count and depth-prior rows are. And the trimming row is now a trim that bites. The propensity score for this contrast spans roughly $[0.50, 0.73]$, so the fixed $[0.10, 0.90]$ trim reported previously removed no cells at all and returned numerically identical results: it was a no-op rather than a passed test. The row now trims to the central 90% of the observed score distribution.

Table 23: Robustness of the low-dose (5–< 100 mGy) city difference to modelling choices. Each row is a separate refit on the standardised footing. m_τ and the depth prior act on the moderating forest only; the leaf concentration a_0 is shared by both forests. The Nagasaki-minus-Hiroshima difference excludes zero in all sixteen specifications, and also on the per-cell footing, which is in the accompanying CSV. These fits use the shorter sensitivity MCMC budget of one chain with 1,000 burn-in and 1,000 retained draws, so the default row reads 0.128 against the 0.132 obtained from the two-chain headline fit.

Specification	Cells	Hiroshima std. ERR	Nagasaki std. ERR	Nagasaki − Hiroshima (95% C
$m_\tau = 10$ (tightest)	8,541	−0.033	0.079	0.112 (0.025, 0.200)
$m_\tau = 25$	8,541	−0.034	0.082	0.116 (0.026, 0.207)
$m_\tau = 50$ (default)	8,541	−0.035	0.093	0.128 (0.038, 0.217)
$m_\tau = 100$	8,541	−0.042	0.101	0.143 (0.051, 0.238)
$m_\tau = 200$ (loosest)	8,541	−0.040	0.103	0.142 (0.050, 0.241)
Depth prior (0.05, 4), tight	8,541	−0.040	0.083	0.123 (0.036, 0.217)
Depth prior (0.5, 2), loose	8,541	−0.042	0.102	0.144 (0.051, 0.243)
Depth prior (0.95, 2), prognostic-like	8,541	−0.043	0.103	0.146 (0.049, 0.245)
$a_0 \times 0.5$ (both forests)	8,541	−0.047	0.052	0.099 (0.028, 0.170)
$a_0 \times 2$ (both forests)	8,541	−0.044	0.088	0.132 (0.039, 0.220)
Propensity score constant	8,541	−0.038	0.089	0.127 (0.034, 0.224)
Propensity score simple logistic	8,541	−0.040	0.096	0.136 (0.044, 0.228)
Propensity score rich logistic	8,541	−0.038	0.087	0.126 (0.040, 0.222)
Propensity score random forest	8,541	−0.028	0.094	0.122 (0.022, 0.227)
Trimmed to the central 90% of $\hat{\pi}$	7,687	−0.047	0.088	0.135 (0.046, 0.224)
Not in city added to control arm	9,826	−0.039	0.103	0.142 (0.050, 0.239)

An independently fitted adjusted Poisson model with dose-band by city interactions gives the same localisation. The interaction term improves the fit at $p = 0.012$ on a likelihood-ratio test with

three degrees of freedom, and the improvement is carried entirely by the low band (Table 24), where the rate-ratio ratio is 1.154 (1.054, 1.264). The finding is therefore not a CountBCF artefact but a feature of the data that a conventional model also sees once it is told to look for it, and it has since been reproduced independently in Poisson regression by a co-author at RERF. What CountBCF adds is that it located the modifier without being told. It should be said plainly, though, what this parametric check does and does not establish. It was run on the same data and the same variable after the forest result was known, so it establishes that the association is in the data rather than in the estimator. It is not independent evidence against multiplicity, and it does not address any of the explanations below, because it shares the same reference group.

Table 24: Dose-band by city interaction terms from an adjusted Poisson model. The interaction is confined to the low band. Entries are rate-ratio ratios relative to Hiroshima with 95% confidence intervals.

Interaction term	Rate-ratio ratio (95% CI)	p
5–< 100 mGy × Nagasaki	1.154 (1.054, 1.264)	0.002
100–< 500 mGy × Nagasaki	1.111 (0.976, 1.263)	0.11
500–< 2000 mGy × Nagasaki	1.001 (0.870, 1.151)	0.99

B.10.5 Reconciliation with the published city analyses

The canonical 1958–1998 analysis reported a significant low-dose dose response but did not foreground a city-specific split [Preston et al., 2007], and the 1958–2009 update found little evidence that city modified the aggregate solid-cancer response, estimating Nagasaki's radiation effect at about 12% below Hiroshima's with an interval covering no difference [Grant et al., 2017]. Those analyses are not in conflict with what we report. They evaluate city differences in the magnitude and shape of the dose response across the full dose range, and they were motivated principally by dosimetry, in particular the larger neutron contribution in Hiroshima. A full-range comparison is dominated by the high-dose contribution, where we also find no city difference at all. What is genuinely under-examined in the published literature is a city comparison confined to the lowest band, and that is the gap this section occupies.

B.10.6 Explanation 1: the composition of the zero-dose reference group

This is the explanation the data support, and it is a known problem with a known name. French et al. [2017] show that in the LSS the choice of reference group among zero-dose survivors materially changes the estimated ERR and the inferred curvature, especially below 1 Gy, and they recommend adjusting for city-specific risk differences among distal survivors rather than pooling them. Cologne et al. [2019] show that heterogeneity in background incidence, including city differences in stomach cancer, *Helicobacter pylori* prevalence and smoking, distorts inference about the aggregate solid-cancer dose response. Neither of those mechanisms is representable by any covariate in the public file.

The geography is the reason the reference group is not exchangeable across cities. Hiroshima's hypocentre was directly above the downtown area, and because the surrounding terrain is relatively flat the damage pattern was largely radial; moving outward from the hypocentre the population becomes progressively more rural in every direction, so Hiroshima's distal survivors beyond 3 km were largely rural at the time of the bombing. Nagasaki's hypocentre was north of downtown, in the Urakami district, and the Urakami valley is enclosed by mountain ranges; most of Nagasaki's

distal survivors, 3 km and beyond to the south and southeast, were urban residents. The same nominal stratum, "zero-dose survivor beyond 3 km", therefore denotes a rural population in one city and an urban one in the other. A further asymmetry comes from cohort construction: after the initial selection of LSS participants, additional distal survivors were added to the Nagasaki cohort, randomly selected to match the original Nagasaki cohort on sex and age, which raises both the number and the proportion of distal survivors in Nagasaki. We note that "distal" is conventionally ≥ 3 km although the original selection used ≥ 2.5 km, because dose estimates were not available when the cohort was assembled and distance was the best available surrogate for separating exposed from virtually unexposed survivors.

Our own reference arm has exactly this structure, and the imbalance is large (Table 25). In Hiroshima 73.9% of reference-arm person-years come from beyond 3 km; in Nagasaki, 55.7%. Meanwhile the distal ring carries a higher adjusted solid-cancer rate than the outer proximal ring in both cities, which is the direction French et al. [2017] report. Pooling the two rings therefore raises the reference rate more in Hiroshima than in Nagasaki, which depresses Hiroshima's ERR and inflates Nagasaki's – which is the sign and, to within the precision available, the size of the observed city contrast.

Table 25: Composition of the 0–< 5 mGy reference arm by city and distance ring, with the 5–< 100 mGy band for comparison. Crude rates are cases per 10^4 person-years; the adjusted rates come from a Poisson model with the same sex-specific spline baseline used throughout. The exposed band contains no cell beyond 2.71 km, so the distal ring has probability zero of low-band membership.

City	Stratum	Ground distance (km)	Subjects	Person-years	Crude rate	Adjusted rate
Hiroshima	Reference, outer proximal	2.47–3.00	5,197	139,394	64.7	60.9
	Reference, distal	3.32–7.58	14,842	394,595	67.0	64.2
	Low band, 5–< 100 mGy	1.60–2.51	21,105	566,405	61.5	–
Nagasaki	Reference, outer proximal	2.52–2.97	6,836	170,049	52.1	56.5
	Reference, distal	3.22–6.97	8,670	214,161	54.8	58.2
	Low band, 5–< 100 mGy	1.63–2.71	6,684	163,199	56.4	–

The arithmetic is worth doing explicitly, because it predicts the size of the effect and not only its direction. Weighting the two adjusted ring rates by each city's own person-year shares gives a pooled reference rate of 63.3 in Hiroshima against 60.9 for the outer proximal ring alone, a ratio of 1.040, and 57.5 against 56.5 in Nagasaki, a ratio of 1.017. Moving from the pooled reference to the outer proximal reference should therefore move the two cities' ERRs apart by roughly $1.040/1.017 - 1 = 0.023$ in Hiroshima's favour, that is, shrink the Nagasaki-minus-Hiroshima difference by about 0.023. The observed shrinkage is 0.029.

Table 26 reports what actually happens. Two constructions are the ones the literature recommends: restricting the reference arm to the outer proximal ring, which is the RERF practice, and keeping the full reference arm but giving the prognostic forest a distal indicator, which is the French et al. [2017] recommendation of adjusting for city-specific risk differences among distal survivors. Under both, the Nagasaki-minus-Hiroshima difference falls from 0.134 (0.041, 0.231) to 0.105 (−0.018, 0.222) and 0.106 (−0.011, 0.223) respectively, and in both the interval covers zero. The same is true on the per-cell footing, where the difference falls from 0.225 (0.090, 0.388) to 0.154 (−0.020, 0.324) and 0.145 (−0.037, 0.322).

Two qualifications on that result. First, the effect is a 22% attenuation of the point estimate combined with a wider interval, and the widening is partly mechanical: the outer-proximal reference arm carries 6,194 cases against 10,009, so precision is lost as well as signal. What the exercise

establishes is not that the difference is zero but that it is not robust to the reference-group choice known to matter most for exactly this question. Second, adding the ring to the propensity score rather than to the prognostic forest attenuates less, to 0.122 (0.009, 0.239) and 0.118 (0.015, 0.233), and those intervals stay marginally clear of zero; but those are also the specifications in which the fitted score runs down to 5×10^{-9}, because the distal reference cells have probability exactly zero of low-band membership. Conditioning on the ring in the score is the wrong instrument for that reason, which is precisely why the recommended constructions handle the distal group through the baseline instead.

The last row of the table is the check Igor asked to see reported in full. Putting continuous ground distance into both the score and the prognostic forest reverses the sign of the city difference, to −0.084 (−0.332, 0.156), and destroys positivity completely: the fitted propensity spans 2×10^{-16} to 1.000. That fit is uninterpretable as a causal contrast and is reported only to document the sign reversal, which was previously mentioned in Section B.2 and nowhere else.

Table 26: The low-dose city difference under alternative treatments of the $< 3/\geq 3$ km distance ring. All entries are on the standardised footing over exposed cells; the per-cell column is given for cross-reference. The first two rows after the baseline are the constructions recommended by RERF practice and by French et al. [2017]. The last row conditions on the continuous assignment variable and is reported only to document the sign reversal and the positivity collapse.

Specification	Cases	$\hat{\pi}$ range	Nagasaki − Hiroshima, std. (95% CrI)	Same, per-cell (95% CrI)	Separates?
Baseline, full 0–< 5 mGy reference	10,009	0.50–0.73	0.134 (0.041, 0.231)	0.225 (0.090, 0.388)	Yes
Reference = outer proximal only (2.5–3.0 km)	6,194	0.69–0.84	0.105 (−0.018, 0.222)	0.154 (−0.020, 0.324)	No
Distal indicator in the prognostic forest	10,009	0.50–0.73	0.106 (−0.011, 0.223)	0.145 (−0.037, 0.322)	No
Reference = distal only (3–10 km)	8,221	0.61–0.85	0.103 (0.014, 0.196)	0.170 (0.049, 0.308)	Yes
City × distance in the score only	10,009	5×10^{-9}–0.84	0.118 (0.015, 0.233)	0.154 (−0.008, 0.324)	Yes
City × distance in score and forest	10,009	5×10^{-9}–0.84	0.122 (0.009, 0.239)	0.182 (−0.024, 0.374)	Yes
Continuous ground distance in score and forest	10,009	2×10^{-16}–1.000	−0.084 (−0.332, 0.156)	−0.063 (−0.289, 0.151)	No

The parametric model reaches the same conclusion, which matters because it is free of any question about the forest and because it is the form a radiation epidemiologist would fit (Table 27). Refitting the dose-band by city interaction of Table 24 under the two recommended reference constructions moves the joint likelihood-ratio test from $p = 0.012$ to $p = 0.106$ and $p = 0.108$, and the low-band interaction from 1.154 (1.054, 1.264) at $p = 0.002$ to 1.131 (1.006, 1.273) at $p = 0.040$ and 1.131 (1.005, 1.272) at $p = 0.041$. Restricting the reference to the distal ring instead, which is the construction that should not help if the mechanism is what we claim, leaves the result essentially intact at $p = 0.029$.

Table 27: The parametric dose-band by city interaction under four reference-group constructions. Entries are rate-ratio ratios relative to Hiroshima with 95% confidence intervals; the final column is the joint likelihood-ratio test on three degrees of freedom for the whole interaction.

Reference construction	5–< 100 mGy × Nagasaki	p	Joint LRT p
Primary, full 0–< 5 mGy reference	1.154 (1.054, 1.264)	0.002	0.012
City-specific distal baseline term	1.132 (1.006, 1.273)	0.040	0.106
Reference = outer proximal only	1.131 (1.005, 1.272)	0.041	0.108
Reference = distal only	1.151 (1.042, 1.272)	0.006	0.029

B.10.7 Explanation 2: a dose effect must have a gradient; this does not

The reference-group explanation and a dose explanation make opposite predictions inside the band, and the prediction is easy to test. A radiation effect must grow with dose: the ERR per Gy should be roughly the same in each sub-band, so the ERR itself should rise as the sub-band mean dose rises. A population offset attached to being in the exposed ring at all makes no such prediction: it produces a step at the 5 mGy boundary and then a flat ERR across the rest of the band, so the implied ERR per Gy collapses as the divisor grows.

Splitting the low band into 5–< 20, 20–< 50 and 50–< 100 mGy and refitting each against the same 0–< 5 mGy reference gives the second pattern, not the first (Table 28). Nagasaki's standardised ERR is 0.118 (0.010, 0.226) at 5–< 20 mGy, 0.044 (−0.077, 0.166) at 20–< 50 mGy and 0.101 (−0.067, 0.287) at 50–< 100 mGy: no rise with dose. The Nagasaki-minus-Hiroshima difference is 0.167 (0.049, 0.282), 0.111 (−0.024, 0.253) and 0.110 (−0.069, 0.298): a level, not a gradient, and credible only in the sub-band where there is essentially no dose at all. The implied ERR per Gy in Nagasaki accordingly falls from 13.0 at 5–< 20 mGy, where the dose contrast is 9.1 mGy, to 1.28 and 1.37 in the two higher sub-bands. A tenfold collapse in the apparent per-Gy slope as one moves from 9 mGy to 73 mGy of contrast is the signature of a fixed offset divided by a growing denominator.

Two honest qualifications. The exposed arm shrinks as the sub-band narrows, from about 2,080 exposed cases at 5–< 20 mGy to about 880 at 50–< 100 mGy, so some of the loss of credibility in the upper sub-bands is power rather than signal. And the point estimate of the city gap does not vanish above 20 mGy; it stays near 0.11. But that is precisely the point: a gap that stays at roughly the same *level* while the dose contrast grows eightfold is not a dose response, whatever else it is.

Table 28: The low band split by dose, each sub-band contrasted against the same 0–< 5 mGy reference. A dose effect predicts a roughly constant ERR per Gy down the last two columns; a population offset predicts a roughly constant ERR in the middle columns and a collapsing ERR per Gy. The data show the second.

Sub-band	Dose contrast (Gy)	Hiroshima std. ERR	Nagasaki std. ERR	Nagasaki − Hiroshima	Sep.?	Nagasaki ERR/Gy
5–< 20 mGy	0.0096	−0.049 (−0.109, 0.011)	+0.118 (0.010, 0.226)	0.167 (0.049, 0.282)	Yes	13.0
20–< 50 mGy	0.0353	−0.068 (−0.129, −0.002)	+0.044 (−0.077, 0.166)	0.111 (−0.024, 0.253)	No	1.28
50–< 100 mGy	0.0743	−0.010 (−0.087, 0.069)	+0.101 (−0.067, 0.287)	0.110 (−0.069, 0.298)	No	1.37

B.10.8 Explanations the data address less well

Cancer-site composition is not the driver. Repeating the low-band contrast on the aggregate with breast and thyroid removed, with non-melanoma skin removed, with stomach removed, and restricted to non-sex-specific sites gives a Nagasaki-minus-Hiroshima difference of 0.129 (0.031, 0.228), 0.124 (0.032, 0.222), 0.142 (0.034, 0.251) and 0.107 (0.016, 0.206) respectively, against the baseline 0.134 (0.041, 0.231); all four still separate. The result is therefore not an artefact of one or two sites with city-specific background structure, in the way that Sasaki et al. [2014] attribute a low-dose Nagasaki excess mainly to lung, liver and gallbladder. It is a property of the aggregate, which is what a reference-group explanation predicts and a site-specific one does not.

Dose assignment remains a live possibility that we cannot settle. The two cities have different shielding histories and different gamma-to-neutron mixtures, so an error in the neutron weighting or in the distal dose imputation would present as a city difference in risk per weighted Gy. Two considerations cut against it being the main explanation: the neutron fraction is smallest at exactly

these doses, and the cities agree at high dose where the neutron component is largest, which is the opposite of what a neutron-weighting error predicts. Residual and fallout dose is a different matter. It is not in DS02 at all, it is city-specific, and if present it would make the result a genuine radiation effect misattributed to city. Nothing in the public file can address it.

Case ascertainment is the explanation we can say least about from the data and most about from domain knowledge. LSS incidence, unlike mortality, depends on the two tumour registry catchment areas, and person-years are adjusted for migration into and out of them [Grant et al., 2017]; that adjustment is city-specific and varies with distance from the city centre, which is the axis of this finding. We are advised by our RERF co-author that migration adjustment, case ascertainment, registry quality, Adult Health Study participation and examination protocols are comparable between the two cities. The one relevant asymmetry is the Nagasaki distal augmentation described in Section B.10.6, which acts on the reference arm along precisely the axis identified there. The decisive check would be to repeat the low-band city contrast on the LSS mortality file, which is free of the catchment and migration problem; that file is not in the public release, and the check is recorded in Section B.15 as the first item for follow-up.

Multiplicity cannot be dismissed either. This is one flagged result among 294 separation tests (Table 15), and $p = 0.012$ from the parametric interaction is not overwhelming, particularly since that test was run on the same data and the same variable after the forest result was known.

B.10.9 What we conclude

The low-dose city contrast is a real feature of the data and is not an artefact of CountBCF: it survives every prior, propensity and trimming perturbation, and a conventional Poisson model sees it too. It is not, on this evidence, a radiation finding. On the standardised footing Nagasaki's low-band ERR is not distinguishable from the canonical model's own prediction; its E-values sit at or below the study's "less robust" benchmark; a dose reading would require a low-dose slope several times the cohort slope while the two cities agree exactly at high dose; the difference is attenuated and loses credibility under the two reference-group constructions the LSS literature recommends for exactly this problem; and it does not show the dose gradient inside the band that a dose effect must produce. What the data support is that the zero-dose reference group is not exchangeable across the two cities, for reasons of geography and cohort construction that are documented in the LSS literature and that no covariate in the public file can represent.

That is the result we report, and it is a better demonstration of the method than a novel city claim would have been. CountBCF was told nothing about city except to let it modify the effect, and it identified the one place where the canonical model class is structurally unable to express what the data contain, because city enters the baseline rate and never the ERR. Following that flag with the standard epidemiological tools then located the structure in the reference group. The division of labour is the point: the estimator says where to look, and the domain analysis says what is there.

B.11 Propensity score and overlap

The primary propensity score is a logistic regression of band membership on city by sex, natural splines with four degrees of freedom in age at exposure and in attained age, and a city by age-at-exposure interaction. Ground distance is deliberately excluded because it is the assignment mechanism rather than a confounder. Four alternatives were run for comparison: a constant score, a plain additive logistic model, a saturated spline model, and a probability random forest.

Overlap is not strained anywhere. For the featured high-dose contrast the primary score spans

0.117 to 0.752, with no cells below 0.05 or above 0.95, and the same holds for the low and medium contrasts and for every logistic specification. Only the random forest, which overfits with an area under the curve of 0.77 against 0.56 for the logistic models, pushes two cells above 0.95. The score is nearly flat because dose was determined by location and shielding rather than by demography, so the discrimination of the logistic model is barely above chance. Figure 15 displays the distributions.

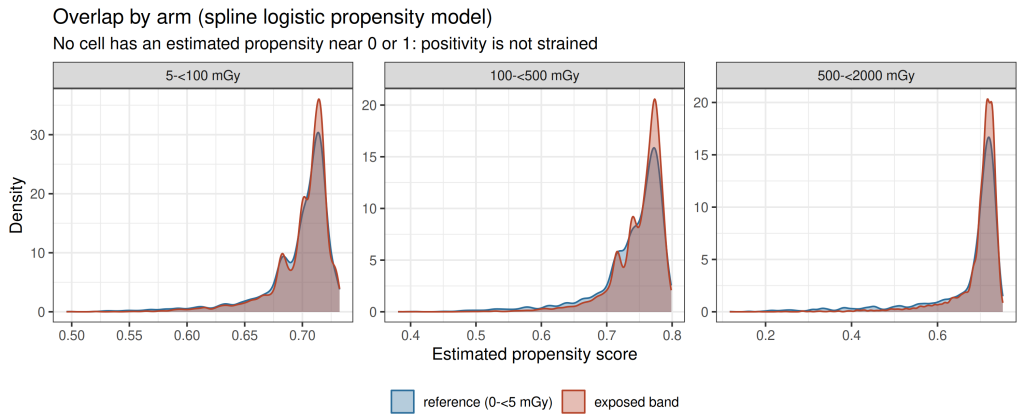


Figure 15: Propensity-score distributions by arm for each dose-band contrast. No cell falls below 0.05 or above 0.95 under the primary specification.

Weighting removes what imbalance is present. Cell-level standardised differences before weighting run from 0.056 to 0.110 for city and reach −0.206 for age at exposure; after inverse-probability weighting with the primary score they are −0.003 to 0.002 for city, −0.002 for age at exposure, −0.003 for attained age, and −0.002 for sex. Trimming is a no-op: because no cell falls outside $[0.10, 0.90]$, trimming at either $[0.05, 0.95]$ or $[0.10, 0.90]$ removes nothing and returns numerically identical results, with all three variants giving 8,249 cells, a standardised ERR of 0.495, and an age-at-exposure ERR difference of 0.795. Capping and overlap weights are therefore unnecessary here.

Table 29: Sensitivity of the featured contrast to the propensity-score specification. Even a constant score reproduces the primary estimate, because the prognostic forest already conditions on city, sex, and both age variables.

Propensity specification	Cells	Standardised ERR	EAR/10^4 PY (95% CrI)	ERR, <20 minus 40+
Constant (no covariates)	8,249	0.497	30.8 (25.1, 36.7)	0.815 (0.473, 1.19)
Simple logistic	8,249	0.500	30.9 (25.1, 36.9)	0.844 (0.494, 1.21)
Spline logistic (primary)	8,249	0.495	30.7 (25.3, 36.5)	0.795 (0.447, 1.14)
Saturated spline logistic	8,249	0.491	30.6 (25.4, 36.2)	0.773 (0.429, 1.13)
Probability random forest	8,249	0.497	30.8 (25.3, 36.4)	0.913 (0.464, 1.39)

The answer does not depend on the score at all (Table 29). Even a constant propensity gives 0.497 against the primary 0.495, because the prognostic forest is already conditioning on city, sex, and both ages, leaving the score little to correct. That is a reassuring finding rather than a disappointing one: the identification here does not hinge on getting the score right.

The featured subgroups are also not in a low-overlap tail. Within the exposed arm of the high-dose contrast, the youngest-at-exposure cells, which carry the large ERR, have the tightest propensity distributions of all, spanning 0.700 to 0.731 for males exposed at ages 0–9 and 0.714 to

0.742 for females, with no cell below 0.05 or above 0.95. The only subgroup with any low-propensity cells is those exposed at age 40 or later, with a minimum of 0.183, and that subgroup sits at the low end of the ERR gradient. The main surface is therefore not an artefact of a low-overlap tail.

One limitation of these diagnostics has to be stated explicitly, because it is the hinge of Section B.10. They are computed from a propensity model that deliberately omits ground distance, the assignment variable. What they establish is that the arms overlap on the demographic covariates, which they do, comfortably. What they cannot establish is overlap on anything distance-correlated, because within a city the dose band is very nearly a distance ring: the 5–< 100 mGy band runs from about 1.6 to 2.7 km from the hypocentre and the reference arm from 2.5 km outward, with no demographic variable in the file distinguishing the two. A flat propensity score and clean balance tables are consistent with a large unadjusted difference in the urban and rural composition of the two rings, and, as Section B.10 shows, that is what the data contain. The sharpest statement of the problem is that the 5–< 100 mGy band contains no cell beyond 2.71 km, while 74% of Hiroshima's reference person-years and 56% of Nagasaki's come from beyond 3.22 km. Those distal reference cells have a probability of low-band membership of exactly zero, which is the same positivity argument that Section B.2 uses to exclude the not-in-city group, applied one ring closer in. Adding the ring to the propensity model makes this visible immediately: the fitted score then runs from essentially zero rather than from 0.50.

B.12 Repeated person-years

Each survivor contributes person-years to many cells as they move through attained-age and calendar-period categories, so an independent-cell likelihood should in principle understate the variance. We quantify this with a frequentist cluster bootstrap of the parametric model, resampling the 1,128 survivor cohorts, defined by city, sex, age at exposure, dose category, and distance category, with replacement over 300 replicates (Table 30).

Table 30: Cluster bootstrap of the parametric model over 300 replicates of the 1,128 survivor cohorts. The design effect is the ratio of the bootstrap sampling variance to the model variance (the square of the ratio of the bootstrap standard error to the model standard error).

Dose band	Coefficient	Model SE	Cluster-bootstrap SE	Design effect
Low	−0.0036	0.0205	0.0211	1.06
Medium	0.1112	0.0259	0.0247	0.91
High	0.4261	0.0325	0.0346	1.14

The clustering penalty is at most about 7% on the standard error, and the point estimates do not move. The reason is structural: each cell already aggregates hundreds of survivors, and the exposure is constant within a survivor cohort, so the between-cell correlation induced by repeated person-years is small relative to the Poisson sampling noise in cells that are 74% zeros. This bounds the precision of the independent-cell likelihood without changing any conclusion [Field and Welsh, 2007, Kish, 1995].

We do not report a CountBCF random-intercept fit alongside the bootstrap. The current implementation accepts a random-effects design and samples the coefficients, but they never enter the count likelihood: the fitted log mean is the offset plus the prognostic term plus the treatment term only. The switch is therefore inert, and this has a consequence worth stating plainly, because it is easy to be misled by it. Any fit made with the random-effects arguments enabled is numerically an independent-cell fit that additionally reports a sampled variance component; the variance com-

ponent is a prior draw and carries no information about the survivor-cohort structure. Two rows of our own sensitivity output were originally labelled "random intercepts" on that basis and were not random-intercept fits. They have been replaced by the mechanism the package documents for group structure, a survivor-cohort indicator supplied to the prognostic forest, which does enter the likelihood. That substitution is a real test and it passes: giving the prognostic forest a 60-level city by sex by age-at-exposure indicator moves the featured standardised ERR from 0.495 (0.401, 0.594) to 0.494 (0.399, 0.592), and giving it the 432-level finest survivor cohort moves it to 0.507 (0.403, 0.616). The credible interval widens by 3% and 6% respectively, which brackets the 7% the cluster bootstrap reports and is the same answer by a different route. Adding a random-effects component that enters the count likelihood is on the development list for the method.

B.13 Low-dose sub-cohorts

Dropkin [2016] re-analysed this same file with generalized additive models restricted to women and to the 0–20, 0–100, and 0–500 mGy sub-cohorts, and reported relative risks at 10 mGy that are ten to forty-five times higher than a linear extrapolation from the full cohort. The same restrictions were run here (Table 31).

Table 31: CountBCF fits on low-dose sub-cohorts. The dose contrast column is the mean weighted-dose separation between the exposed and reference groups.

Sub-cohort	Cells	Cases	Dose contrast (Gy)	Standardised ERR (95% CrI)	EAR/10^4 PY (95% CrI)
5–< 100 mGy, women	4,360	5,474	0.029	0.0006 (−0.061, 0.059)	0.01 (−3.43, 3.13)
5–< 100 mGy, men	4,181	4,535	0.031	−0.047 (−0.110, 0.012)	−3.60 (−8.61, 0.90)
5–< 20 mGy, both sexes	3,839	7,687	0.0096	−0.005 (−0.057, 0.049)	−0.28 (−3.36, 2.90)
5–< 20 mGy, women	1,938	4,239	0.0096	−0.004 (−0.075, 0.065)	−0.25 (−4.09, 3.47)

CountBCF finds no low-dose excess in any of them. This is not a refutation of Dropkin [2016] and should not be presented as one, because the two designs answer different questions. His model fits a smooth dose response in a transformed dose coordinate, the empirical cumulative distribution function of dose, and reads a risk off it at 10 mGy, borrowing strength across the whole low-dose range. Ours contrasts a 5–< 20 mGy band against a 0–< 5 mGy band,

a mean dose separation of only 9.6 mGy, with no functional form to borrow from. At that separation the design has essentially no power: the credible interval on the ERR, at roughly ±0.07, corresponds to ±7 per Gy, which cannot discriminate a null from any of the low-dose hypotheses in play. The honest statement is that the categorical-contrast design, adopted for clarity of the causal estimand, is the wrong instrument for the low-dose question, and that a continuous-dose extension of CountBCF is what would be needed to engage with that result directly.

B.14 Excess cases and attributable fractions

Converting the exposed-arm effect back to a case count gives the comparison in Table 32. At medium and high dose the two approaches agree closely, at 372 against 402 excess solid cancers in the 0.5–2 Gy range and 178 against 254 in the 0.1–0.5 Gy range. The gap widens at low dose, where CountBCF returns a null with a slightly negative point estimate while the parametric model assigns 81 excess cases. That is the linear dose-response assumption doing work: the parametric model borrows the slope from the whole 0–2 Gy range and applies it at 30 mGy, whereas the band contrast must find the signal within the band itself. Neither is wrong, but they answer different questions, and the difference illustrates concretely what a nonparametric causal estimator will and will not deliver.

Table 32: Excess cases and attributable fractions from CountBCF, compared with the corresponding published estimates. AF denotes the attributable fraction among the exposed.

Endpoint	Band	Observed cases	CountBCF excess (95% CrI)	CountBCF AF	Published excess	Published AF
All solid	5–< 100 mGy	4,406	−86 (−274, 103)	−2.0%	81	1.8%
	100–< 500 mGy	2,112	178 (68, 286)	8.4%	254	12.0%
	500–< 2000 mGy	1,148	372 (300, 438)	32.4%	402	35.0%
Stomach	5–< 100 mGy	1,230	−58 (−165, 48)	−4.7%	14	1.2%
	100–< 500 mGy	555	0.3 (−57, 60)	0.0%	46	8.2%
	500–< 2000 mGy	269	54 (21, 89)	20.0%	69	25.9%

B.15 Caveats and open items

Several limitations qualify the results above and are stated here in full.

The low-dose city difference is robust to modelling and not robust to the reference group. It survives every prior, propensity and trimming perturbation we applied and is independently visible in a parametric interaction model, so it is not an artefact of the estimator. It does not survive the reference-group construction that RERF uses for exactly this problem, and it does not behave like a dose effect when the band is split by dose. Its E-values sit at or below this study's own "less robust" benchmark. It should be reported as what it is: a real feature of the data that the domain analysis attributes to the composition of the zero-dose reference group, not evidence of a city difference in radiosensitivity. Two explanations that we cannot rule out with the public file remain open, namely residual and fallout dose, which is city-specific and absent from DS02, and multiplicity.

The high-dose sex effect should be reported as prior-sensitive in magnitude and as carried by breast and thyroid. On the standardised footing its credible interval excludes zero under all twelve regularisation settings, so its existence is prior-robust, but its size falls ninefold across that sweep, and it does not survive the data perturbations: its interval covers zero once breast and thyroid are removed, once stomach is removed, and in the non-sex-specific fit. Its agreement with the published female-to-male ERR ratio is nevertheless close on the standardised footing, at 1.54 against 1.6. The medium-dose sex effect does not separate at all on the standardised footing, separates marginally on the per-cell footing, and has not been put through the regularisation sweep; it is not a finding.

The band-contrast design is deliberately blunt at low dose. The 5–< 100 mGy contrast has a mean dose separation of 30 mGy, so nothing about the broader low-dose controversy can be resolved with it. This cuts both ways, and it is the single most important quantitative caveat on Section B.10: any ERR expressed per Gy in that band is divided by a number close to 0.03, so a small difference in baseline composition between two groups is magnified roughly thirtyfold when it is read as a dose effect.

CountBCF's flat stomach effect surface is a defensible consequence of differential shrinkage on a weak signal, and it is stable across dose banding. The twelve-setting sweep in Table 20 now tests the regularisation concern directly: all eleven non-extreme rows, including the default repeats, produce no subgroup separation on either ERR footing, with a small surface standard deviation. The lone exception is the extreme shared $a_0 \times 2$ setting, which produces an unstable surface with 9 of 49 standardised flags and 15 of 49 per-cell flags. That pathological fit is not evidence for a stomach modifier; the ordinary sweep and the published significance tests agree that the stomach data do not support one.

Every unit of analysis is a cell rather than a person, so all weighting is by person-years and no individual-level covariate such as smoking, diet, or *Helicobacter pylori* status is available. The stomach results in particular are confounded by unmeasured causes known to matter a great deal

for that site in Japan.

The public table carries no information about case ascertainment. LSS incidence, unlike mortality, depends on the two tumour registry catchment areas, and the published person-years are adjusted for migration into and out of them [Grant et al., 2017]. That adjustment is city-specific and varies with distance from the city centre, which is the same axis as the low-dose city contrast. We are advised by the RERF co-author on this paper that migration adjustment, case ascertainment, registry quality, the proportion of Adult Health Study participants, and examination protocols are comparable between Hiroshima and Nagasaki, with one exception: after the initial selection of LSS participants, additional distal survivors were later added to the Nagasaki cohort, randomly selected to match the original Nagasaki cohort on sex and age. That augmentation raises both the number and the proportion of distal survivors in Nagasaki, and it therefore acts on the reference arm along exactly the axis Section B.10 identifies. None of this is representable in the public file.

Ground distance is excluded from the primary propensity model and from both forests. That exclusion is correct, since distance is the dose-assignment mechanism rather than a pre-exposure confounder, but within a city the dose band is nearly the same object as a distance ring. Anything distance-correlated and city-specific is therefore unadjustable by construction in the primary specification, and the overlap diagnostics of Section B.11 do not speak to it, because they come from a score that omits the assignment variable. The additional fits reported in Section B.10, which do use the coarse $< 3/\geq 3$ km ring, are the study's answer to this, and they change the low-dose city result.

The dose is a person-year-weighted cell mean of an adjusted, truncated DS02 organ dose with an assumed 35% dose error already corrected in the RERF release. Dose-measurement error beyond that correction is not modelled. Residual and fallout dose is not in DS02 at all and is city-specific, so a component of it would appear in our analysis as a city difference rather than as a dose effect.

Finally, one check we would have liked to run is not available in the public data. Replicating the low-band city contrast on the LSS mortality file would be free of the registry catchment and migration problem entirely, and would separate an ascertainment explanation from a baseline-rate explanation. That file is held by RERF and the check is left as the first item for a follow-up analysis.

B.16 Summary

On the full LSS incidence table, CountBCF reproduces the established average solid-cancer dose response and the main age-at-exposure and attained-age gradients without imposing their functional form. Its agreement with the refitted canonical surface is close over most of the observed covariate space, agreeing to four significant figures on the person-year-weighted mean excess rate at high dose, while localized departures at young attained age and short latency identify regions in which parametric extrapolation deserves scrutiny. The age-at-exposure gradient survives every regularisation setting and every data perturbation, whereas the high-dose sex effect survives the first but not the second, being carried by breast and thyroid, and its magnitude varies ninefold across the regularisation sweep even where its interval stays clear of zero, although on the standardised footing it agrees closely with the published female-to-male ratio. The model appropriately declines to create stomach-cancer heterogeneity unsupported by the site-specific data. Demographic positivity is not strained, the answer does not depend on the propensity specification, and the clustering induced by repeated person-years costs at most about 7% on the standard error. Positivity on the distance axis is a different matter, and it is the study's main identification limitation: ground distance is the dose-assignment mechanism and is therefore excluded from the score and the forests, but within a city the dose band is nearly a distance ring, so conditioning on distance collapses overlap and reverses the sign of the estimate, and anything distance-correlated and city-specific is unadjustable

by construction in the primary specification.

The low-dose city contrast is best read as a demonstration of what the method is for rather than as a new radiation finding. CountBCF located, without being told to look, a modifier that the canonical model class structurally cannot express, since city enters the baseline rate and never the ERR. The association is robust to every modelling choice we varied and is independently visible in a conventional Poisson interaction model. It is not robust to the reference-group construction that RERF uses for precisely this problem, and it does not behave like a dose effect when the band is split by dose. What the forest found is real structure in the data, and the domain follow-up locates that structure in the composition of the zero-dose reference group rather than in radiation. That is the intended division of labour: the estimator identifies where the canonical model is being asked to do something it cannot, and the epidemiology then says what is actually there. Taken together, these results show that CountBCF can recover established radiation-epidemiology structure, quantify its uncertainty, and expose interpretable departures for further investigation.